Rules, Discretion, and Soft Information in Small Business Lending*

Jun Yoo†

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Abstract

Rules eliminate discretion, but discretion can embed information. I study this tradeoff using the January 2014 SBA reform, which replaced lender-discretionary collateral appraisals with standardized haircuts for the largest federal small business lending program. Exploiting cross-state variation in homestead exemptions, which determine the friction lenders face in accessing home equity upon default, I find that spreads fall by approximately 14 basis points where business assets are the lender's primary recovery channel, with a sharp binary comparison of unlimited- versus low-exemption states yielding 10.6 basis points. A novel match of SBA loans to Uniform Commercial Code filings reveals asymmetric lender adaptation: where home equity is protected, lenders increase information acquisition to preserve private valuation knowledge, partially offsetting the spread reduction; where home equity is accessible, lenders shift toward generic blanket liens at minimal cost, passing through the full uncertainty reduction as lower spreads. The net pricing effect thus depends on the lender's endogenous behavioral response, not the policy change alone. Credit growth declines by 1.5 percentage points for experienced repeat borrowers but is unchanged for first-time borrowers, consistent with an irrecoverable relationship-information channel distinct from the recoverable collateral-information channel.

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†Baruch College, City University of New York. Email: hyunjun.yoo@baruch.cuny.edu.

1 Introduction

A recurring question in financial regulation is whether to constrain lender discretion. Discretion allows lenders to embed private information into credit decisions (Berger and Udell, 1995; Petersen and Rajan, 1994), but it also introduces conservative bias, rationing, and inconsistency (Stiglitz and Weiss, 1981; Manove et al., 2001). Reforms that standardize credit assessment aim to reduce these costs (Liberti and Petersen, 2019), yet discretion and information are co-produced: when a lender exercises judgment to assess a borrower's collateral, the resulting valuation embeds the lender's private knowledge of asset quality and the firm's creditworthiness. When a regulator replaces that judgment with a rule, what private information is lost, who bears the cost, and how large is that cost?

This paper answers these questions using a collateral valuation reform and cross-state variation in homestead exemptions. In January 2014, the SBA replaced discretionary valuations on business assets with preset haircuts based on net book value for its 7(a) program, the largest federal small business lending program. The homestead exemption determines the lender's exposure to this reform: in low-exemption states, home equity serves as an implicit collateral buffer that partially insulates the lender from valuation changes, while in unlimited-exemption states, the borrower's home is protected from involuntary seizure and business collateral becomes the lender's primary recovery channel. A model of lending with explicit and implicit collateral motivates the empirical analysis and generates testable predictions about pricing, lender documentation behavior, and credit allocation.

The reform reduced loan spreads, and the reduction is larger where lenders depend more on business collateral for recovery. A continuous difference-in-differences across all states (103,506 loans) estimates a spread reduction of 0.92 basis points per log-unit of dollar exemption (roughly 14 bp for a zero-to-unlimited shift). The sharp binary comparison of seven unlimited-exemption versus twelve low-exemption states yields 10.6 basis points, confirming the continuous gradient. For the median treated-state loan ($1.17 million), this implies approximately $1,240 in annual interest savings. The effect is stable across loan-size bins within the $350K+ tier, suggesting that the reform's pricing impact operates primarily through the homestead exemption channel.

The reduction may seem counterintuitive when preset rules destroy private valuation advantages, but the pre-reform "orderly liquidation value" baseline was conservative, so the preset haircuts raised recognized collateral value on average even as they eliminated the relationship lender's private judgment. This average pools heterogeneous collateral types and borrower responses, motivating the mechanism analysis below.

To investigate the mechanisms underlying this spread change, I match SBA loans to

Uniform Commercial Code filings across California, Florida, and Colorado, the three states where I completed the collateral classification pipeline. Florida has an unlimited homestead exemption; California and Colorado have limited exemptions ($75,000). Because UCC Article 9 covers only personal (movable) property, matched loans proxy for movable-collateral borrowers, while unmatched loans are more likely backed by real estate. The filings reveal both what the lender claimed (the extensive margin of collateral coverage) and how much effort the lender invested in specifying individual asset categories (the intensive margin of information acquisition). Three major findings emerge from this matched sample.

First, lenders adapted asymmetrically on two margins. In unlimited-exemption states, where business assets are the lender's primary recovery channel, lenders maintained or increased collateral information acquisition post-reform, investing in detailed asset-specific descriptions to reduce residual valuation uncertainty. In limited-exemption states, the reform reduced the information cost of claiming collateral credit to near zero, and lenders responded on the extensive margin: limited-exemption lenders expanded UCC filing by 13.8 percentage points more than Florida lenders post-reform (from a pooled CA+CO baseline of 45% versus Florida's 84%), and the new filings predominantly adopted generic blanket lien language, the cheapest path to perfecting a nominal claim. Second, within the movable-collateral sample the average FL-versus-CA+CO spread effect is near zero, masking offsetting heterogeneity in lender effort: Florida lenders who increased information acquisition post-reform show a net spread change of essentially zero, while those who did not saw increases of nearly 20 basis points. The effort-change interaction is directionally consistent with information acquisition costs offsetting the uncertainty-reduction benefit for movable-collateral loans, though imprecise ($p = 0.13$). Third, the fraction of loans requiring collateral declined by 4.9 percentage points more in unlimited-exemption states, indicating that the reform altered collateral requirements, not just pricing.

These information acquisition responses may also reshape credit allocation. Within the three UCC states, experienced repeat borrowers in Florida see higher credit growth than those in California and Colorado post-reform, reversing the all-states gradient. A channel consistent with this reversal is collateral encumbrance: California and Colorado lenders expanded blanket lien coverage post-reform, encumbering business assets that were previously available to support subsequent borrowing, while Florida lenders, already filing at 84% pre-reform, imposed no additional encumbrance. This evidence is suggestive given the small sample, but the direction is consistent with the model's prediction that the filing-cost channel can dominate the relationship-information channel where lenders expand their collateral base.

Falsification tests on charge-off rates, the probability of a second SBA loan, and an Express-program placebo yield precise nulls, consistent with the reform altering valuation methodology rather than borrower selection or screening quality. Inference is validated via wild cluster bootstrap at the county level (MacKinnon et al., 2023) and HonestDiD sensitivity analysis (Rambachan and Roth, 2023).

This paper makes three contributions. First, it advances the collateral valuation literature beyond the question of *what* can be pledged (Cerqueiro et al., 2016; Ongena et al., 2025; Calomiris et al., 2017) to how the *assessment method* shapes both pricing and lender conduct. Second, a novel SBA-to-UCC match linking collateral descriptions to loan-level outcomes reveals that average treatment effects of standardization can mask offsetting behavioral responses—the reform's pricing impact depends on the endogenous effort response, not the exogenous policy change alone. Third, the all-states credit growth decline combined with the within-UCC-sample reversal shows that the credit consequences of standardization depend on how lenders respond: standardization affects credit allocation through both the elimination of the relationship-information premium and the endogenous expansion of collateral claims.

Related papers study collateral pledgeability (Cerqueiro et al., 2016), asset redeployability (Benmelech and Bergman, 2009), eligibility (Ongena et al., 2025), the shadow value of different collateral types (Luck and Santos, 2024), legal reforms that reshape the collateral menu (Calomiris et al., 2017; Campello and Larrain, 2016; Aretz et al., 2020), and borrower responses to collateral regime changes (Vig, 2013). These papers establish that both the *type* of asset pledged and the *rules* governing its seizure and valuation carry distinct pricing and behavioral implications; this paper exploits a reform that held asset eligibility fixed while replacing the valuation method, isolating how the loss of discretionary assessment reshapes both pricing and lender conduct. The theoretical framework builds on the collateral structure of Rampini and Viswanathan (2025), who treat recovery rates as exogenous, and the information channel of Inderst and Mueller (2007), in which lenders use private information to reduce inefficient project rejection. I endogenize the recovery rate through the government's choice between discretionary and standardized valuation rules, and provide causal evidence that the reform both displaces and induces private information production. On regulatory discretion, Behn et al. (2022) and Plosser and Santos (2018) show that banks game discretionary risk inputs to reduce regulatory capital charges, and Agarwal et al. (2017) document heterogeneous intermediary response to a mandated-standardization program in mortgage renegotiation. I show that removing valuation discretion triggers a distinct behavioral response: lenders invest in observable information acquisition to partially recover the private valuation advantage, with measurable pricing feedback.

The lending relationships literature (Petersen and Rajan, 2002; Liberti and Petersen, 2019; Blickle et al., 2025) predicts that standardization eliminates the soft-information premium; I confirm this through the credit growth decline for experienced borrowers, validated by the null result on charge-offs. Mann (2018), Gopal and Schnabl (2022), and Gopal (2021) provide the methodological precedent for using public collateral registries (USPTO patent assignments and UCC filings) to study lender behavior. The paper also contributes to research on homestead exemptions and small business credit (Gropp et al., 1997; Berkowitz and White, 2004; Cerqueiro et al., 2017; Ersahin et al., 2021; Cole et al., 2025). These papers treat the exemption as a treatment; I use it as a moderator of a collateral reform that applies uniformly to all lenders, eliminating selection into treatment. On government-guaranteed lending (Brown and Earle, 2017; Bachas et al., 2021; Collier et al., 2025; Pan et al., 2025), I study a distinct dimension of the program: how the rules for valuing collateral shape pricing, credit allocation, and lender behavior. Hackney (2023) provides complementary evidence on the SBA 7(a) program's institutional structure and lender behavior.

The remainder of the paper proceeds as follows. Section 2 describes the SBA 7(a) program and the January 2014 collateral valuation reform. Section 3 develops a theoretical framework of small business lending with explicit and implicit collateral, effort choice, and two channels of soft information; the static equilibrium generates the pricing and adaptation predictions, and a two-period extension generates the credit growth prediction. Section 4 describes the data sources and sample construction, including the SBA-to-UCC match. Section 5 presents the empirical strategy. Sections 6–7 report the pricing results and lender adaptation evidence. Section 8 discusses credit growth, the two channels of soft information, and policy implications. Section 9 concludes.

2 Institutional Background

2.1 The SBA 7(a) Loan Program

The SBA 7(a) program is the largest federal small business lending program, approving \$15–\$25 billion annually across 40,000–60,000 loans during my sample period (2011–2019).[1] The SBA does not lend directly; it guarantees 75% of each loan, with the lender

[1]See Section 4 for data sources. The 7(a) program has two delivery channels: Standard Guaranty (full collateral documentation, 75–85% guarantee) and SBA Express (streamlined, 50% guarantee). My sample is restricted to Standard Guaranty loans above \$350K, where the guarantee rate is uniformly 75%.

bearing the remaining 25% of any net loss after collateral liquidation (pro rata loss sharing, 13 CFR 120.545(d)). The lender makes the credit decision and sets the interest rate; the SBA reviews the file and authorizes the guarantee. The lender's collateral valuation methodology therefore directly shapes expected recovery on its retained share and the spread it charges.

In addition to business collateral, the SBA requires a *personal guarantee* from all owners with 20% or greater ownership (13 CFR 120.160(a)). The personal guarantee is an unsecured legal obligation (a mandatory program requirement, not a negotiated term), exposing the guarantor's personal assets to creditor claims upon default (Berkowitz and White, 2004). The practical reach of the personal guarantee depends on the state homestead exemption, which I discuss in Section 2.4.

Collateral is required "to the maximum extent possible" (13 CFR 120.150). For loans above \$350K, the lender must collateralize to the maximum extent, using available business assets first; only if a shortfall remains is the lender required to take available equity in the personal real estate of the principals.[2] This requirement is unchanged by the reform and applies throughout the sample period. Whether the loan is deemed "fully secured" depends on the valuation methodology applied to these assets, the subject of the January 2014 reform.

2.2 Collateral Valuation Before 2014

Before January 2014, the SBA's collateral provisions did not specify any asset-type-specific valuation rules. The governing collateral requirement, set forth in the SBA's Standard Operating Procedure (SOP),[3] defined "fully secured" as follows:

> SBA considers a loan as "fully secured" if the lender has taken security interests in all available assets with a combined "**liquidation value**" up to the loan amount. "**Liquidation value**" is the amount expected to be realized if the lender took possession after a loan default and sold the asset after conducting a reasonable search for a buyer and after deducting the costs of taking possession, preserving and marketing the asset, less the value of any existing liens. (SBA collateral rules, pre-reform)

This definition delegated the entire valuation to the lender. The collateral rules provided no asset-type-specific haircuts, no reference schedule, and no formula linking asset

[2]The SBA's collateral provisions specify that personal real estate of the principals is reached only when business assets are insufficient to satisfy the "fully secured" threshold. The lender takes a security interest in available home equity to close the collateral gap, not as a first-resort requirement.

[3]The SBA issues SOPs to implement the regulations in 13 CFR Part 120. Collateral requirements are in the SOP's Subpart B, Chapter 4. I refer to these requirements as the "collateral rules" throughout.

characteristics to collateral value. "Liquidation value"[4] was whatever the lender judged it to be: an estimate of hypothetical proceeds under hypothetical default conditions, net of hypothetical costs. Two lenders evaluating identical equipment could arrive at different liquidation values, and the rules imposed no constraint on the gap.

The scope for lender-specific discretion was amplified by a further provision: for business assets whose valuation exceeded their depreciated value (net book value), the lender needed only to "obtain an independent appraisal by a qualified individual" to support the higher figure (pre-reform collateral rules). A relationship lender familiar with the borrower's maintenance practices, the local secondary market for the borrower's equipment, or the borrower's track record of preserving asset value could justify a valuation above book value through a targeted appraisal, thereby embedding private information into the collateral assessment. The assessed value of business collateral was therefore a lender-specific quantity, varying across lenders for the same borrower and across borrowers at the same lender depending on the depth of the relationship. This variation is the soft-information premium formalized in the model (Section 3).

2.3 The January 2014 Collateral Standardization

Effective January 1, 2014, the SBA replaced discretionary liquidation value with a schedule of preset haircuts based on net book value (NBV) for business assets. The revised collateral rules redefined "fully secured" as follows:

> SBA considers a loan as "fully-secured" if the lender has taken security interests in all available fixed assets with a combined "**net book value**" as adjusted below up to the loan amount. For 7(a) loans, the term "fixed assets" means real estate, including land and structures, machinery and equipment owned by the business or an EPC [Eligible Passive Company]. "**Net book value**" is defined as an asset's original price minus depreciation and amortization. (SBA collateral rules, post-reform)

The shift from "liquidation value" to "net book value as adjusted" is the core of the reform. Under the pre-reform regime, the valuation was a lender's subjective estimate of what assets would fetch in liquidation, an inherently discretionary quantity. Under the post-reform regime, the valuation is anchored to an accounting quantity (original price minus depreciation) with preset percentage adjustments. Table 1 reports the specific haircut schedule.

[4]The SOP's definition describes proceeds from "a reasonable search for a buyer," net of costs. This corresponds to what appraisal standards term "orderly liquidation value" (OLV), not "forced liquidation value" (a fire-sale scenario). I use "orderly liquidation value" throughout when referring to this pre-reform concept.

The reform's impact differs sharply by asset type. For **business assets** (machinery, equipment, furniture and fixtures), the reform replaced lender-discretionary liquidation value with formula-based haircuts anchored to net book value (original price minus depreciation), an accounting quantity recorded on the borrower's balance sheet. For new equipment, the haircut is fully mechanical; for used equipment, the lender retains a limited option to obtain an OLV appraisal for a more favorable rate (80% vs. the 50% default), but even this residual discretion operates within a narrow, preset range rather than the unconstrained judgment of the pre-reform regime. The reform therefore sharply narrows the soft-information channel: a relationship lender who knew the borrower's equipment was well-maintained can no longer translate that knowledge into an arbitrarily generous valuation.

For **real estate**, the reform introduced an 85% haircut on market value, but the underlying valuation process, an appraisal of market value functionally equivalent to the pre-reform orderly liquidation value for real property, was maintained. Real estate appraisals remain inherently discretionary: the appraiser selects comparables, adjusts for condition and location, and exercises professional judgment. The 85% haircut standardizes the *discount applied to the appraisal* but does not standardize the *appraisal itself*. Lender discretion over real estate collateral valuation is therefore preserved.

The reform therefore narrows lender discretion over business-asset valuation while preserving it for real estate.[5] The empirical strategy exploits cross-state variation in homestead exemptions to measure exposure to this change: lenders in unlimited-exemption states, who depend entirely on business collateral, are more affected than lenders in zero-exemption states, who retain a home equity buffer.

2.4 State Homestead Exemptions

Homestead exemptions are state-law provisions that protect a debtor's equity in a primary residence from creditor claims. They vary dramatically across states: from zero protection (Delaware, Maryland, New Jersey, Pennsylvania) to unlimited protection (Arkansas, Florida, Iowa, Kansas, Oklahoma, South Dakota, Texas), with the remaining states spanning \$5,000 to \$550,000 as of 2013.[6] This cross-state variation is the moderator exploited

[5]Two contemporaneous policy changes are irrelevant to my sample. In October 2013, the SBA waived both the upfront and annual guarantee fees for loans at or below \$150K. In July 2014, the SBA mandated credit score screening for Standard Guaranty loans at or below \$350K (SBA Notice 5000-1314, codified in SOP 50 10 5(G)). My sample restriction to loans above \$350K excludes both (Section 5).

[6]Exemption amounts are drawn from a monthly state-level panel constructed from state statutes and secondary sources. Three states (California, New York, Washington) have county-level variation. See Appendix C for construction details and the full distribution.

in the empirical design.

The exemption matters for SBA lending not primarily through bankruptcy but through its effect on the lender's outside option in default.[7] When an SBA borrower defaults and business collateral is insufficient to cover the net loss, the lender pursues the mandatory personal guarantee by seeking a deficiency judgment against the guarantor's personal assets. The homestead exemption determines how far this claim reaches. In zero-exemption states, the borrower's home equity is fully exposed, providing the lender with a substantial additional recovery source that effectively extends the collateral base beyond the business. In unlimited-exemption states, the primary residence is shielded, and the deficiency judgment can reach only non-exempt personal assets (financial accounts, non-homestead real estate, vehicles above state limits), which for most small business owners are modest (Gropp et al., 1997; Berkowitz and White, 2004).

The economic consequence is a sharp difference in how lenders in the two regimes respond to collateral valuation rules. A lender in a zero-exemption state operates with a two-layer recovery structure: business assets valued under the SBA's collateral rules, plus the borrower's home equity accessible through deficiency judgment. Even if the collateral reform compresses business-asset valuations, the home equity buffer partially insulates the lender's expected recovery. A lender in an unlimited-exemption state, by contrast, has no such buffer: recovery depends almost entirely on the business assets whose valuation the reform standardizes. The same policy shock therefore generates a larger change in expected recovery, and hence a larger pricing response, where the exemption removes the lender's fallback. This asymmetry in exposure to collateral valuation rules is the source of the identifying variation.

Among the unlimited-exemption states, Texas is unique: the Texas Constitution (Article XVI, Section 50) renders home equity liens for business purposes constitutionally unenforceable, so Texas borrowers cannot voluntarily pledge home equity even if they wish to.[8] The extreme-exemption classifications are time-invariant throughout the sample period: no state moved into or out of the unlimited or zero category between 2011 and 2019. The variation is therefore a permanent feature of the legal environment, not a consequence

[7]Bankruptcy filings are uncommon among SBA borrowers because the personal guarantee makes bankruptcy unattractive: filing discharges the business debt but triggers creditor claims against the guarantor's non-exempt personal assets. The more common resolution is negotiated workout or collateral liquidation outside of bankruptcy court. The exemption's relevance is therefore its effect on deficiency judgment recovery, not on bankruptcy discharge.

[8]In other unlimited-exemption states, the exemption protects against involuntary seizure but does not prohibit voluntary pledging. A borrower in Florida could, in principle, offer a second lien on the home as additional collateral. This attenuates the treatment in those states relative to Texas, where the constitutional prohibition makes the lender's exclusive dependence on business collateral exact.

of the reform or of contemporaneous policy changes.

The empirical design exploits this variation in two forms. The primary specification uses the continuous exemption amount across all states, from $0 to unlimited. The secondary specification restricts to seven unlimited- and twelve low-exemption states, producing coefficients that map to the model's propositions (Section 5).

3 Theoretical Framework

I develop a model of small business lending in which the SBA lender's recovery upon default depends on both *explicit collateral* (business assets, valued via a Uniform Commercial Code filing) and *implicit collateral* (unprotected home equity). Drawing on Rampini and Viswanathan (2025), I show how the January 2014 reform, which replaced lender-discretionary valuations with preset haircuts, simultaneously reduces valuation uncertainty and eliminates the soft-information premium that differentiated relationship borrowers from new borrowers. The static equilibrium generates pricing and adaptation predictions (Propositions 1 and 2), and a two-period extension generates the credit growth prediction (Proposition 3).

A numerical example fixes ideas. Consider a $500,000 SBA loan backed by $300,000 in new equipment and a 75% guarantee. Because the guarantee is pro-rata, the lender bears 25% of every dollar of shortfall, so every dollar of recognized collateral directly reduces the lender's expected loss. Pre-reform, the lender valued equipment using subjective "orderly liquidation value." A conservative lender might assess 50 cents on the dollar ($150,000); a relationship lender who has inspected the equipment might assess 70 cents ($210,000). The collateral shortfall is $500,000 minus the assessed value: $350,000 for the conservative lender, $290,000 for the relationship lender. At a 25% loss share, the lender's expected loss is $87,500 versus $72,500, a $15,000 difference priced into the spread. The borrower knows the incumbent's assessment but not what a competing lender would assign: the $60,000 valuation range is opaque, limiting the ability to shop and generating an informational rent for the incumbent.

Post-reform, a preset haircut assigns 75% of purchase price ($225,000), identically for all lenders. Two effects operate. A *level effect*: recognized collateral rises for both lender types (shortfall falls to $275,000, loss share to $68,750), reducing expected losses. A *dispersion effect*: cross-lender variation collapses to zero, the borrower can verify and shop, and competition compresses informational rents. The relationship lender loses the collateral-valuation advantage but retains private information on default risk and borrower character, soft information that operates through channels other than the collateral assessment

(Section 8). For real-estate-backed loans, the level effect is larger (the 85% preset is generous relative to the conservative baseline), but the dispersion effect is smaller, because appraisals were already partially standardized through USPAP methodology.

3.1 Setup

Two periods, $t \in \{1, 2\}$. Three agents: a risk-neutral borrower, a risk-neutral lender, and the government (SBA). The borrower has initial wealth w_1, business collateral C_1 (tangible assets pledgeable via a UCC filing), and home equity E. The borrower operates a decreasing-returns project: invest I_t to produce $f(I_t)$ with probability $(1 - \pi)$ and 0 with probability π, where $f' > 0$, $f'' < 0$, and $f'(0) > 1 + \rho$. The risk-free rate is ρ.

As described in Section 2, the borrower is subject to a mandatory personal guarantee, exposing all personal assets to creditor claims upon default. The homestead exemption limits the lender's reach: $E_a = \max(E - H, 0)$. In unlimited-HE states ($H = \infty$), the personal guarantee cannot reach the borrower's primary residence, producing effective limited liability over business assets only.[9] Because the personal guarantee is an unsecured legal obligation, the borrower cannot adjust this exposure in response to collateral valuation changes, reinforcing the separation between default risk (π) and recovery terms (θ_c, θ_u).[10]

Upon default, the lender recovers:

$$\text{Recovery} = \theta_c \cdot C_t + \theta_u \cdot E_a + g \cdot \max\big(L_t - \theta_c \cdot C_t - \theta_u \cdot E_a,\ 0\big), \tag{1}$$

where θ_c is the assessed valuation rate on business collateral, $\theta_u < \theta_c$ is the recovery rate on unprotected home equity, and $g = 0.75$ is the SBA guarantee rate.[11]

Business collateral evolves across periods as a fraction of cumulative investment: $C_2 = C_1 + \varphi I_1$, where $\varphi \in (0, 1)$ is the share of period-1 investment that becomes pledgeable.[12]

[9] E_a reflects effective availability: $E_a = 0$ exactly in Texas (home equity business liens are constitutionally unenforceable), approximately zero in other unlimited-exemption states (voluntary pledging possible but involuntary seizure blocked), and $E_a = E$ in zero-exemption states. See Section 2 for institutional details.

[10] This separation follows screening models with collateral (Bester, 1985; Besanko and Thakor, 1987) and pledgeable-income models (Holmström and Tirole, 1997). In contrast to Holmström and Tirole (1997), where limited pledgeability arises from borrower moral hazard, θ_c here is a regulatory parameter set by the SBA's haircut schedule.

[11] The 75% guarantee applies to Standard 7(a) loans above \$150,000. Because my identification relies on the collateral-review threshold at \$350,000, all loans in the estimation sample carry the 75% rate. The ordering $\theta_u < \theta_c$ reflects legal priority: the UCC filing grants a perfected security interest, whereas recovery on home equity requires an unsecured deficiency judgment.

[12] The restriction $\varphi < 1$ ensures that the collateral shortfall (Assumption 1) holds in both periods; see Appendix A.

Assumption 1 (Collateral shortfall)**.** $\theta_c \cdot C_t + \theta_u \cdot E_a < L_t$. *Total collateral never fully covers the loan; the guarantee is always triggered upon default.*[13]

Collateral valuation and soft information. Under the pre-reform collateral rules, the lender valued business assets using "orderly liquidation value" (OLV), a subjective assessment. I decompose the pre-reform valuation into three components:

$$\theta_c^{\text{pre}}(i,l) = \bar{\theta} + \delta(i,l) + \varepsilon(i,l), \tag{2}$$

where $\bar{\theta}$ is the baseline conservative valuation, $\delta(i,l) \geq 0$ is the *soft-information premium* (the lender's private assessment of collateral quality above the baseline), and $\varepsilon(i,l)$ is a mean-zero valuation error with variance σ^2 that captures the lender's residual uncertainty about asset recovery.[14]

The lender prices the loan using the expected valuation $\bar{\theta}+\delta$ but charges a risk premium for the residual uncertainty σ^2, because valuation errors translate into uninsured losses on the lender's unguaranteed 25% share.[15]

I distinguish two components of soft information:

Assumption 2 (Two separable channels of soft information)**.** *The soft-information premium decomposes as $\delta = \delta_c + \delta_r$, where:*

(i) Collateral-channel soft information *(δ_c): the lender's private knowledge of asset quality, condition, maintenance, and redeployability. This knowledge is embedded in the lender's collateral assessment; a lender who knows the equipment is well-maintained can describe it precisely in the UCC filing, specifying individual assets, their condition, and their market depth. Post-reform, this knowledge is partially recoverable through information acquisition.*

(ii) Relationship-channel soft information *(δ_r): the lender's private knowledge of borrower reliability, business viability, and management quality, acquired through repeated interactions. This knowledge resides in the lending relationship, not in any document. Post-reform, a more detailed UCC filing cannot encode the lender's trust in the borrower.*

[13] Empirically reasonable at the loan sizes in my sample. Under the shortfall condition, recovery simplifies to $g \cdot L_t + (1-g)(\theta_c \cdot C_t + \theta_u \cdot E_a)$. This additive form is used in all subsequent derivations.

[14] Three properties of δ: (i) *Relationship-dependent:* $\delta > 0$ for repeat borrowers at the same lender; $\delta = 0$ for new borrowers. (ii) *Lender-specific:* different lenders assign different δ to the same borrower. (iii) *Unverifiable:* δ is the lender's private information.

[15] The soft-information premium complements the private-signal mechanism of Blickle et al. (2025): δ captures private information about *collateral* quality (a loss-given-default channel) rather than borrower quality (a default-probability channel). The uncertainty premium $\kappa\sigma^2$ also reflects guarantee repurchase risk: under SBA servicing rules, the agency can deny or reduce a guarantee claim if the lender's collateral valuation at origination is found to be inadequate. Pre-reform, the lender bore the risk that the SBA would later disagree with the subjective OLV. Post-reform, the preset schedule eliminates this repurchase exposure.

The maintained assumption is that the two channels have different recovery technologies: δ_c can be partially recovered through effort (equation 9), while δ_r cannot ($\tilde{\delta}_r(e) = 0$ for all e). The experienced-versus-first-time split provides indirect evidence for this separation: if $\delta_r > 0$ drives the credit growth decline, the effect should concentrate among experienced borrowers regardless of lender information acquisition (Section 8).

The reform. The January 2014 reform replaced discretionary OLV with preset haircuts that differ by asset type:

$$\theta_c^{\text{post}}(j) = \bar{\theta}_{\text{std}}(j) \qquad (\delta = 0,\ \sigma^2 = 0 \text{ for all borrowers and lenders}), \tag{3}$$

where $j \in \{\text{movable}, \text{immovable}\}$ indexes asset type. The preset haircuts are $\bar{\theta}_{\text{std}}^{\text{imm}} = 0.85$ for improved real estate and $\bar{\theta}_{\text{std}}^{\text{mov}} \in \{0.50, 0.75\}$ for used and new equipment, respectively.[16]

The reform eliminates valuation uncertainty ($\sigma^2 \to 0$) and the soft-information premium ($\delta \to 0$). The preset is a deterministic function of verifiable inputs (purchase price minus depreciation minus prior liens), producing a collateral value with no residual uncertainty and no role for private information.[17]

Assumption 3 (Differential pre-reform uncertainty by asset type)**.** *Two conditions hold:*

(i) Valuation uncertainty: $\sigma^2_{mov} > \sigma^2_{imm}$. *Immovable assets (improved real estate) were already valued using comparable-sales appraisals, a standardized methodology with low cross-lender dispersion. Movable assets (equipment, inventory, intangibles) lacked comparable benchmarks; lenders relied on subjective assessments of condition, redeployability, and market depth, producing high cross-lender dispersion.*

(ii) Preset generosity: *The preset haircuts differ in their relationship to the pre-reform baseline:*

$$\bar{\theta}_{std}^{imm} > \bar{\theta}^{imm} \qquad \text{and} \qquad \bar{\theta}_{std}^{mov} \gtreqless \bar{\theta}^{mov} + \delta.$$

For immovable assets, the 85% preset exceeds the pre-reform conservative baseline, producing an unambiguous collateral-valuation improvement. For movable assets, the 50–75% preset

[16] The full haircut schedule from SOP 50 10 5(F): improved real estate at 85% of market value, new machinery and equipment at 75% of price, used machinery and equipment at 50% of net book value. See Section 2 for details.

[17] This contrasts with pre-reform OLV, which required the lender to estimate what a willing buyer would pay under orderly disposition, a subjective assessment with no standardized methodology, generating substantial cross-lender variation ($\sigma^2 > 0$).

may or may not exceed the pre-reform valuation inclusive of the soft-information premium; the net effect depends on the magnitude of δ.

This assumption motivates the UCC analysis (Sections 7–7.2): effort adaptation operates on movable-collateral loans, where pre-reform uncertainty was highest and the preset is least generous relative to δ.

3.2 Static Equilibrium

Pre-reform equilibrium. The competitive lender sets the spread to break even in expectation. Under Assumption 1, the pre-reform spread reflects both expected recovery, through effective collateral $\Omega \equiv (\bar{\theta} + \delta)C + \theta_u E_a$, and a risk premium for residual valuation uncertainty σ^2:

$$s = \underbrace{\frac{\pi}{1-\pi}\left[(1+\rho) - g - (1-g) \cdot \frac{\Omega}{L}\right]}_{\text{break-even spread from expected recovery}} + \underbrace{\frac{\pi}{1-\pi}(1-g) \cdot \kappa\sigma^2 \cdot \frac{C}{L}}_{\text{valuation uncertainty premium}}, \tag{4}$$

where $\kappa > 0$ scales the uncertainty premium.[18]

The sensitivity to the collateral valuation rate is:

$$\frac{\partial s}{\partial \theta_c} = -\frac{\pi}{1-\pi}(1-g) \cdot \frac{C}{L} < 0, \tag{5}$$

proportional to C/L. The SBA guarantee dampens the pass-through: with $g = 0.75$, only $(1-g) = 0.25$ of any collateral valuation change flows through to the spread.[19]

Equation (4) has two empirical implications for the pre-reform regime. First, the uncertainty premium $\kappa\sigma^2(C/L)$ represents the information cost embedded in spreads: lenders charge more when they know less about what collateral is worth. This premium is larger for movable assets (where pre-reform OLV was most subjective and σ^2 was highest) and larger where business collateral is the primary recovery channel (higher C/L). Second, the

[18] The proportionality $\lambda(\sigma^2) = \kappa\sigma^2$ is a simplification. Appendix A derives the exact form under normality: λ arises from the convexity of the loss function in valuation errors. The nonlinear closed form is approximately linear in σ^2 when $\theta_u E_a/C$ is small, which holds in unlimited-HE states ($E_a = 0$). The linear approximation simplifies the effort first-order condition without affecting comparative statics.

[19] For a loan with $C/L = 0.60$ and $\pi = 0.10$, the sensitivity is $[0.10/0.90] \times 0.25 \times 0.60 = 0.0167$ per unit of θ_c. The full schedule change from discretionary OLV ($\sim$50%) to preset NBV ($\sim$75%) is approximately 25 percentage points, implying a predicted spread reduction of $0.0167 \times 0.25 \approx 0.0042$, or approximately 42 basis points. The actual binary estimate of 10.6 basis points is approximately one-quarter of this prediction, consistent with voluntary home equity pledging in non-Texas states attenuating the effective treatment to roughly 25% of the full schedule change. See Appendix A for the calibration.

soft-information premium δ creates a relationship discount: lenders with private knowledge of asset quality charge less because their expected recovery is higher. Both components are eliminated by standardization.

Post-reform equilibrium without effort. With $\delta = 0$ and $\sigma^2 = 0$, and no information acquisition, the post-reform spread is:[20]

$$s^{\text{post}} = \frac{\pi}{1-\pi}\Big[(1+\rho) - g - (1-g)\cdot\frac{\bar{\theta}_{\text{std}}C + \theta_u E_a}{L}\Big]. \tag{6}$$

The net spread change is:

$$\Delta s = -\frac{\pi}{1-\pi}(1-g)\frac{C}{L}\Big[\underbrace{(\bar{\theta}_{\text{std}} - \bar{\theta})}_{\text{baseline shift}} + \underbrace{\kappa\sigma^2}_{\text{uncertainty reduction}} - \underbrace{\delta}_{\text{information loss}}\Big]. \tag{7}$$

The bracket contains three terms. The first two are positive (the preset exceeds the conservative baseline, and uncertainty vanishes), reducing spreads. The third is negative (the soft-information premium is eliminated), increasing spreads. The net effect is negative (spreads fall) when the baseline improvement and uncertainty reduction together exceed the information loss.

The elimination of σ^2 also reduces the information cost of claiming collateral credit. Pre-reform, obtaining the SBA's collateral adequacy credit required the lender to acquire costly information about asset values (inspect equipment, estimate redeployability, produce an OLV appraisal). Post-reform, the preset provides credit based on verifiable accounting records, and a blanket UCC filing is sufficient for legal perfection. The lender can claim collateral credit without acquiring information about what the assets are actually worth. This reduction in information cost should be observable on two margins: lenders who previously did not file UCC (because the information cost exceeded the return) should begin filing, and the filing response should be larger where lenders had been under-investing in business-asset documentation.

Post-reform equilibrium with effort. The preset eliminates regulatory uncertainty ($\sigma^2 = 0$ for the SBA's assessment), but the lender may still face uncertainty about actual liquidation recovery. Equipment that the SBA credits at 50% of NBV might recover more or less depending on condition, market depth, and redeployability. Let σ_0^2 denote the pre-reform

[20] The distinction between regulatory assessment ($\sigma^2 = 0$) and actual recovery ($\sigma^2_{\text{recovery}} > 0$) matters for the effort choice below. The preset eliminates uncertainty about what the SBA will credit; it does not eliminate uncertainty about what the lender would actually recover in liquidation. Effort targets the latter.

baseline recovery uncertainty (distinct from the regulatory assessment uncertainty set to zero above). I allow the lender to invest effort $e \in [0,1]$ in collateral information acquisition to reduce this residual uncertainty:

$$\sigma^2_{\text{recovery}}(e) = \sigma_0^2 \cdot (1-e). \tag{8}$$

At $e = 0$, the lender accepts the preset with no additional investigation; at $e = 1$, the lender fully resolves the recovery uncertainty through detailed asset inspection and documentation. Effort also partially recovers the collateral-channel soft-information premium:

$$\tilde{\delta}_c(e) = \delta_c \cdot h(e), \qquad h(0) = 0,\ h(1) = 1,\ h' > 0,\ h'' < 0. \tag{9}$$

Crucially, effort does *not* recover the relationship-channel premium: $\tilde{\delta}_r(e) = 0$ for all e. A more detailed UCC filing can describe asset condition and redeployability (collateral-channel information) but cannot encode the lender's trust in the borrower or knowledge of the borrower's business viability (relationship-channel information).

The information acquisition cost is $c(e)$ with $c' > 0$ and $c'' > 0$. The cost reflects the expense of investigating each additional asset category: standard equipment with serial numbers and purchase invoices is cheap to verify; specialized machinery, intangibles, and accounts receivable aging require progressively more investigation.[21]

The lender's optimal effort e^* equates the marginal benefit of recovery-uncertainty reduction and partial δ_c recovery with the marginal cost:

$$\underbrace{\frac{\pi}{1-\pi}(1-g)\frac{C}{L}\left[\kappa\sigma_0^2 + \delta_c \cdot h'(e^*)\right]}_{\text{marginal benefit}} = \underbrace{c'(e^*)}_{\text{marginal cost}}. \tag{10}$$

The marginal benefit is proportional to C/L, which is higher in unlimited-HE states ($E_a = 0$ implies smaller equilibrium L^* for given C). The marginal cost does not depend on the homestead exemption. The asymmetry is entirely on the incentive side: where business assets are the primary recovery channel, each dollar of recovery-uncertainty reduction is worth more to the lender. Therefore:

$$e^*\big|_{H=\infty} > e^*\big|_{H=0}. \tag{11}$$

Lenders in unlimited-HE states invest more because business collateral is their primary re-

[21]The convexity of c arises from the ordering of assets by information acquisition cost: the first blanket lien category documented faces low marginal cost, but extending coverage to illiquid or hard-to-value asset classes requires substantially more due diligence per category.

covery channel. The information acquisition differential should be observable in collateral filing data: unlimited-HE lenders should file more detailed, asset-specific descriptions, while limited-HE lenders should shift toward generic blanket claims.

3.3 Predictions from the Static Framework

The static equilibrium generates two testable predictions about pricing and lender behavior. I treat these as a stylized framework that organizes the empirical analysis, not as a structural model.

Assumption 4 (Similar business collateral)**.** *Business collateral C is sufficiently similar across HE regimes that C/L differences are driven primarily by the equilibrium loan size channel. The empirical specifications control for collateral share; Section 6 reports balance tests.*

Proposition 1 (Spread Reduction)**.** *Under Assumptions 1–4, the spread difference-in-differences is negative:*

$$\text{Spread DD} = \Delta s\big|_{H=\infty} - \Delta s\big|_{H=0} < 0.$$

Spreads fall on average post-reform, with a larger reduction where homestead exemptions are higher. For immovable-collateral loans, the DD is unambiguous ($\bar{\theta}^{imm}_{std} > \bar{\theta}^{imm}$, low pre-reform σ^2). For movable-collateral loans, the average effect depends on the balance of uncertainty reduction and information loss, but the DD remains negative because unlimited-HE states have higher C/L (since $E_a = 0$ reduces the equilibrium loan size for given business collateral), amplifying the sensitivity to any valuation change.

The C/L asymmetry arises because implicit collateral relaxes the borrowing constraint in zero-HE states ($\partial L^*/\partial E_a > 0$; Appendix A), lowering C/L there and amplifying the spread sensitivity (5).

Remark 1 (Effort attenuation). In unlimited-HE states, effort creates a tradeoff for the observed spread: it reduces the lender's residual recovery uncertainty and recovers part of the collateral-channel soft-information premium—both of which *lower* the spread—but effort is costly, and the convex cost schedule $c(\cdot)$ is passed through to the borrower. From the post-reform spread and equations (8)–(9), the change in the post-reform spread relative to the no-effort benchmark is:

$$s^{\text{post}}(e^*) - s^{\text{post}}(0) \;=\; -\frac{\pi}{1-\pi}(1-g)\frac{C}{L}\Big[\kappa\sigma_0^2 \cdot e^* + \delta_c \cdot h(e^*)\Big] \;+\; \frac{c(e^*)}{L}.$$

The bracket contains effort's total benefits, which enter negatively (reducing the spread): $\kappa\sigma_0^2 \cdot e^*$ is the cumulative reduction in residual recovery uncertainty, and $\delta_c \cdot h(e^*)$ is the partial recovery of the collateral-channel premium. The final term $c(e^*)/L$ is effort's cost, which raises the spread. At the interior optimum (10), marginal benefit equals marginal cost, but the *average* cost pass-through to the spread can be substantial when $c(\cdot)$ is convex. The testable implication is that within the UCC-matched movable-collateral sample, lenders who increase their effort more at the reform boundary should exhibit smaller net spread reductions—effort's cost pass-through offsets part of the uncertainty-reduction benefit. This is testable using the within-lender pre-to-post effort change from UCC filing data.

Proposition 2 (Lender Adaptation by HE Regime). *From* (11)*, lenders in unlimited-HE states invest more in collateral information acquisition post-reform ($e^*_{H=\infty} > e^*_{H=0}$). This manifests as:*

(i) Higher information acquisition (more asset categories listed, more precise descriptions) in unlimited-HE states.

(ii) A faster shift toward generic blanket liens in limited-HE states, where effort is less valuable because home equity buffers the lender's recovery.

(iii) Within unlimited-HE states, lenders who increase effort see smaller spread reductions, reflecting the pricing feedback from information acquisition costs.

The reform does not change the economic value *of collateral recovery; it changes the lender's* uncertainty about that value *and the information acquisition required to reduce it.*

3.4 Dynamic Extension: Credit Growth

The static framework addresses pricing; a two-period extension generates predictions about credit growth for repeat borrowers who straddle the reform boundary (first loan pre-reform, second loan post-reform). Because the first-period spread was set under the pre-reform regime, the straddling borrower's internal equity is unaffected by the policy change, and period-2 credit growth depends on the post-reform valuation alone.

Proposition 3 (Credit Growth and the Two Channels of Soft Information). *Holding lender filing behavior fixed, the reform reduces credit growth for repeat borrowers, with a larger reduction in unlimited-HE states relative to low-HE states:*

$$\Delta \log(L_2/L_1) \propto -\Big[\delta_r + \delta_c \cdot (1 - h(e^*))\Big] \cdot \frac{C}{L}. \tag{12}$$

The effect is larger for borrowers with prior SBA experience ($\delta_r > 0$) than for first-time borrowers ($\delta_r \approx 0$). Effort partially recovers δ_c but cannot recover δ_r: even lenders who increase information *acquisition cannot encode the relationship premium through better UCC filings. This separates the two channels empirically:* δ_c *affects pricing and is partially recoverable through effort;* δ_r *affects credit access and is irrecoverable.*[22]

The model also requires that default probability π, loan counts, and borrower composition are unchanged at the reform boundary (Section 5). The maintained assumption that π is exogenous to the valuation methodology implies no first-order effect on charge-off rates; I test this as a falsification exercise in Section 6. Violation of Assumption 1 attenuates all predicted effects toward zero. Appendix A provides proofs and discusses alternative interpretations.

4 Data

4.1 SBA 7(a) Loan Records

The primary dataset consists of administrative loan-level records from the SBA's 7(a) program, obtained via the Freedom of Information Act.[23] Each record contains the gross approval amount, interest rate (fixed or variable), loan term, guarantee percentage, approval and disbursement dates, borrower name, address, state, two-digit NAICS code, lender identity and location, and loan status (active, paid in full, cancelled, or charged off). I restrict to loans approved during 2011Q1–2019Q4, providing a balanced pre- and post-reform window around the January 2014 collateral standardization.

Loan spreads are constructed using daily Treasury yields (3-month T-bill and 1-, 2-, 3-, 5-, 7-, 10-, 20-, and 30-year maturities) from FRED and the U.S. Treasury. For variable-rate loans, the spread equals the stated interest rate minus the 3-month T-bill rate on the approval date. For fixed-rate loans, the spread equals the stated rate minus the piecewise-linearly interpolated Treasury yield curve rate at the loan's maturity.

[22] See Appendix A for the full derivation, including retained earnings and collateral accumulation channels. When lender filing behavior is endogenous, the credit growth prediction can reverse: if limited-HE lenders expand collateral claims post-reform, encumbering previously undocumented assets, repeat borrowers face tighter borrowing constraints regardless of the relationship channel. Section 8 examines this mechanism.

[23] SBA 7(a) loan data are publicly available through the SBA Open Data portal (`data.sba.gov`). Brown and Earle (2017) provide a comprehensive analysis of 7(a) program effects on firm employment using similar data.

4.2 UCC Filings and Collateral Classification

UCC Article 9 governs security interests in *personal property* (that is, movable assets such as equipment, inventory, accounts receivable, chattel paper, and general intangibles).[24] A UCC filing does not, and legally cannot, perfect a security interest in real estate. This statutory scope makes UCC match status a natural proxy for collateral type: loans successfully matched to a UCC filing are backed primarily by movable business assets, while unmatched loans are more likely backed by real estate or a mixed collateral base that includes immovable property. This distinction is central to testing Proposition 1, which predicts a larger spread reduction for immovable-collateral loans (generous preset haircut, low pre-reform valuation uncertainty) than for movable-collateral loans (less generous haircut, high pre-reform uncertainty, effort-dependent residual risk).

Each filing records the debtor, the secured party, and a free-text collateral description detailing what assets the lender has a lien against. This collateral-level detail is unavailable in the SBA administrative data, which records only whether collateral was pledged, not its composition. Linking SBA loans to their UCC filings therefore enables direct observation of what the lender actually claimed as collateral, how broadly the lien was drafted, and how much effort the lender invested in specifying individual asset categories, the variation needed to test the adaptation mechanism (Proposition 2).

UCC filings are maintained at the state level and are not available through any centralized federal source. I obtain filings from three states: California and Florida, where I web-scrape the state filing office portals, and Colorado, where filings are publicly available through the state open data portal.[25] The restriction to three states reflects data availability, not sample design: no centralized UCC database exists, and most states do not provide bulk electronic access to filing records. These three states account for approximately 20% of SBA 7(a) lending volume and span diverse economic environments (California's technology and agriculture sectors, Florida's services and tourism, Colorado's energy and professional services), providing reasonable coverage despite the constraint.

I link SBA loans to UCC filings through an industrial-scale fuzzy matching pipeline that combines name-based blocking, geographic verification, and a restricted time window around the loan approval date. Each candidate pair receives a composite score in-

[24] UCC Article 9, §9-109(a)(1): "this article applies to a transaction, regardless of its form, that creates a security interest in personal property." Real property (land, buildings, fixtures affixed to land) falls outside Article 9 and is governed by state real estate law. The one exception is fixtures: goods that become attached to real property may be perfected under Article 9 (§9-334), though they are classified as movable property for UCC purposes.

[25] Colorado UCC filings are published at `data.colorado.gov`. California and Florida filings are obtained through automated extraction from the respective Secretaries of State online filing systems. See the replication package for extraction scripts.

corporating name similarity and geographic proximity, with tiered acceptance gates that balance precision against recall. Match quality is evaluated through repeated manual verification of random samples across all three states; precision exceeds 95%, with the majority of retained matches classified as high confidence. The estimation sample for the collateral composition analysis comprises 18,341 classified loans across California, Florida, and Colorado. Appendix C.3 provides the full algorithmic details.

Collateral classification. Each matched filing's collateral description is classified into one of 14 Article 9 asset categories. Blanket liens (filings that claim "all assets" of the debtor) dominate the regression sample (approximately 80% of >$350K loans), consistent with the SBA's collateral adequacy requirement. Within blanket liens, I distinguish generic filings ("all assets of debtor") from itemized filings that specify individual asset categories, a distinction that captures the lender's information acquisition.

Effort index and effort change. The composite effort index standardizes three measures within the blanket lien subsample: an inverse blanket scope indicator (rewarding itemized over generic filings), a specificity score (asset-specific identifiers normalized by filing length), and the number of distinct Article 9 categories covered. Higher values reflect more targeted information acquisition. See Appendix C.2 for construction.

To illustrate the variation these metrics capture within blanket liens, consider two anonymized filings from the matched sample:

> **Filing A (generic):** "All assets of debtor, now owned and hereafter acquired, including all proceeds of the foregoing."
>
> **Filing B (itemized):** "This financing statement covers the following collateral: One (1) **New** 2015 [Manufacturer Model] 40 Ton Crane S/N #XXX mounted on One (1) **New** 2015 [Manufacturer Model] truck VIN #XXX. All personal property listed above, together with any and all attachments, accessions, appurtenances, additions, replacements, improvements, modifications and substitutions thereto; all proceeds in the form of goods, accounts, commercial deposit accounts, inventory, equipment, and general intangibles now owned or hereafter acquired."

Both filings are blanket liens: both legally perfect a security interest in all Article 9 collateral of the borrower under UCC §9-504. They differ in what the lender reveals about its knowledge of the borrower's asset base. Filing A establishes a nominal legal claim with no asset-specific information. Filing B identifies the condition ("New"), year, make, model, and serial number of specific equipment, revealing that the lender has examined the borrower's collateral in detail. In the effort index, Filing A scores low on all three components, while Filing B scores high on specificity (asset-level identifiers).

The effort index measures the lender's investment in publicly observable collateral documentation on the UCC financing statement—that is, the breadth and specificity of the collateral description on the public notice document. It does not measure enforceable collateral scope (which depends on the unobserved security agreement), private valuation knowledge (which the lender may hold without encoding in the filing), or the content of the security agreement itself.[26] The mapping from observable filing effort to the theoretical δ_c—the lender's private knowledge of asset quality that the model treats as partially recoverable through documentation—is assumed monotone but cannot be validated with filing data alone. Section 3.2 formalizes how standardization shifts the marginal value of this documentation effort from reducing discretionary valuation uncertainty (pre-reform, when the lender's private information flowed directly into the OLV appraisal) to reducing residual-recovery uncertainty on the lender's 25% unguaranteed exposure (post-reform, when the SBA's standardized schedule is fixed but the actual liquidation recovery remains uncertain).

The *effort change* variable Δe_l measures each lender's pre-to-post shift in mean effort index within the blanket lien subsample, for lenders with filings in both periods. Exploiting the *change* rather than the level sidesteps the fact that a generic pre-reform filing is uninformative about collateral composition: a lender who shifts from generic to itemized filings at the reform boundary reveals a behavioral response to the policy change even if the pre-reform filing was uninformative. This variable serves as the moderator in the effort–pricing interaction (Remark 1): the triple interaction $\text{Treated}_s \times \text{Post}_t \times \Delta e_l$ is interpretable as a conservative test of effort attenuation under the maintained assumption that filing noise is orthogonal to the homestead-exemption treatment—plausible because overfiling and template conventions are not state-specific. The effort–pricing interaction

[26] Two documents govern a secured transaction. The *financing statement* (UCC-1) is a public notice filing governed by UCC §9-504, which requires only that the filing "reasonably identifies" the collateral—a filing that says "all assets" satisfies this standard. The *security agreement* is a private contract governed by §9-108(c), which requires a description "sufficient to identify what is described"—a security agreement that says "all assets" is unenforceable. The financing statement perfects the lien; the security agreement creates it. I observe only the financing statement. The gap between the two documents is one-directional: lenders routinely overfile, claiming "all assets" on the UCC-1 while the underlying security agreement covers a narrower set, because the $20 filing cost is trivial relative to its option value of blocking subsequent creditors. Itemized filings are less ambiguous, because a 200-word collateral description with serial numbers and model years is almost certainly drawn from the same collateral schedule used in the loan documents. Neither source of error threatens identification because the paper's estimand is the *differential change* in filing behavior across homestead exemption regimes, not the level of collateral coverage; overfiling and template conventions are plausibly time-invariant within lender and absorbed by lender fixed effects.

is a descriptive decomposition, not a causal mediation estimate (Section 8).

Matched versus unmatched loans as a collateral-type proxy. Because UCC Article 9 covers only movable property, unmatched loans (those for which no UCC filing links to the borrower) likely have an immovable (real estate) collateral component. Summary statistics within the CA/CO/FL sample support this interpretation. In the pre-reform subsample, unmatched loans carry spreads that are 32 bp lower on average (4.65 versus 4.96 percentage points), have substantially longer maturities (250 versus 210 months), and are more likely to carry a fixed rate (24% versus 19%). These patterns are consistent with real-estate-backed loans carrying the longer terms, lower spreads, and fixed-rate structures typical of commercial mortgages, and with movable-collateral loans carrying a higher risk premium due to greater asset-specific valuation uncertainty.

4.3 Homestead Exemption Panel

State-level homestead exemption limits are constructed from the panel compiled by Indarte (2023), which I verify against primary state statutes and extend to cover the full sample period.[27] For the continuous treatment specification, I use the dollar exemption amount for all states; for the binary design, I restrict to the seven unlimited-exemption states (AR, FL, IA, KS, OK, SD, TX) and twelve low-exemption states (exemption $\leq$ \$23K: AL, GA, IL, IN, KY, MD, MO, NJ, PA, TN, VA, WY). All twelve states have exemptions below \$23K, which protects less than a quarter of typical home equity for SBA borrowers in this size tier, leaving the lender with substantial recovery through deficiency judgment. The \$23K threshold coincides with the federal bankruptcy homestead exemption (\$21,625–\$25,150 during the sample period, 11 USC §522(d)(1)), which serves as a benchmark: states at or below the level that Congress determined to be minimum adequate protection offer minimal homestead shielding from the lender's perspective. Results are robust to modest shifts in the threshold.

Lenders are classified as banks (matched to FDIC via institution number), credit unions (via NCUA number), or non-bank lenders (neither). Multi-state lenders are identified as those operating in both unlimited-exemption and low-exemption states within the sample.

[27] The base panel is publicly available at `sashaindarte.github.io/public_goods/`. I verify each state's exemption amount against the relevant statute and extend the panel to include monthly observations through 2019, county-level variation for California, New York, and Washington, and the federal bankruptcy exemption option for states that permit it (Appendix C).

4.4 Key Variables

The binary treatment indicator Treated$_s = 1$ for the seven unlimited-exemption states and 0 for the twelve low-exemption states. Post$_t = 1$ for loans approved on or after January 1, 2014. For the continuous specification, I use $\log(1 + \text{exemption}_s)$ and a rank-based measure of the state exemption amount across all states.[28] Controls include log loan amount, term (months), a fixed-rate indicator, a relationship lending indicator (equal to one if the borrower has a prior SBA loan from the same lender; see Appendix C.1 for construction), previous SBA experience, a collateral indicator, a corporation indicator, and a franchise indicator.

Borrower identities are constructed via union-find fuzzy record linkage on standardized names within state (Appendix C.1). Credit growth is $\log(L_2/L_1)$ for borrowers with at least two SBA loans within the \$350K–\$1M tier. Post-entry status is defined by the first loan's approval date. The indicator PreviousSBA$_i$ equals one if the borrower has any prior SBA loan from any lender, serving as the empirical proxy for the relationship premium δ_r (Assumption 2). The *second-loan indicator* equals one if a first-loan borrower in the \$350K–\$1M tier obtains any subsequent SBA loan, testing for differential selection into the repeat-borrower sample.

Collateral composition outcomes. Five variables measure collateral coverage. The *blanket lien indicator* equals one if the filing pledges all business assets. The *perfection scope score* (0–5) is a weighted count of breadth clauses: after-acquired property, proceeds, and location breadth (Appendix C.2). The *blanket scope indicator*, defined within the blanket lien subsample, equals one if the filing uses generic "all assets" language without itemizing specific categories; a decrease means fewer filings use generic language. The *composite effort index* is described in Section 4.2.

Falsification outcomes. Falsification tests use six outcomes: state-quarter log loan counts and lending volume, loan-level cancellation rates, the repeat-borrower share, charge-off rates, and Express-to-Express credit growth (placebo). Section 6.4 reports the results.

[28] Unlimited-exemption states are coded at \$5,000,000 following Indarte (2023), yielding $\log(1 + 5{,}000{,}000) \approx 15.4$. The rank specification (column 4 of Table 7) assigns these states to rank 1.0, providing a coding-independent robustness check: the rank coefficient of -9.9 bp is consistent with the log specification.

4.5 Sample Construction and Summary Statistics

The full SBA 7(a) dataset contains approximately 117,000 Standard Guaranty loans above $350K approved during 2011Q1–2019Q4. The $350K threshold is economically motivated: it ensures a uniform guarantee rate ($g = 0.75$), a uniform collateral adequacy standard (the "fully secured" requirement with preset haircuts), and eliminates two contemporaneous policy changes that affect smaller loans (the October 2013 fee waiver for loans ≤$150K and the July 2014 credit scoring mandate for Standard Guaranty loans ≤$350K; see Section 5).

Data cleaning. I drop loans in U.S. territories (GU, MP, VI, PR, AS) and geographically non-contiguous states (AK, HI), which face distinct lending environments and housing markets. Spreads are winsorized at the 1st and 99th percentiles to limit the influence of extreme values. To ensure that lender fixed effects are estimated from a sufficient number of observations, I retain only lenders with at least six loans per period (pre- and post-reform), yielding 409 lenders accounting for 86% of loans in the estimation sample.

Estimation samples. After applying the cleaning protocol, the primary spread analysis uses the full-state sample (103,879 loans, 103,506 after singleton removal) with the continuous exemption measure. For the binary comparison, I restrict to nineteen states (seven unlimited-exemption, twelve low-exemption with exemption ≤ $23K), yielding 44,034 loans (43,770 after singletons). For the credit growth analysis, I identify repeat borrowers within the $350K–$1M tier who obtained at least two SBA loans during the sample period. This within-tier restriction ensures that credit growth $\log(L_2/L_1)$ reflects changes in borrowing capacity rather than migration across loan-size categories. The all-states repeat-borrower panel contains 3,210 borrower pairs (3,131 after singletons); within the nineteen-state binary sample, 1,153 pairs (911 after singletons). The small size of this panel reflects the relative rarity of repeat SBA borrowing: approximately 4.7% of first-loan borrowers in the $350K–$1M tier obtain a second loan within the sample window (Table IA.9), with similar rates in treated and control states.

For the collateral composition analysis (Design 2), the UCC-matched sample covers 18,341 loans across California, Florida, and Colorado. The all-states sample ($N = 103{,}506$ after singleton removal) is used for the continuous treatment specification.

Summary statistics. Table 2 reports summary statistics for the full-state sample, split at the median homestead exemption. Panel A describes the loan-level sample (103,879 loans across 48 contiguous states and the District of Columbia). Mean loan amounts are approximately $1.1 million in both high- and low-HE groups, with similar term lengths (approximately 206–214 months). These are substantial loans to established businesses, not microenterprise credit; the typical borrower is an incorporated entity (87–95% corporation share) financing a major capital expenditure. Pre-reform spreads average 4.93 pp in high-HE states and 5.12 pp in low-HE states, a 19-basis-point level difference that may reflect compositional differences across state groups; identification relies on parallel trends in spreads around the reform, not level equality. Both groups experience spread compression post-reform, but the compression is differential, the pattern examined in the event study. Collateral rates are high in both groups (88–93%), confirming that the "fully secured" requirement binds: nearly all loans in this size tier are collateralized, so the reform's haircut schedule applies to the vast majority of the sample. The homestead exemption row confirms the sharp treatment contrast: high-HE states average $1.6–1.7 million in exemption (driven by unlimited-exemption states), while low-HE states average approximately $23K.

Panel B describes the repeat-borrower subsample (3,210 pairs, all states). First-loan characteristics are broadly similar across groups: mean loan amounts of approximately $610K, spreads of 5.0–5.2 pp, and term lengths of 185–196 months. The pre-reform first-loan spread is 17 basis points lower in high-HE states than in low-HE states, a level difference that does not threaten identification because the DiD relies on parallel *trends* around the reform, validated by the event study in Section 6. Covariates that differ in levels (corporation share, collateral share) are included as controls in all specifications. Appendix Table IA.1 reports the corresponding statistics for the sharp binary design (seven unlimited- versus twelve low-exemption states), and Appendix Table 3 provides a formal comparison of pre-reform borrower characteristics.

Figure 1 plots mean spreads for the binary subsample (seven unlimited- versus twelve low-exemption states) over the sample period. Both groups decline from approximately 5.2–5.4 pp in 2011 to 4.9–5.1 pp in 2013, moving in parallel before the reform. After January 2014, a visible gap emerges: spreads in unlimited-exemption states fall more steeply, consistent with the model's prediction that the reform reduces pricing where business collateral dominates the lender's recovery. Loan volumes trend similarly in both groups (Appendix Figure IA.1). The formal event study in Section 6 confirms that pre-reform coefficients are jointly insignificant (Wald $F = 1.96$, $p = 0.142$ under NAICS2 $\times$ quarter; Table IA.3).

5 Empirical Strategy

The primary analysis uses a difference-in-differences comparing Standard Guaranty loans above \$350K across homestead exemption regimes, before and after the January 2014 collateral standardization. This design estimates the differential effect of the reform on loan pricing and credit growth, with heterogeneity tests by loan size, collateral type, and borrower experience. The homestead exemption moderates a bundle of three related features: the lender's exposure to business-collateral standardization, the availability of a home-equity backstop for recovery, and the scope for substitution toward real estate collateral. The model treats these as jointly determined by the exemption; the empirical design estimates their combined effect. A secondary design examines collateral composition and information acquisition using UCC filing data, providing direct evidence on the lender adaptation channel. Section 5.3 explains how the two designs connect despite using different control groups and different samples.

5.1 Design 1: Homestead Exemption DiD

Treatment variation. The collateral reform replaced discretionary valuations with preset haircuts for all SBA 7(a) loans. As shown in Section 3, the reform's impact on spreads and credit growth depends on the composition of the lender's effective collateral base, which varies continuously with the state homestead exemption: borrowers in high-exemption states are more sensitive to θ_c changes because business collateral constitutes a larger share of the lender's recovery (Propositions 1–3).

The estimand is the differential effect of collateral standardization on loans in high-exemption states relative to low-exemption states: under the continuous specification, the average marginal effect of exemption intensity; under the binary specification, the average treatment effect on the treated (ATT) for unlimited-exemption states. The primary specification exploits this continuous variation across all states using $\log(1 + \text{exemption}_s)$ and rank-based measures, interacted with a post-reform indicator. This continuous treatment design uses all Standard Guaranty loans above \$350K across 49 states ($N \approx 104{,}000$), does not depend on any binary classification, and identifies the gradient from the full exemption distribution. For interpretability, I also restrict to an extreme binary comparison between seven unlimited-exemption states (AR, FL, IA, KS, OK, SD, TX; $\text{Treated}_s = 1$) and twelve low-exemption states (exemption $\leq$ \$23K; $\text{Treated}_s = 0$), which produces the coefficients that map directly to the model's propositions.[29]

[29] Unlimited-exemption states fully protect the borrower's primary residence from involuntary seizure, though borrowers may voluntarily pledge home equity, attenuating the treatment. Texas is the exception:

Sample restriction. I restrict to *collateralized Standard Guaranty loans above \$350K*. The \$350K threshold ensures a uniform guarantee rate ($g = 0.75$), a uniform collateral adequacy requirement ("fully secured" under the preset haircut schedule), and eliminates two contemporaneous policy changes affecting smaller loans (the October 2013 fee waiver for loans ≤\$150K and the July 2014 credit scoring mandate for loans ≤\$350K). The sample period is 2011Q1–2019Q4, providing a balanced pre- and post-reform window.

Spread specification. The primary specification exploits the continuous exemption variation across all states:

$$\text{Spread}_{ilst} = \alpha + \beta \left(\log(1 + \text{exemption}_s) \times \text{Post}_t \right) + X'_{ilst}\phi + \gamma_c + \mu_l + \tau_t + \epsilon_{ilst}, \tag{13}$$

where $\log(1 + \text{exemption}_s)$ is the log dollar homestead exemption in state s, Post_t equals one for loans approved on or after January 1, 2014, X_{ilst} is a vector of loan-level controls (log amount, term, fixed-rate indicator, relationship lending, previous SBA experience, collateral indicator, corporation, franchise, and revolver indicators), γ_c denotes county fixed effects, μ_l denotes lender fixed effects, and τ_t denotes year-quarter fixed effects. The coefficient of interest is β: the marginal effect of exemption intensity on the post-reform spread change. Proposition 1 predicts $\beta < 0$. For the binary comparison (Appendix Table IA.2), $\text{Treated}_s \in \{0, 1\}$ replaces $\log(1 + \text{exemption}_s)$ in (13), where $\text{Treated}_s = 1$ for the seven unlimited-exemption states and county fixed effects absorb the state-level main effect. The preferred specification replaces additive time effects with NAICS2 $\times$ quarter interactions (see the fixed effects strategy below).

Credit growth specification. For repeat borrowers (those with at least two SBA loans identified by borrower ID), I estimate:[30]

$$\text{Growth}_b = \alpha + \beta \left(\log(1 + \text{exemption}_s) \times \text{Post}_t \right) + X'_b\phi + \mu_l + \eta_t + \nu_j + \epsilon_b, \tag{14}$$

the Texas Constitution (Article XVI, Section 50) prohibits business-purpose liens on homestead property entirely (see the identification discussion below). Low-exemption states provide minimal homestead protection (≤\$23K), leaving the vast majority of home equity accessible through deficiency judgment.

[30] Borrower identities are constructed via union-find fuzzy record linkage on standardized names within state (Appendix C.1). The repeat borrower panel includes both same-lender and lender-switching pairs (approximately 19% of repeat borrowers obtain their second loan from a different lender). The primary selection concern is survivorship: borrowers who were rationed out entirely, receiving no second loan after the reform, are excluded. I test for differential selection directly in Appendix Table IA.9: the DiD on P(second loan) is a precise zero (-0.003, $p = 0.73$), confirming that the reform did not differentially alter the probability of obtaining a second loan across treatment groups.

where $\text{Growth}_b = \log(L_2/L_1)$ is credit growth for borrower b with both loans within the \$350K–\$1M tier, Post_t is defined by the entry date of the first loan, X_b controls for first-loan characteristics (log amount, term, fixed-rate indicator, relationship lending, previous SBA experience, collateral indicator, corporation and franchise indicators), μ_l denotes the first-loan lender's fixed effect, η_t denotes entry-quarter fixed effects (distinct from the approval-quarter effects τ_t in the spread equation), and ν_j denotes two-digit NAICS industry fixed effects.[31] Proposition 3 predicts $\beta < 0$. For the binary comparison (Internet Appendix), Treated_s replaces $\log(1 + \text{exemption}_s)$ in (14).

Heterogeneity tests. The model generates two primary heterogeneity tests. First, the experienced-versus-first-time test: Proposition 3 predicts that the credit growth decline is larger for borrowers with prior SBA experience ($\delta_r > 0$) and absent for first-time borrowers ($\delta_r \approx 0$). I split the repeat-borrower sample by PreviousSBA_i at entry: *experienced* borrowers ($\text{PreviousSBA} = 1$) held an SBA lending relationship before their first loan in the sample, while *first-time* borrowers ($\text{PreviousSBA} = 0$) entered the SBA system with no prior relationship. Both subsamples contain borrowers with at least two observed SBA loans (required to compute credit growth $= \log(L_2/L_1)$); the distinction is whether the borrower had accumulated relationship capital at entry, not whether the borrower has multiple loans. I estimate (14) separately on each subsample. The comparison is cross-sectional: within each subsample, the DiD compares credit growth for borrower pairs whose first loan was approved before versus after the reform, across high- versus low-exemption states. If δ_r drives the credit growth decline, the effect should concentrate among experienced borrowers regardless of lender information acquisition (Assumption 2). A pooled triple interaction $\log(1 + \text{exemption}) \times \text{Post} \times \text{PreviousSBA}$ on the full continuous sample formally tests whether the experienced-versus-first-time difference is statistically significant. Second, a loan-size heterogeneity test: I estimate (13) separately by loan-size bin (\$350K–\$1M, \$1M–\$2M, above \$2M) to assess whether the reform's pricing effect varies with the collateral-to-loan ratio.

Table 4 summarizes the mapping from model predictions to empirical tests.

Identification. Identification hinges on the parallel trends assumption: absent the reform, spreads and credit growth would have followed the same trajectory across homestead exemption regimes. I test this directly with an event study that plots year-by-year Treated $\times$ Year coefficients, with 2013 as the reference period. Pre-reform coefficients

[31] County fixed effects are omitted from the credit growth specification because the small repeat-borrower panel cannot support the additional saturation without substantial singleton removal.

should be jointly indistinguishable from zero. I supplement with the sensitivity framework of Rambachan and Roth (2023), which bounds the post-treatment effect under calibrated violations of parallel trends (Internet Appendix, Table IA.12). The SBA announced the reform in August 2013 with a January 2014 effective date; as a robustness check, I exclude the transition quarters (Q4 2013 and Q1 2014) to guard against anticipatory adjustment.

Two concerns bear on identification. First, outside Texas, borrowers in unlimited-exemption states may voluntarily pledge home equity, attenuating the effective treatment. Texas is the only state where the model's assumption $E_a = 0$ holds exactly—the Texas Constitution (Article XVI, §50) prohibits business-purpose liens on homestead property—so the Texas-only binary DiD in Section 6 provides a clean upper bound while the continuous specification estimates the gradient without relying on any binary classification. Second, the binary design uses nineteen states (seven treated, twelve control), raising small-cluster inference concerns. The continuous specification with ~ 50 state clusters, wild cluster bootstrap (MacKinnon et al., 2023), HonestDiD sensitivity analysis (Rambachan and Roth, 2023), and falsification tests reported in Section 6 address these concerns.

5.2 Design 2: Collateral Documentation

The model predicts that the reform changes both the information acquisition and composition of pledged collateral (Proposition 2). I test this directly using UCC filing data matched to SBA loans across California, Florida, and Colorado (18,341 classified loans in the estimation sample). UCC filings record security interests in personal (movable) property under Article 9 of the Uniform Commercial Code; the filing descriptions reveal what assets the lender perfected a lien against and how much the lender invested in information acquisition about those assets.

I estimate a general DiD specification across multiple collateral outcomes y_{ilst}:

$$y_{ilst} = \alpha + \beta\,(\text{Treated}_s \times \text{Post}_t) + X'_{ilst}\phi + \mu_l + \tau_t + \nu_j + \gamma_c + \epsilon_{ilst}, \tag{15}$$

where $\text{Treated}_s = 1$ for Florida (unlimited homestead exemption) and 0 for California and Colorado (limited exemption, $\sim\$75\text{K}$), ν_j denotes two-digit NAICS industry fixed effects, and γ_c denotes county fixed effects.[32] I cluster standard errors at the lender level.

I examine three sets of outcomes. First, *collateral composition*: the blanket lien probability, the perfection scope score (a weighted count of breadth clauses including after-

[32] I also estimate a continuous treatment specification replacing Treated_s with $\log(1 + \text{exemption}_s)$ across all three states, which exploits the within-limited-HE variation between California and Colorado.

acquired property and proceeds language), and asset-type indicators (equipment, fixtures). Proposition 2 predicts that the shift toward generic blanket liens is smaller in unlimited-HE states, where friction in accessing home equity makes business assets the primary recovery channel and lenders retain an incentive to maintain targeted coverage. The perfection scope score should increase more in unlimited-HE states as lenders invest in precise documentation of their primary recovery channel.

Second, *information acquisition*. Within the blanket lien subsample, I distinguish generic filings from itemized filings and measure effort using an inverse blanket scope indicator, a specificity score, and a composite effort index (see Section 4 and Appendix C.2 for construction and interpretation). Conditioning on blanket liens holds legal scope constant and isolates an information-acquisition margin: because all blanket filings perfect the same security interest under UCC §9-504, variation in specificity proxies for the lender's underlying due diligence rather than legal coverage. Proposition 2 predicts that unlimited-HE lenders maintain higher information acquisition post-reform because friction in accessing home equity makes business collateral their primary recovery channel ($E_a \approx 0$), while limited-HE lenders can substitute toward implicit real estate collateral and reduce effort.

Third, *effort–pricing feedback*. I estimate the triple interaction

$$\text{Treated}_s \times \text{Post}_t \times \text{EffortChange}_l,$$

where EffortChange_l is the pre-to-post change in lender-level effort within blanket liens. This test operationalizes Remark 1: lenders who increased information acquisition post-reform should exhibit smaller spread reductions as information acquisition costs pass through to borrowers. The test connects the collateral composition evidence to pricing outcomes and distinguishes the collateral-channel component δ_c (recoverable through effort) from the relationship-channel component δ_r (irrecoverable; Assumption 2).

5.3 Linking the Designs

Designs 1 and 2 examine different margins of the same reform using different samples and control groups. The continuous specification $\log(1+\text{exemption}_s)$ bridges across both by estimating the gradient from the full exemption distribution without relying on any binary grouping.

Different control groups by necessity. Design 1 uses low-exemption states (HE $\leq$ \$23K) as the benchmark; Design 2 compares Florida (treated) against California and Colorado (control), the three states where I have classified UCC filing data. This difference reflects data availability, not a design choice. The continuous coefficient has the same sign and economic interpretation whether estimated on the all-states sample or the three-state UCC sample.

UCC-matched loans as a movable-collateral proxy. UCC Article 9 covers personal (movable) property; real estate liens are recorded separately. A UCC-matched loan is therefore backed primarily by movable business assets. As documented in Section 4.2, unmatched loans display characteristics consistent with real-estate-backed lending (Table 8).

Effort as the mediating channel. Design 1 estimates the average pricing effect; Design 2 measures the information acquisition response and tests whether it feeds back into pricing (Section 7.2). This effort–pricing feedback is a descriptive decomposition, not a causal mediation estimate. The FL-versus-CA+CO comparison was chosen for data availability. The NAICS2 $\times$ quarter and county fixed effects absorb industry-specific and local trends, and the pre-trend tests pass (blanket scope $F = 1.00$, $p = 0.369$; effort index $F = 1.22$, $p = 0.294$). The continuous specification within the three-state sample bridges to the all-states design without depending on Florida-specific shocks.

5.4 Inference

Standard errors are clustered at the lender level throughout, reflecting lender-level variation in collateral valuation practices and pricing.[33] I report wild cluster bootstrap p-values at the county level using the Webb six-point distribution (MacKinnon et al., 2023), which captures geographic correlation in local housing and lending markets beyond what lender-level clustering absorbs. I also report county-clustered and two-way (lender $\times$ county) standard errors, which capture geographic correlation driven by local housing market conditions. Full results for all inference procedures, together with HonestDiD sensitivity analysis on quarterly event studies, are reported in the Internet Appendix (Tables IA.11–IA.13).

Fixed effects strategy. For spreads (Design 1), treatment varies at the state level, making industry-specific time shocks the primary confound. Oklahoma and Texas account for a

[33]Lender-level clustering provides a substantially larger number of clusters and captures the relevant dimension of heterogeneity in the pricing decision. The choice is conservative: lender-clustered standard errors are comparable to or larger than county-clustered standard errors for all headline specifications (Internet Appendix, Table IA.13).

large share of the treated group, and the oil price collapse (late 2014 through 2016) coincides with the post-reform window, so the spread specification uses NAICS2 $\times$ quarter interactions to absorb industry-specific business cycle shocks while preserving the cross-state identifying variation. Lender $\times$ time interactions would restrict identification to multi-state lenders operating in both treatment groups simultaneously, a small and unrepresentative subset (136 of 887 lenders, predominantly large nationals where the soft-information premium $\delta = \delta_c + \delta_r$ is predicted to be weakest on both components); the attenuation under this specification is therefore interpretable under the model rather than indicative of confounding.

The binary repeat-borrower panel ($N = 911$) cannot support industry $\times$ quarter interactions without substantial singleton removal. The continuous all-states panel ($N = 3{,}131$) can: with 622 NAICS2 $\times$ entry-quarter cells (~ 5 observations per cell), the pre-trend Wald test passes ($F = 1.47$, $p = 0.230$), confirming that the credit growth result is robust to industry-specific shocks under the preferred specification. Mining-sector borrowers (NAICS 21) constitute fewer than 1% of the repeat-borrower sample, so the oil-industry channel cannot mechanically drive the credit growth result. For spreads, any oil-related confound would bias the estimate toward zero (higher credit risk in oil states would raise, not lower, spreads), making the negative spread finding conservative.

Table 5 summarizes the mapping between the model's theoretical variables and the empirical proxies used throughout the analysis.

For collateral composition (Design 2), the nature of the confound is different. Collateral documentation is an internal lender choice, not disciplined by competition: two lenders in the same state and quarter can have different filing conventions due to template updates, new legal counsel, or system migrations unrelated to the reform. Time-varying lender heterogeneity in documentation practices is therefore a first-order concern that additive lender fixed effects do not address. I report results under the tightest available specification, lender $\times$ quarter plus NAICS2 $\times$ quarter plus county fixed effects, which absorbs all time-varying lender and industry heterogeneity simultaneously by restricting identification to within-lender, within-quarter variation across states. The coefficients are stable or increase in magnitude relative to the additive baseline, confirming that the collateral adaptation pattern is not driven by documentation drift (Section 7).

6 Results: Pricing

6.1 Average Spread Reduction

Using all Standard Guaranty loans above \$350K across all states (Table 7), the continuous DiD estimates a spread coefficient of -0.92 bp per unit of $\log(1 + \text{exemption})$ in dollars (significant at 1%). A 10% increase in the dollar exemption reduces post-reform spreads by approximately 0.09 bp; moving from a zero-exemption state to one with a \$100,000 exemption reduces spreads by approximately 11 bp, and the move to an unlimited-exemption state yields approximately 14 bp—consistent with the binary 10.6 bp in Table IA.2. A complementary rank specification (column 4), scaling each state's percentile in the exemption distribution to $[0, 1]$, yields -9.9 bp (significant at 1%).

Spreads fall more in states where home equity is more protected and business collateral is the lender's primary recovery channel. These magnitudes are consistent with Proposition 1: the 75% guarantee absorbs three-quarters of any collateral revaluation, so only 25% of the valuation improvement passes through to the spread. The gradient holds across the full exemption distribution (Figure 2).

Restricting to the extreme HE regimes (seven unlimited-exemption versus twelve low-exemption states) produces the coefficients that map directly to Proposition 1. The baseline specification estimates a spread reduction of 10.6 bp (significant at 1%; Table IA.2). Column 3 replaces additive time effects with NAICS2 $\times$ quarter interactions, absorbing industry-specific business cycle shocks including the oil price collapse that disproportionately affected Texas and Oklahoma. The estimate is -11.0 bp (significant at 1%), indicating that neither differential house price trends nor industry composition drives the result.

The spread event study (Figure 3) is consistent with parallel pre-trends under the preferred NAICS2 $\times$ quarter specification (Wald $F = 1.96$, $p = 0.142$); the additive specification shows a marginally significant pre-trend ($F = 2.68$, $p = 0.068$; Table IA.3) driven by 2012 industry composition differences that the NAICS2 $\times$ quarter effects absorb. Post-reform, coefficients become negative and grow in magnitude, consistent with portfolio rotation from pre- to post-reform pricing as the reform applies only to new approvals.[34]

For the median treated-state loan (\$1.17 million), the 10.6 bp spread reduction implies approximately \$1,240 in annual interest savings. Across the approximately 15,100 treated-state loans in the post-reform sample, the aggregate annual reduction is approximately \$19 million. The modest per-loan magnitude is consistent with the model's prediction

[34] The reform was announced August 2013 and effective January 2014. Using 2012 as the event study reference year (rather than 2013, which overlaps with the announcement window) yields qualitatively similar results.

that the 75% guarantee absorbs most of any collateral revaluation, limiting the lender's exposure to 25% of any valuation change.

The remainder investigates what drives this average: heterogeneity by filing status (Section 6.2), lender adaptation on both the extensive and intensive margins (Section 7), and the effort-pricing feedback connecting documentation behavior to loan pricing (Section 7.2).

6.2 Heterogeneity by UCC Filing Status

The spread reduction varies by whether the loan has a matched UCC filing. UCC Article 9 covers security interests in personal (movable) property; loans matched to a UCC filing involve documented business-asset collateral, while unmatched loans may involve real estate, mixed collateral, or movable assets without a filed lien. Summary statistics within the CA/CO/FL sample support a systematic difference (Table 8): UCC-matched loans carry spreads that are 32 bp higher on average (4.96 versus 4.65 percentage points), have shorter maturities (210 versus 250 months), and are less likely to carry a fixed rate (19% versus 24%). These patterns are consistent with UCC-matched loans carrying higher valuation uncertainty pre-reform (movable assets lacked standardized appraisal methods), while unmatched loans display characteristics typical of real-estate-backed lending (longer terms, lower spreads, more fixed-rate). UCC filing is itself a lender decision that responds to the reform (Section 7.1), so this heterogeneity reflects both collateral-type differences and endogenous lender information acquisition responses.

Panel B of Table 8 reports spread DiD estimates by collateral type. Unmatched loans (the immovable-collateral proxy) show a spread reduction of -15.7 bp (significant at 1%), exceeding the full-sample estimate of -10.7 bp. This is consistent with the generous 85% real estate preset producing a clean spread reduction where pre-reform valuation uncertainty was lowest. The FL-versus-low-HE triple interaction with UCC status is positive but statistically insignificant, reflecting the offsetting effort-adaptation mechanism for movable-collateral loans.

These systematic differences motivate the UCC-based analysis in Section 7, which decomposes the matched/unmatched distinction into collateral-type differences and endogenous information acquisition responses.

6.3 Loan-Size Heterogeneity

The model implies that the spread sensitivity to collateral valuation changes is proportional to C/L (equation 5), predicting potentially larger effects for smaller loans where business-asset valuations represent a larger share of the collateral adequacy calculation. Table 9 reports the 7v12 binary DiD estimated separately by loan-size range.

The effect is stable across loan-size bins: -11.2 bp for \$350K–\$1M ($p < 0.01$, $N =$ 24,851), -9.0 bp for \$1M–\$2M ($p = 0.06$, $N =$ 11,068), and -10.1 bp above \$2M ($p = 0.06$, $N =$ 6,888). The uniformity across bins suggests that the reform's pricing effect operates primarily through the homestead exemption channel rather than the collateral-to-loan ratio: the expected recovery differential between unlimited- and low-exemption states is similar regardless of loan size within the \$350K+ tier.

6.4 Robustness

I summarize the robustness exercises here; full results are in the Appendix.

Alternative estimators. The doubly robust inverse probability weighted (DRDID) estimator of Sant'Anna and Zhao (2020), residualized by lender and industry × quarter fixed effects with 999 lender block-bootstrap replications, yields an ATT of -13.4 bp for the 7v12 binary sample ($p = 0.014$ under 999-replication lender block bootstrap, 95% CI: $[-24.8, -2.2]$; untabulated). The larger DRDID magnitude suggests that the OLS DiD may be slightly attenuated by compositional imbalance between treatment groups.

Symmetric windows. Restricting to a symmetric three-year window (2011Q1–2016Q4) yields -10.1 bp ($p = 0.006$, $N =$ 33,596; Table IA.8). The stability across windows rules out a long-run secular trend masquerading as a treatment effect.

Lender × year fixed effects. Replacing additive lender fixed effects with lender × year interactions attenuates the 7v12 estimate to -7.8 bp ($p = 0.35$)—approximately 60% of the baseline. Most SBA lenders operate in a single state, so lender × year effects approximate state × year effects and absorb the identifying variation. For loans between \$350K and \$1M, the coefficient is -11.5 bp ($p = 0.21$), closer to the baseline magnitude.

Sample and inference robustness. Table IA.4 examines sensitivity to sample definition. The binary credit growth estimate is null across alternative samples (-3.0 pp for \$350K–\$2M, -1.1 pp for multi-state lenders; both statistically insignificant), consistent with the diluted contrast in the 7v12 binary design. The continuous all-states specification remains significant excluding any single state (Tables 7 and 17). Wild cluster bootstrap at the county level (Internet Appendix, Table IA.11) provides a robustness check that captures geographic correlation beyond lender-level clustering. HonestDiD sensitivity analysis (Internet Appendix, Table IA.12) shows that the continuous credit growth result is robust to moderate violations of parallel trends ($\bar{M} \leq 0.5$).

Texas sensitivity. Table IA.5 reports leave-one-state-out sensitivity. Dropping Texas renders the binary DiD insignificant (-1.4 bp), reflecting Texas's unique status as the only state where $E_a = 0$ holds exactly. A Texas-only binary DiD (Texas versus the twelve low-exemption states; untabulated) yields the largest estimates: spread -18.0 bp under the baseline specification and -18.1 bp under NAICS2 $\times$ quarter with home price controls (both significant at 1%). Credit growth is -5.3 pp ($p = 0.27$, 878 repeat borrower pairs). Pre-trend Wald tests fail to reject parallel pre-reform trajectories (spread $F = 1.70$, $p = 0.18$; credit growth $F = 1.44$, $p = 0.24$). The continuous specification remains significant excluding Texas. Excluding all seven unlimited-exemption states, the continuous gradient remains significant at 1% ($\beta = -0.0075$, $p = 0.009$; Figure 5), showing that the result is driven by the interior of the exemption distribution, not by the Texas-versus-zero-HE contrast. A full continuous leave-one-state-out analysis (Appendix Table IA.6) shows that no single state drives the continuous gradient: all 49 state exclusions produce significant negative coefficients, with the range $[-0.0133, -0.0051]$.[35]

Within-borrower test. Adding borrower fixed effects to the spread DiD yields a precise zero ($+0.03$ pp, $p = 0.84$ for the 7v12 binary; Table IA.14), indicating that the spread reduction is cross-sectional rather than a within-borrower repricing event. This result supports the two-channel separation in Assumption 2. New borrowers ($\delta_r = 0$) receive the full benefit of the collateral-valuation improvement and experience lower spreads. Repeat borrowers, who held a relationship premium ($\delta_r > 0$) under the pre-reform regime, do not: the collateral benefit is offset by the loss of the relationship-information advantage, producing a net zero within-borrower spread change. The collateral channel (δ_c) thus operates cross-sectionally through new matches, while the relationship channel (δ_r) manifests

[35] The Texas Constitution (Article XVI, Section 50) enumerates the only permissible liens on homestead property, and business-purpose liens are not among them, rendering any such lien constitutionally unenforceable. Texas Property Code §41.001 specifies that the homestead exemption is unlimited in value. In the other six unlimited-exemption states, the exemption protects against involuntary seizure but does not prohibit voluntary pledging. When borrowers voluntarily offer home equity as supplementary collateral, the lender's effective recovery base broadens ($E_a > 0$ in practice), attenuating the reform's bite on equilibrium pricing. The Texas result therefore provides an upper bound on the treatment effect, while the continuous specification, which estimates the gradient across the full exemption distribution weighted by dollar amounts rather than statutory labels, captures the average marginal effect inclusive of this attenuation.

on the extensive margin as a credit growth decline for experienced borrowers (Section 8).[36]

Falsification. Table 10 examines whether the reform changed borrower selection rather than collateral valuation methodology. The Express-to-Express credit growth estimate is statistically insignificant (-1.7 pp, $p = 0.81$), consistent with Express lenders being unaffected by the reform. The Standard-to-Standard estimate for loans below $350K is negative but statistically insignificant (-6.1 pp, $p = 0.53$), consistent with the July 2014 credit scoring mandate contaminating this subsample. State-quarter log loan counts ($+0.037$) and aggregate lending volume (-0.021) show no differential change. Loan-level cancellation rates and the repeat-borrower share are similarly unchanged. The log number of distinct lenders per state-quarter is unchanged in both the binary ($+0.035$, $p = 0.38$) and continuous ($+0.006$, $p = 0.20$) specifications, ruling out differential lender entry or exit as a confound. Charge-off rates show no differential change at the reform boundary (Appendix Table IA.7), supporting the model's maintained assumption that default probability is exogenous to the collateral valuation methodology.

The pricing results establish that the reform lowered average spreads where business collateral dominates recovery, with the effect stable across loan sizes and concentrated in immovable-collateral loans. I now turn to how lenders adapted their collateral information acquisition in response.

7 Results: Lender Adaptation

7.1 Collateral Composition Shift

Proposition 2 predicts that lenders in unlimited-HE states invest more in collateral information acquisition post-reform because friction in accessing home equity makes business assets the primary recovery channel, while limited-HE lenders shift to generic blanket liens where home equity provides a recovery buffer. I test this using UCC filing data matched to SBA loans across California, Florida, and Colorado.

Table 11 reports summary statistics for the UCC-matched sample. Panel A shows reasonable loan-level balance (all normalized differences below 0.25 except the corporation indicator (0.40) and the fixed-rate indicator (0.25)). Panel B reveals large pre-reform

[36] An alternative interpretation is that lenders simply do not reprice existing relationships regardless of the reform. Under pure pricing stickiness, however, the credit growth decline for experienced borrowers (Section 8) is difficult to explain: stickiness preserves existing terms but does not predict reduced subsequent lending. The combination of the within-borrower spread zero and the experienced-borrower credit growth decline favors the δ_r interpretation.

level differences in collateral filing practices (blanket scope ND $= 1.03$, effort index ND $= -1.00$), reflecting the distinct legal environments that motivate the research design. The event studies in Appendix Figures IA.2–IA.3 and Figure 4 show that pre-reform *trends* were parallel despite these level gaps.

Table 12 reports the collateral composition results from estimating equation (15) on the UCC-matched subsample (18,341 classified loans; approximately 530 unique lender clusters). The sample comprises Florida loans (treated) and California and Colorado loans (control). Because this design compares one unlimited-exemption state (FL) against two limited-exemption states, I present the continuous treatment specification as the primary result and the binary comparison as a robustness check.[37]

Collateral composition (extensive margin). Table 12 reports the extensive-margin response on the full UCC-classified sample ($N = 18{,}341$). Two DiD coefficients characterize the shift. First, the blanket lien indicator rises by $+5.4$ pp in Florida relative to CA+CO (col 3, significant at 5%): Florida lenders used blanket-form UCC statements more frequently post-reform. Second, the perfection scope score—a 0–5 count of breadth clauses such as after-acquired property, proceeds, and all-locations language—rises by $+0.28$ in the binary comparison (col 7, significant at 1%) and $+0.058$ per unit of log-exemption in the continuous specification (col 8, significant at 1%). The aggregate scope-score increase is driven mechanically by the extensive-margin shift toward blanket-form filings: blanket liens have higher scope scores than itemized filings by construction, so Florida's higher blanket-form rate translates directly into a higher aggregate scope score. These extensive-margin results are consistent with Proposition 2: where the homestead exemption shields real property ($E_a \approx 0$), lenders broaden their business-asset claim perimeter to maintain collateral coverage. The remaining columns (N categories, equipment, fixtures/RE-related) are directionally consistent but statistically insignificant.

The within-R^2 values (0.004–0.014) reflect substantial noise in individual filing-level data; the significant coefficients are identified from the large sample (18,341 loans) rather than strong model fit. Under the tightest specification (lender $\times$ quarter plus NAICS2 $\times$ quarter plus county fixed effects; untabulated), the coefficients are stable or increase in magnitude, indicating that the collateral adaptation pattern reflects the reform rather than lender-level drift.

[37]The continuous specification exploits $\log(1 + \text{exemption})$ variation across all three states, including the within-control-group margin between California and Colorado. The control group here (CA and CO at $\sim$\$75K exemption) differs from the low-exemption control in Design 1 (12 states with exemption $\leq$ \$23K); the continuous specification, which does not depend on any binary grouping, provides the bridge between the two designs.

Information acquisition (intensive margin within blanket liens). Table 13 reports effort results within the blanket lien subsample ($N = 14{,}688$). Three DiD coefficients capture different dimensions of within-blanket information acquisition. First, the blanket scope indicator—which equals one when a filing uses maximum-breadth language (all assets, after-acquired property, proceeds, at all locations) and zero otherwise—falls by -18.4 pp in Florida relative to CA+CO (col 1, significant at 1%): the maximum-breadth share rose in both groups post-reform, but by 18.4 pp less in Florida.[38] Second, the specificity score DiD is $+0.145$ (col 3, significant at 5%): Florida's blankets use asset-specific language (serial numbers, make/model, condition descriptors) more densely than CA+CO's. Third, the composite effort index DiD is $+0.28$ standard deviations (col 5, significant at 1%). Taken together, Florida lenders *filed blanket liens more often* (extensive margin; preceding paragraph) *while keeping those blankets tightly defined rather than maximally broad* (intensive margin; this paragraph), whereas CA+CO lenders shifted the blankets they did file toward maximum-breadth boilerplate. This joint pattern is the core signature of heterogeneous lender adaptation: where home equity provides no recovery buffer, lenders broadened their legal claim perimeter *and* invested in precise asset-level descriptions; where home equity provides a buffer, lenders accepted maximum-breadth boilerplate as sufficient.

Under the tightest specification (untabulated), both results strengthen. The effort index event study (Figure 4) shows pre-2013 coefficients close to zero and a sharp break at the reform date ($F = 1.22$, $p = 0.294$).[39]

Robustness to sample selection. The intensive-margin coefficients in Table 13 are estimated within the blanket lien subsample, which is itself a post-treatment outcome (blanket lien indicator DiD $+5.4$ pp; Table 12). A mechanical concern is that the subsample is endogenously recomposed by the treatment. Two checks address this. First, restricting to the 135 lenders who filed blanket liens in *both* the pre- and post-reform periods ($N = 14{,}145$) leaves all three intensive-margin coefficients essentially unchanged: blanket scope DiD -0.179 vs -0.184 baseline ($p = 0.009$); specificity DiD $+0.140$ vs $+0.145$ baseline ($p = 0.022$); effort index DiD $+0.268$ SD vs $+0.278$ baseline ($p = 0.008$). The findings reflect incumbent behavioral change within the blanket lien subsample, not compositional entry into it. Second, estimating the specificity-score DiD on the full UCC-classified sample ($N = 18{,}341$, not conditional on blanket lien status) yields $+0.176$ ($p = 0.007$), stronger than the within-blanket estimate of $+0.145$. The same-sign, larger-magnitude full-sample coefficient rules out selection into the blanket lien pool as the driver of the within-blanket

[38] Regression-adjusted conditional shares of maximum-breadth-scope filings within the blanket lien subsample: Florida +4.5 pp versus CA+CO +22.4 pp.

[39] Wald F-statistics: blanket scope $F = 1.00$, $p = 0.369$; scope score $F = 0.69$, $p = 0.504$.

specificity finding.

Filing behavior (extensive margin). The pooled CA+CO UCC match rate rose from 45% to 59% post-reform (14 percentage points), while Florida's remained stable at 84–86%. The DiD is -13.8 percentage points (significant at 1%; Table 12, cols 1–2): limited-exemption lenders expanded UCC filing significantly more than unlimited-exemption lenders. The pattern is consistent with limited-HE lenders under-filing pre-reform, when home equity provided a recovery buffer that reduced the return to business-asset perfection, and then filing broadly once the reform eliminated the information cost of claiming collateral credit. On the intensive margin (conditional on filing), unlimited-HE lenders invest in detailed asset descriptions because the preset schedule rewards knowing the asset composition, while limited-HE lenders file generic blanket liens because the nominal claim is sufficient given their home equity backstop.

7.2 Effort–Pricing Feedback

Remark 1 predicts that within the UCC-matched sample, lenders who increase information acquisition post-reform should exhibit smaller spread reductions or spread increases, because information acquisition costs are passed through to borrowers. I test this using triple interactions within the multi-state lender subsample (lenders operating in at least two of FL, CA, and CO), restricting to loans in the $350K–$1M tier.[40]

Table 14 reports three specifications. First, the loan-level blanket lien indicator interacted with the FL × Post DiD yields a directionally consistent triple interaction of -13.9 bp ($p = 0.27$): among Florida loans, those with blanket lien filings see a smaller spread increase than non-blanket loans. Second, the pre-to-post change in lender-level effort produces a larger but imprecise coefficient. Lenders who increased effort post-reform (effort_change > 0) are associated with a 19.9 bp smaller spread increase than lenders who did not ($p = 0.13$). The base FL × Post coefficient for lenders who did not increase effort is $+19.8$ bp ($p = 0.03$), so FL lenders who increased effort experienced a net spread change of essentially zero (-0.1 bp, joint SE 8.5 bp; $p = 0.99$). The triple interaction is directionally consistent with the effort-attenuation mechanism of Remark 1, but the wide standard error means the data are equally consistent with a large offset and with no offset at all. The effort-change test isolates the reform-boundary margin by comparing the

[40]The $1 million upper bound narrows the sample to the dense mass of SBA Standard Guaranty loans and avoids noise from the small number of very large loans ($1M–$5M) whose pricing differs from the typical 7(a) loan. Table 12 (collateral composition) uses the full >$350K sample without the upper bound. All sample restrictions and variable definitions (effort index, winsorization, lender activity filter) are inherited from the canonical `full_state_sample.rds` used in the main pricing specifications.

same lender's behavior before and after January 2014, rather than comparing structurally different lender types cross-sectionally.

These results connect the collateral composition evidence (Section 7.1) to pricing outcomes. The cross-sectional null from the previous framework, that the Treated $\times$ Post $\times$ HighEffort triple interaction is zero in the full blanket lien sample (Table 15), is not contradicted. The cross-sectional test compares high- versus low-effort loans *within treatment groups*, where effort levels reflect both pre-reform lender type and post-reform adjustment. The dynamic test here isolates the *change* in effort at the reform boundary, which is the margin relevant for effort attenuation. Lenders who maintained or increased effort post-reform absorbed the uncertainty-reduction benefit into information acquisition costs; those who reduced effort (predominantly in CA and CO) passed it through as lower spreads.

7.3 Florida versus California and Colorado

A natural question is why the FL versus CA+CO spread DiD is positive ($+6.3$ bp, $p = 0.23$; untabulated) when Proposition 1 predicts a negative effect. The raw spread changes reveal the source: unconditional pre-post mean spreads (all loans, not regression-adjusted) fell by 49 bp in California and 43 bp in Colorado, but only 8 bp in Florida. This pattern does not reflect a failure of the reform in Florida. It reflects the adaptation mechanism formalized in Proposition 2.

The UCC-matched sample consists entirely of movable-collateral loans, where the preset haircuts (50–75$\%$) are less generous than the real estate haircut (85$\%$) and where pre-reform valuation uncertainty was higher. In this setting, the net spread change depends on the balance between uncertainty reduction and effort cost. California and Colorado lenders, who benefit from a home-equity recovery buffer, reduced information acquisition post-reform (pooled effort index: $+0.24 \rightarrow +0.08$; Table 11), accepting preset valuations and passing through the uncertainty reduction as lower spreads. Florida lenders, who rely primarily on business collateral for recovery, maintained effort (effort index: $-0.29 \rightarrow -0.26$), investing in precise asset-level information acquisition to preserve their private valuation advantage. The information acquisition cost partially offset the uncertainty-reduction benefit, producing a near-zero net spread change.

The FL vs CA+CO comparison is therefore *consistent with* the adaptation mechanism formalized in Remark 1, not evidence against the spread prediction (Section 7.2 reports the effort-pricing decomposition).

7.4 Collateral Requirement Shift

The reform's extensive margin provides a final piece of evidence on the credit channel. If preset haircuts change what loans require collateral, this affects the composition of the borrower pool reaching the collateral adequacy review.

The fraction of loans with collateral pledged declines by 4.9 percentage points more in unlimited-HE states post-reform ($p < 0.05$) in the 7v12 binary specification, and by 0.37 pp per unit of log-exemption ($p = 0.03$) in the continuous specification. Post-reform, some loans in high-HE states no longer trigger the collateral adequacy requirement under the preset haircut schedule, either because the standardized valuation pushes the loan below the fully-secured threshold or because the preset rules reduce the information acquisition burden, allowing marginal cases to proceed without formal collateral pledges. Loan counts show no differential change ($+0.037$, $p = 0.69$ in the 7v12 binary), indicating that the reform altered collateral requirements rather than loan approval rates. This is consistent with the model's prediction that the reform changed the terms of credit, not access to it.

Taken together, the pricing and adaptation results form a consistent picture of a reform that simultaneously reduced valuation uncertainty and destroyed relationship-specific information. Spreads fell where business collateral dominates recovery, with the effect concentrated in immovable-collateral loans and attenuated among movable-collateral loans where lenders invested in information acquisition. Lenders in unlimited-HE states maintained targeted collateral descriptions while limited-HE lenders shifted toward generic liens, and the effort differential feeds back into pricing: high-effort Florida lenders are associated with near-zero net spread changes while low-effort lenders saw increases. The collateral requirement shift indicates that the reform altered the terms of credit. Section 8 examines the credit growth implications and the two channels of soft information.

8 Discussion

8.1 Credit Growth and Lender Adaptation

The pricing and lender-adaptation results establish the reform's effects on spreads and collateral information acquisition. A separate question is whether the reform also affected credit allocation for repeat borrowers.

Two sets of evidence bear on this question and point in different directions. In the all-states continuous specification (3,131 borrower pairs), credit growth declines by -0.58 pp per unit of log-exemption (significant at 5%; Table 17), with the effect concentrated among experienced borrowers (-1.50 pp, $N = 777$) and absent for first-time borrowers (-0.26 pp, statistically insignificant). A pooled triple interaction with PreviousSBA yields -0.011 ($p = 0.098$). Pre-trend Wald tests pass ($F = 1.47$, $p = 0.230$) and wild cluster bootstrap confirms the experienced-borrower result. The binary 7v12 panel (911 pairs after singleton removal) yields a null (-0.3 pp, $p = 0.95$), so the experienced-versus-first-time split should be interpreted as suggestive given the small subsamples. This pattern is consistent with the model's prediction that the relationship premium δ_r is eliminated by standardization and irrecoverable through effort.[41]

However, within the three UCC states (FL vs CA+CO, 894 repeat borrowers), the pattern reverses directionally. Experienced FL repeat borrowers see *higher* credit growth than experienced CA+CO borrowers post-reform ($+20.9$ pp, $p = 0.24$, $N = 188$ after singleton removal), while the pooled DiD is null ($+0.8$ pp, $p = 0.89$). The coefficient is economically large but statistically imprecise due to the small number of experienced FL repeat borrowers (41 in FL versus 147 in CA+CO before singleton removal). This pattern is suggestive about mechanism. The all-states result uses low-exemption states as controls, where the reform's effect operates through the elimination of the relationship premium in states with no home-equity backstop. The FL-versus-CA+CO comparison captures a different margin: the endogenous expansion of collateral claims.

The UCC analysis provides the explanation. California lenders expanded UCC filing by 16 pp post-reform, encumbering previously undocumented business assets. For repeat borrowers at these lenders, the broader collateral claims may tighten the effective borrowing constraint: assets that were previously unencumbered and could support a second loan are now formally pledged to the first. Florida lenders, who were already filing at 84% pre-reform, imposed no additional encumbrance. The credit growth reversal within the UCC states thus reflects the filing-cost channel rather than the relationship-information channel: where lenders expanded their collateral base, repeat borrowers face tighter constraints regardless of relationship quality.

This evidence suggests that the credit consequences of valuation standardization depend on *how* lenders respond, not only on the exemption level. The all-states result captures the average effect of both relationship-information destruction (operating through the zero-to-unlimited HE contrast) and collateral-base expansion (operating through the lender filing response). The UCC-matched analysis separates these channels, revealing

[41] The same-lender subsample ($N = 723$; Appendix Table IA.10) yields a positive but insignificant coefficient ($+2.2$ pp, $p = 0.64$), consistent with relationship continuity partially offsetting the δ_r loss for borrowers who stay with their original lender.

that the filing-cost mechanism can reverse the credit growth effect within the subset of states where lender behavior is observable.

8.2 Policy Implications

Collateral valuation standardization achieves its stated goals: it reduces uncertainty, improves consistency, and lowers average pricing. The 85% real estate haircut and the 50–75% equipment haircuts replaced subjective, heterogeneous assessments with known, verifiable rules. Average spreads fell by 10.6 basis points in the most affected states.

The cost falls on two margins. First, relationship borrowers lose the soft-information premium that rewarded repeat interactions with known lenders. This cost is irrecoverable: no amount of lender discretion can substitute for the lending relationship. Second, lenders in unlimited-exemption states, where friction in accessing home equity limits the recovery buffer, face higher information acquisition costs as they invest in detailed collateral descriptions to reduce residual uncertainty on movable assets. This cost is passed through to borrowers as higher spreads, partially offsetting the standardization benefit.

These two margins have different policy implications. Permitting lender discretion (e.g., flexible valuation rules or supplementary appraisals) can partially offset the pricing cost by reducing the need for costly information acquisition. It cannot, however, restore the credit-access benefit of lending relationships. The credit-growth cost to experienced borrowers is a structural consequence of replacing judgment with rules, regardless of how the rules are implemented.

8.3 External Validity

The SBA 7(a) program carries a 75% federal guarantee, which dampens the pass-through of collateral valuation changes to spreads by three-quarters. In unguaranteed commercial lending, where the lender bears the full residual risk, the pricing effects of collateral standardization could be substantially larger. However, the guarantee is precisely what makes the SBA setting informative: because the SBA sets the collateral adequacy rules, the reform provides a clean policy experiment in which the treatment is the valuation methodology, not the underlying asset values. SBA 7(a) lenders are predominantly community banks and CDFIs, whose relationship intensity may exceed that of large commercial banks; the dual-channel mechanism may be less pronounced where lending is already standardized. The mechanism should nonetheless generalize to other standardization episodes in financial regulation, including Basel risk-weight standardization and automated credit scoring, where similar tradeoffs between consistency and lender-specific information arise.

8.4 Limitations

The analysis conditions on approved loans. Borrowers who were denied credit under the new rules are unobserved, and compositional shifts in the approval pool may contribute to the estimated effects. The SBA guarantee structure limits the scope for extensive-margin selection (loan counts show no differential change), but the guarantee does not eliminate it.

The binary specification is sensitive to Texas, the only state where the constitutional prohibition on home equity business liens makes the treatment exact. The continuous specification, which exploits the full exemption distribution and does not depend on any binary classification, provides the more robust evidence.

The UCC evidence covers three states (California, Colorado, Florida). The movable/immovable proxy relies on UCC match status rather than direct observation of pledged asset types. Within the three-state sample, the proxy is supported by summary statistics: matched loans carry higher spreads, shorter maturities, and lower fixed-rate shares. However, the proxy cannot be validated outside these states.

The effort-change moderator is endogenous to the reform. Lenders choose how much to invest in information acquisition in response to the new rules, and that choice may correlate with unobserved lender characteristics that also affect pricing. The effort–pricing interaction should be interpreted as a descriptive decomposition of the treatment effect, not a causal mediation estimate.

A related limitation concerns the gap between the effort index and the theoretical object it proxies. The model's δ_c represents the lender's private assessment of asset recovery value—an intensive-margin information object about the quality of specific assets. The effort index captures extensive-margin properties of the filing: how many Article 9 categories are listed and whether the filing uses generic or itemized language. The two coincide when lenders who know more about their collateral produce more granular filings, but the correspondence cannot be tested within filing data alone. To the extent that private knowledge and filing granularity diverge—for instance, if lenders with deep asset knowledge file generically because the $20 filing cost is trivially low regardless of private knowledge—the effort index understates the actual δ_c recovery and the effort–pricing interaction understates the true attenuation. The significant effort–pricing coefficient is therefore consistent with the monotone mapping assumption but does not test it.

9 Conclusion

This paper studies the pricing, credit, and information-acquisition consequences of replacing lender discretion with standardized collateral valuation rules. The January 2014 SBA reform generated three sets of findings. First, spreads fell where homestead exemptions are higher and business collateral is the lender's primary recovery channel, with the reduction driven by immovable (real estate) collateral and attenuated among movable-collateral loans where lenders invested in information acquisition. Second, lenders in unlimited-exemption states maintained targeted collateral descriptions and increased information acquisition, while lenders with a home-equity recovery buffer shifted toward generic blanket claims; the effort-pricing feedback is consistent with information acquisition costs partially offsetting the uncertainty reduction for movable assets. Third, the credit-growth consequences depend on how lenders respond: the all-states continuous specification shows lower credit growth for experienced repeat borrowers in high-exemption states, consistent with the loss of relationship-specific information, while within the three UCC states the pattern reverses—a reflection of the endogenous expansion of collateral claims where lenders had been under-filing pre-reform.

These results show that valuation standardization affects not only pricing but the full ecosystem of lender responses. Collateral-channel soft information, the lender's knowledge of asset quality, is partially recoverable through information acquisition and manifests as a pricing effect attenuated by lender adaptation. The credit consequences, however, depend on the lender's filing response: where lenders expanded collateral claims, repeat borrowers face tighter constraints. Standardization's costs thus depend on the capacity of regulated institutions to adapt, and that adaptation itself has distributional consequences for borrowers.

Section 8 discusses limitations, including selection into approved loans, Texas sensitivity in the binary design, the three-state scope of the UCC evidence, and the endogeneity of the effort moderator. Matching SBA loans to firm-level outcomes via Census data would reveal whether the credit growth reduction translates into real effects on business activity. Other standardization episodes in financial regulation (Basel risk weights, automated underwriting, algorithmic credit scoring) may generate similar tradeoffs between consistency and information; cross-market comparisons would test the generality of the dual-channel mechanism documented here.

References

Sumit Agarwal, Gene Amromin, Itzhak Ben-David, Souphala Chomsisengphet, Tomasz Piskorski, and Amit Seru. Policy intervention in debt renegotiation: Evidence from the Home Affordable Modification Program. *Journal of Political Economy*, 125(3):654–712, 2017. doi: 10.1086/691701.

Kevin Aretz, Murillo Campello, and Maria-Teresa Marchica. Access to collateral and the democratization of credit: France's reform of the Napoleonic security code. *Journal of Finance*, 75(1):45–90, 2020. doi: 10.1111/jofi.12846.

Natalie Bachas, Olivia S. Kim, and Constantine Yannelis. Loan guarantees and credit supply. *Journal of Financial Economics*, 139(3):872–894, 2021.

Markus Behn, Rainer Haselmann, and Vikrant Vig. The limits of model-based regulation. *Journal of Finance*, 77(3):1635–1684, 2022.

Efraim Benmelech and Nittai K. Bergman. Collateral pricing. *Journal of Financial Economics*, 91(1):11–26, 2009.

Allen N. Berger and Gregory F. Udell. Relationship lending and lines of credit in small firm finance. *Journal of Business*, 68(3):351–381, 1995.

Jeremy Berkowitz and Michelle J. White. Bankruptcy and small firms' access to credit. *RAND Journal of Economics*, 35(1):69–84, 2004.

David Besanko and Anjan V. Thakor. Collateral and rationing: Sorting equilibria in monopolistic and competitive credit markets. *International Economic Review*, 28(3):671–689, 1987.

Helmut Bester. Screening vs. rationing in credit markets with imperfect information. *American Economic Review*, 75(4):850–855, 1985.

Kristian Blickle, Zhiguo He, Jing Huang, and Cecilia Parlatore. Information-based pricing in specialized lending. *Journal of Financial Economics*, 2025. Forthcoming.

J. David Brown and John S. Earle. Finance and growth at the firm level: Evidence from SBA loans. *Journal of Finance*, 72(3):1039–1080, 2017.

Charles W. Calomiris, Mauricio Larrain, José Liberti, and Jason Sturgess. How collateral laws shape lending and sectoral activity. *Journal of Financial Economics*, 123(1):163–188, 2017. doi: 10.1016/j.jfineco.2016.09.005.

Murillo Campello and Mauricio Larrain. Enlarging the contracting space: Collateral menus, access to credit, and economic activity. *Review of Financial Studies*, 29(2):349–383, 2016. doi: 10.1093/rfs/hhv069.

Geraldo Cerqueiro, Steven Ongena, and Kasper Roszbach. Collateralization, bank loan rates, and monitoring. *Journal of Finance*, 71(3):1295–1322, 2016.

Geraldo Cerqueiro, Deepak Hegde, María Fabiana Penas, and Robert C. Seamans. Debtor rights, credit supply, and innovation. *Management Science*, 63(10):3311–3327, 2017.

Rebel A. Cole, Tatyana Damm, John Hackney, and Mammadali Suleymanov. It's not (only) personal, it's business: Personal bankruptcy exemptions and business credit. *Review of Finance*, 2025. Forthcoming.

Benjamin L. Collier, Cameron M. Ellis, and Benjamin J. Keys. The cost of consumer collateral: Evidence from bunching. *Econometrica*, 93(3):779–819, 2025. doi: 10.3982/ECTA22303.

Nuri Ersahin, Rustom M. Irani, and Katherine Waldock. Can strong creditors inhibit entrepreneurial activity? *Review of Financial Studies*, 34(4):1661–1698, 2021.

Manasa Gopal. How collateral affects small business lending: The role of lender specialization. Working Paper, Center for Economic Studies, U.S. Census Bureau (CES-WP-21-22), 2021.

Manasa Gopal and Philipp Schnabl. The rise of finance companies and FinTech lenders in small business lending. *Review of Financial Studies*, 35(11):4859–4901, 2022.

Reint Gropp, John Karl Scholz, and Michelle J. White. Personal bankruptcy and credit supply and demand. *Quarterly Journal of Economics*, 112(1):217–251, 1997.

John Hackney. Small business lending in financial crises: The role of government-guaranteed loans. *Review of Finance*, 27(1):247–287, 2023.

Bengt Holmström and Jean Tirole. Financial intermediation, loanable funds, and the real sector. *Quarterly Journal of Economics*, 112(3):663–691, 1997.

Sasha Indarte. Moral hazard versus liquidity in household bankruptcy. *Journal of Finance*, 78(5):2421–2464, 2023. Homestead and wildcard exemption panel available at https://sashaindarte.github.io/public_goods/.

Roman Inderst and Holger M. Mueller. A lender-based theory of collateral. *Journal of Financial Economics*, 84(3):826–859, 2007.

José M. Liberti and Mitchell A. Petersen. Information: Hard and soft. *Review of Corporate Finance Studies*, 8(1):1–41, 2019.

Stephan Luck and João A. C. Santos. The valuation of collateral in bank lending. *Journal of Financial and Quantitative Analysis*, 59(5):2038–2067, 2024. doi: 10.1017/S0022109023000704.

James G. MacKinnon, Morten Ørregaard Nielsen, and Matthew D. Webb. Cluster-robust inference: A guide to empirical practice. *Journal of Econometrics*, 232(2):272–299, 2023.

William Mann. Creditor rights and innovation: Evidence from patent collateral. *Journal of Financial Economics*, 130(1):25–47, 2018. doi: 10.1016/j.jfineco.2018.07.001.

Michael Manove, A. Jorge Padilla, and Marco Pagano. Collateral versus project screening: A model of lazy banks. *RAND Journal of Economics*, 32(4):726–744, 2001.

Steven Ongena, Walid Saffar, Yuan Sun, and Lai Wei. Movables as collateral and corporate credit: Loan-level evidence from legal reforms across Europe. *Journal of Banking and Finance*, 170:107331, 2025.

Guangqian Pan, Zheyao Pan, and Kairong Xiao. The shadow cost of collateral. *Review of Financial Studies*, 38(5):1419–1463, 2025. doi: 10.1093/rfs/hhae073.

Mitchell A. Petersen and Raghuram G. Rajan. The benefits of lending relationships: Evidence from small business data. *Journal of Finance*, 49(1):3–37, 1994.

Mitchell A. Petersen and Raghuram G. Rajan. Does distance still matter? the information revolution in small business lending. *Journal of Finance*, 57(6):2533–2570, 2002. doi: 10.1111/1540-6261.00505.

Matthew C. Plosser and João A. C. Santos. Banks' incentives and inconsistent risk models. *Review of Financial Studies*, 31(6):2080–2112, 2018. doi: 10.1093/rfs/hhy028.

Ashesh Rambachan and Jonathan Roth. A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591, 2023.

Adriano A. Rampini and S. Viswanathan. Collateral and secured debt. NBER Working Paper 33875, 2025.

Pedro H. C. Sant'Anna and Jun Zhao. Doubly robust difference-in-differences estimators. *Journal of Econometrics*, 219(1):101–122, 2020.

Joseph E. Stiglitz and Andrew Weiss. Credit rationing in markets with imperfect information. *American Economic Review*, 71(3):393–410, 1981.

Vikrant Vig. Access to collateral and corporate debt structure: Evidence from a natural experiment. *Journal of Financial Economics*, 109(1):272–291, 2013. doi: 10.1016/j.jfineco.2013.02.009.

Figure 1: Mean Spread by Treatment Group and Year. Shaded bands show 95% confidence intervals around annual group means.

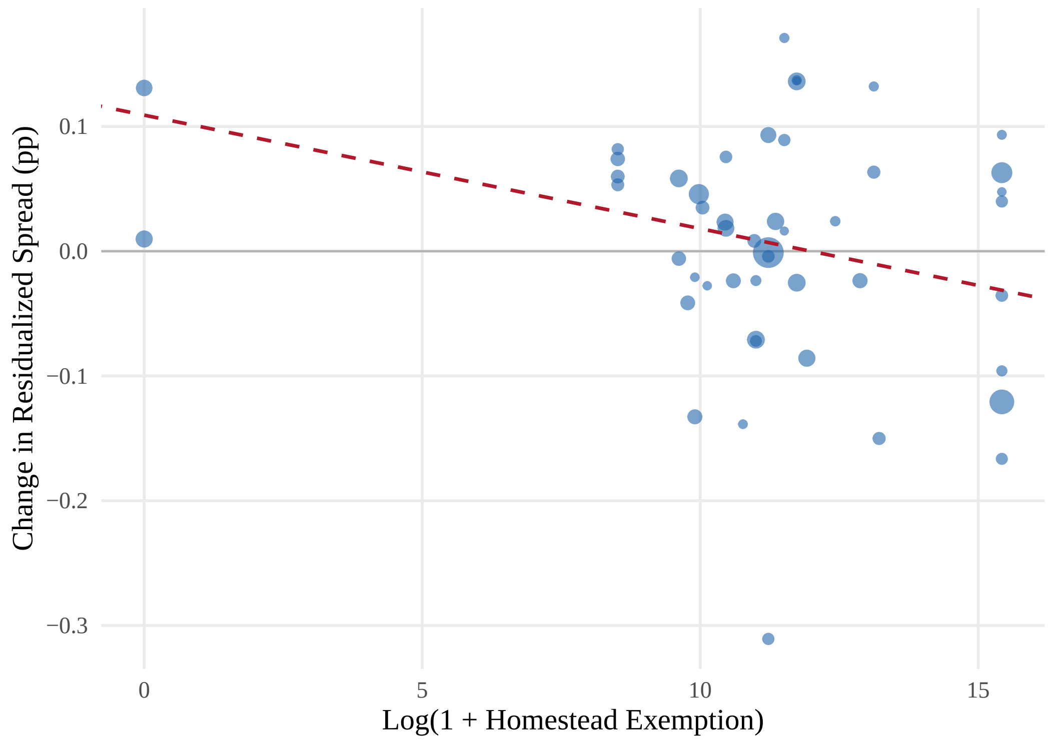

Figure 2: Pre-to-post reform change in residualized spread by state, plotted against log homestead exemption. Spreads are residualized on loan-level controls, lender, NAICS2 × quarter, and county fixed effects. Each point is a state; size reflects sample size. The dashed line shows the slope from the continuous DiD (Table 7, column 3).

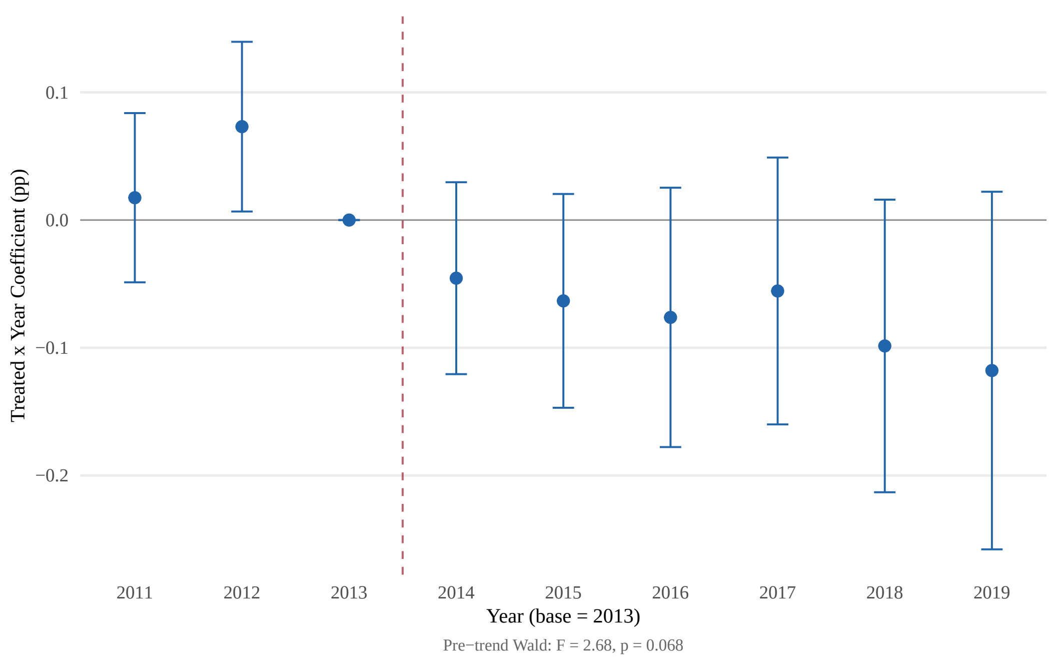

Figure 3: Spread Event Study: Treated × Year Coefficients

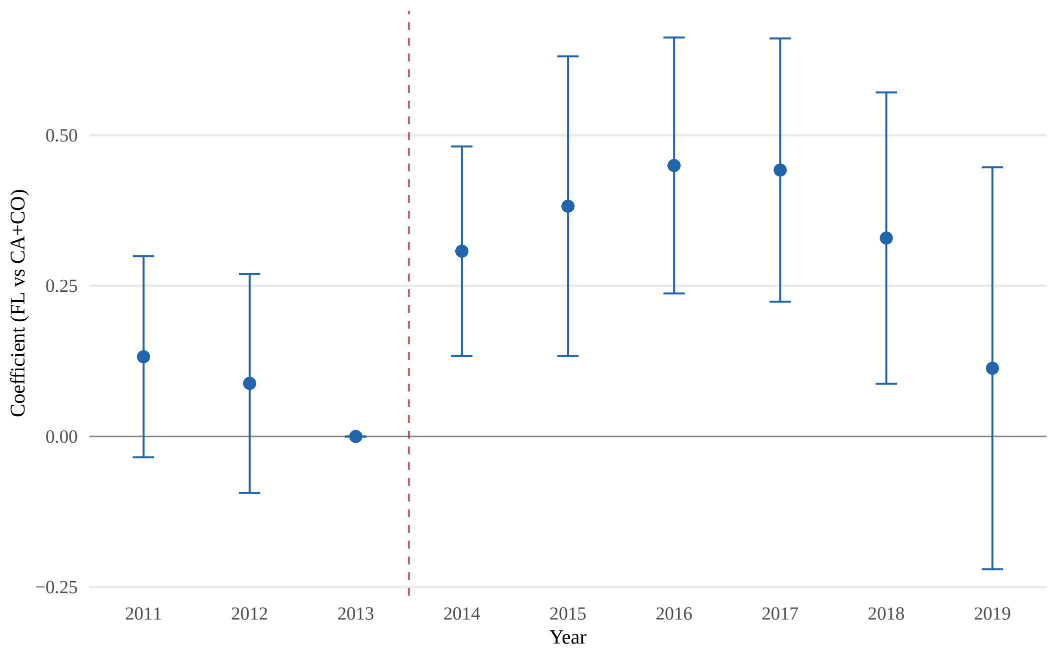

Figure 4: Event Study: Lender Effort Index within Blanket Lien Filings. Reference year is 2013. Wald test for pre-trends: $F = 1.22$, $p = 0.294$.

Table 1: SBA Collateral Haircut Schedule (Post-January 2014)

Asset Type	Valuation Rule	Lender Discretion
New machinery & equipment[a]	75% of purchase price, minus prior liens	None (mechanical)
Used machinery & equipment[a]	50% of NBV (80% with OLV appraisal), minus prior liens	Limited (OLV option)
Real estate[b]	85% of market value	Preserved (appraisal)

Notes: NBV = net book value (original price minus depreciation). OLV = orderly liquidation value. Source: SBA SOP 50 10 5(F), effective January 1, 2014. The business-asset haircuts remained unchanged through SOP 50 10 5(K), April 2019, covering the entire sample period. For real estate, the 85% haircut is applied to a market-value appraisal that is functionally equivalent to the pre-reform OLV; lender discretion over real estate valuation is therefore preserved. [a] The SOP refers to "machinery and equipment"; furniture and fixtures were implicitly included under this category during the sample period and were carved out as a separate asset class in SOP 50 10 6 (October 2020, post-sample). [b] The SOP refers to "real estate" without distinguishing improved from unimproved; the distinction was formalized in SOP 50 10 6 (October 2020, post-sample). Unimproved real estate was addressed in later SOPs and is excluded from this table.

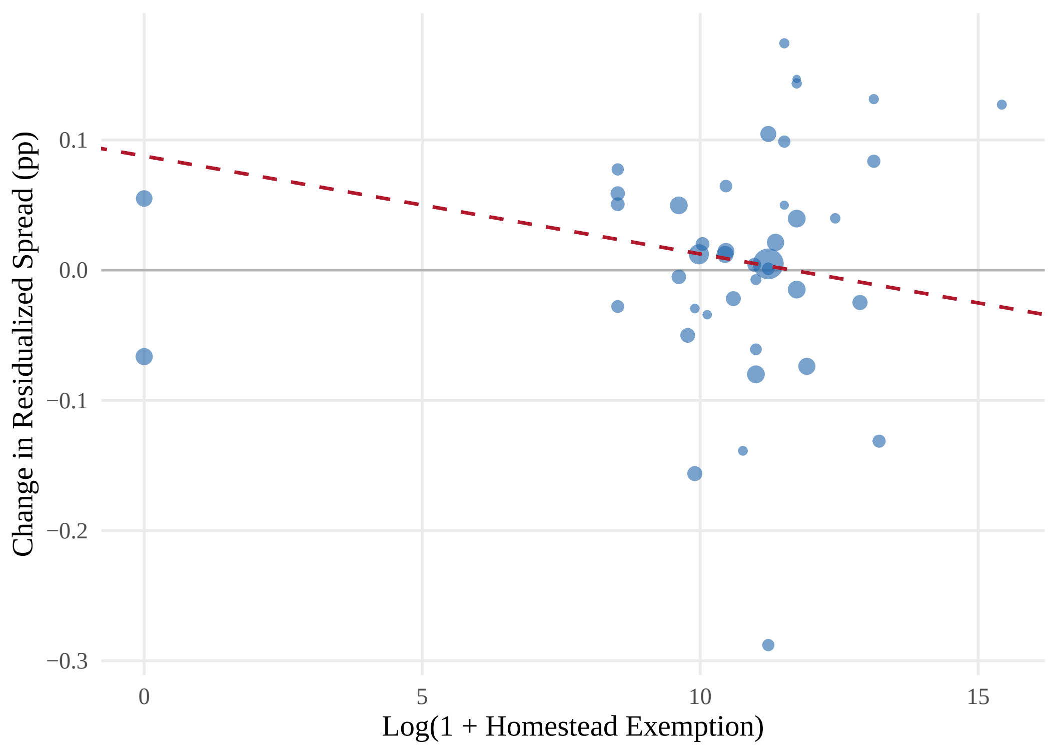

Figure 5: Pre-to-post reform change in residualized spread by state, excluding the seven unlimited-exemption states. Spreads are residualized on loan-level controls, lender, NAICS2 $\times$ quarter, and county fixed effects. The dashed line shows the continuous DiD slope ($\beta = -0.0075$, $p = 0.009$, 42 states, 83,841 loans). The gradient is significant at 1% from the interior of the exemption distribution alone.

Table 2: Summary Statistics

	High HE (above median)		Low HE (below median)		Norm.
	Pre	Post	Pre	Post	Diff.
Panel A: All loans (>$350K Standard Guaranty, all states)					
Loan amount (in $1,000)	1112.65	1229.36	1108.75	1232.64	0.005
	(882.74)	(995.24)	(868.28)	(993.61)	
Term (months)	214.21	214.20	206.33	207.40	0.079
	(91.35)	(90.81)	(90.68)	(89.33)	
Spread (pp)	4.93	4.55	5.12	4.81	-0.226
	(1.11)	(1.21)	(1.00)	(1.09)	
Fixed rate	0.206	0.256	0.183	0.208	0.103
Relationship	0.088	0.079	0.085	0.078	0.005
Previous SBA	0.151	0.135	0.141	0.130	0.017
Collateral	0.925	0.923	0.880	0.924	0.045
Corporation	0.872	0.906	0.934	0.948	-0.171
Franchise	0.082	0.161	0.107	0.198	-0.080
Homestead exemption ($1,000)	1580.4	1651.0	23.2	22.6	1.012
	(2232.2)	(2257.7)	(18.1)	(16.8)	
Observations	15,748	48,761	11,518	27,852	
Panel B: Repeat borrowers ($350K–$1M, all states)					
First loan amount (in $1,000)	605.57	619.43	605.16	614.66	0.022
	(182.97)	(182.81)	(181.54)	(181.90)	
First loan term (months)	189.88	196.01	184.94	187.15	0.084
	(94.53)	(91.84)	(91.43)	(89.13)	
First loan spread (pp)	5.04	4.75	5.21	4.93	-0.188
	(1.03)	(1.19)	(0.90)	(1.09)	
Fixed rate	0.175	0.228	0.148	0.219	0.055
Relationship	0.165	0.196	0.219	0.210	-0.073
Previous SBA	0.251	0.260	0.304	0.285	-0.082
Collateral	0.906	0.906	0.822	0.909	0.111
Corporation	0.892	0.895	0.963	0.962	-0.268
Franchise	0.043	0.154	0.083	0.158	-0.031
Credit growth	0.02	-0.01	0.01	-0.00	-0.001
	(0.32)	(0.29)	(0.30)	(0.29)	
Number of loans	2.22	2.10	2.12	2.10	0.074
	(0.63)	(0.36)	(0.42)	(0.32)	
Observations	702	1,283	540	685	

Notes: Panel A reports loan-level statistics for all >$350K Standard Guaranty loans across all states with non-missing homestead exemption data, 2011Q1–2019Q4. High HE = states with exemption at or above the sample median; Low HE = states below the median. Panel B reports first-loan characteristics and credit growth for repeat borrowers ($350K–$1M). Normalized differences computed as $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$.

Table 3: Balance Table: Pre-Reform Repeat Borrowers ($350K–$1M)

	Treated		Control		Norm.	
	Mean	SD	Mean	SD	Diff.	Balance
First loan amount ($1,000)	606.71	183.00	598.76	177.32	0.044	
First loan term (months)	171.19	91.08	185.06	91.13	-0.152	
First loan spread (pp)	5.02	1.05	5.32	0.83	-0.319	†
Fixed rate	0.23	0.42	0.11	0.31	0.323	†
Relationship	0.14	0.34	0.21	0.41	-0.192	
Previous SBA	0.20	0.40	0.31	0.46	-0.263	†
Collateral	0.92	0.27	0.82	0.38	0.292	†
Corporation	0.93	0.26	0.94	0.23	-0.066	
Franchise	0.05	0.21	0.10	0.30	-0.201	
Number of loans	2.17	0.55	2.11	0.39	0.131	
Observations	176		278			

Notes: Pre-reform (entry before 2014Q1) repeat borrowers, $350K–$1M. Normalized differences computed as $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$. † indicates $|\text{Norm. Diff.}| > 0.25$.

Table 4: Prediction-to-Test Mapping

Prediction	Test	Table	Result
Prop. 1: $\beta < 0$ (spread)	Continuous HE $\times$ Post DiD	7	-0.92 bp/unit
Prop. 3: $\beta < 0$ (credit)	Repeat borrower DiD	17	-0.58 pp/unit
Prop. 2: effort $\uparrow$ in high HE	UCC composition DiD	12–13	Effort $+0.28$ SD
Remark 1: effort attenuates Δs	Effort $\times$ spread triple	14	-19.9 bp ($p = 0.13$)

Notes: Each row maps a model prediction to its primary empirical test. Results column reports the point estimate from the preferred specification.

Table 5: Variable Definitions: Theoretical

Variable	Definition
θ_c	Business collateral valuation rate; the assessed recovery rate on pledged business assets. Pre-reform, determined by lender discretion ("orderly liquidation value"); post-reform, set by SBA preset haircuts.
θ_u	Home equity recovery rate; the recovery rate on unprotected home equity ($\theta_u < \theta_c$), reflecting the lower legal priority of unsecured deficiency judgments relative to perfected UCC security interests.
δ	Soft-information premium ($= \delta_c + \delta_r$); the lender's private assessment of collateral quality above the conservative baseline $\bar{\theta}$. Eliminated by standardization ($\delta \to 0$ post-reform).
δ_c	Collateral-channel soft information; the lender's private knowledge of asset quality, condition, maintenance, and redeployability. Partially recoverable through documentation effort e post-reform.
δ_r	Relationship-channel soft information; the lender's private knowledge of borrower reliability and business viability, acquired through repeated interactions. Irrecoverable post-reform ($\tilde{\delta}_r(e) = 0$ for all e).
σ^2	Valuation uncertainty; the variance of the mean-zero valuation error ε in pre-reform collateral assessments. Higher for movable than immovable assets. Eliminated by preset haircuts ($\sigma^2 \to 0$).
e	Lender documentation effort ($e \in [0, 1]$); post-reform investment in detailed collateral documentation that reduces residual valuation uncertainty and partially recovers δ_c. Cost $c(e)$ is convex.
H	Homestead exemption (dollar amount); the statutory limit on home equity that creditors can seize. $H = \infty$ in unlimited-exemption states.
E_a	Available home equity; $E_a = \max(E - H, 0)$. Equals zero in unlimited-HE states.
g	SBA guarantee rate ($= 0.75$ for Standard 7(a) loans above $150K).
C/L	Collateral-to-loan ratio; higher in unlimited-HE states because $E_a = 0$ constrains equilibrium loan size.
π	Default probability; assumed exogenous to the collateral valuation methodology.
κ	Uncertainty premium coefficient; $\lambda(\sigma^2) = \kappa\sigma^2$.

Notes: Variables defined in Section 3.

Table 6: Variable Definitions: Empirical

Variable	Definition
Spread	Interest rate minus matched-maturity Treasury yield (percentage points). Variable-rate: rate − 3-month T-bill. Fixed-rate: rate − piecewise-linearly interpolated Treasury yield at the loan's maturity.
Treated	$= 1$ for unlimited-HE states (AR, FL, IA, KS, OK, SD, TX); $= 0$ for low-HE states (AL, GA, IL, IN, KY, MD, MO, NJ, PA, TN, VA, WY; exemption $\leq$ \$23K).
Post	$= 1$ for loans approved on or after January 1, 2014.
$\log(1 + \text{exemption})$	Log dollar homestead exemption; continuous treatment in the all-states specification.
Growth	$\log(L_2/L_1)$ for repeat borrowers with at least two SBA loans within the \$350K–\$1M tier.
PreviousSBA	$= 1$ if borrower has any prior SBA loan from any lender; proxy for $\delta_r > 0$.
UCC Matched	$= 1$ if loan matched to a UCC filing. Because UCC Article 9 covers only personal (movable) property, serves as a movable collateral proxy.
Blanket Lien	$= 1$ if UCC filing pledges "all assets" of the debtor.
Scope Score	Weighted count of breadth clauses in the UCC filing (0–5 scale).
Effort Index	Standardized composite within blanket liens; combines inverse scope, specificity score, and number of distinct Article 9 categories.
Effort Change	Pre-to-post change in lender-level mean effort index within the blanket lien subsample.
Collateral Pledged	$= 1$ if the SBA record indicates collateral was pledged.

Notes: Variables constructed from SBA administrative loan records, UCC filing data, and the homestead exemption panel (Section 4). The effort index and effort change variables are defined only for the UCC-matched blanket lien subsample.

Table 7: Spread Difference-in-Differences: Continuous Homestead Exemption (All States)

	Panel A: Full Sample				Panel B: By Loan Size		
	(1) Baseline	(2) + HPrice	(3) NAICS×Qtr	(4) Rank	(5) \$350K–500K	(6) \$500K–1M	(7) \$1M–2M
HE × Post	-0.0092^{***}	-0.0089^{***}	-0.0088^{***}	-0.0985^{***}	-0.0124^{***}	-0.0086^{**}	-0.0099^{**}
	(0.0034)	(0.0034)	(0.0034)	(0.0375)	(0.0044)	(0.0034)	(0.0045)
Observations	103,506	102,328	102,301	103,506	22,033	38,116	26,281
Within R^2	0.4178	0.4175	0.4162	0.4157	0.4330	0.4180	0.3649
Lender FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
NAICS2 FE	No	No	×Qtr	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Home price control	No	Yes	Yes	No	No	No	No

Notes: Dependent variable is the loan spread (percentage points over matched-maturity Treasury). Sample: >\$350K Standard Guaranty loans across all states with non-missing homestead exemption, 2011Q1–2019Q4. Columns (1)–(3) interact $\log(1 + \text{exemption})$ with a post-reform indicator; column (4) uses a rank-based measure (0–1 scale). Column (3) replaces additive FEs with NAICS2 × quarter interactions. All columns include controls for log loan amount, term, fixed-rate indicator, relationship lending, previous SBA, collateral, corporation, and franchise indicators. Standard errors clustered at the lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 8: Collateral-Type Heterogeneity: Summary Statistics and Spread DiD (CA/CO/FL)

	(1) Full Sample 7v12 Binary	(2) Unmatched (Immovable Proxy)	(3) FL vs Zero-HE UCC × DiD
Panel A: Summary Statistics (CA/CO/FL, pre-reform)			
	UCC-Matched	Unmatched	Difference
Spread (pp)	4.964	4.645	0.319
Loan amount ($1,000)	1,086	1,128	-41
Term (months)	210.146	249.871	-39.725
Fixed rate	0.189	0.236	-0.046
Collateral pledged	0.920	0.941	-0.020
Corporation	0.885	0.779	0.106
Relationship	0.076	0.072	0.004
Previous SBA	0.148	0.133	0.014
Observations	3,961	3,675	
Panel B: Spread DiD			
	(1) Full 7v12	(2) Unmatched 7v12	(3) FL vs Zero-HE
Treated × Post	-0.1072^{***}	-0.1566^{***}	0.0387
	(0.0389)	(0.0445)	(0.0478)
Observations	43,733	38,306	30,436
Controls	Yes	Yes	Yes
FE: Quarter + Lender + Industry×Quarter + County	Yes	Yes	Yes
Cluster		Lender	

Notes: Pre-reform (2011Q1–2013Q4) summary statistics for loans in California, Colorado, and Florida, split by UCC match status. UCC Article 9 covers only personal (movable) property, so matched loans are backed primarily by movable business assets and unmatched loans are more likely backed by real estate or mixed collateral. Difference column reports the matched minus unmatched gap.

Table 9: Loan-Size Heterogeneity: Spread DiD by Loan Amount (7v12 Binary)

	(1) \$350K–\$1M	(2) >\$1M–\$2M	(3) >\$2M
Treated × Post	-0.1115^{***}	-0.0898^{*}	-0.1009^{*}
	(0.0426)	(0.0468)	(0.0537)
Observations	24,851	11,068	6,888
Within R^2	0.3911	0.3423	0.3479
Sample	7v12 Binary	7v12 Binary	7v12 Binary
Controls	Yes	Yes	Yes
FE: Quarter + Lender + Industry×Quarter + County	Yes	Yes	Yes
Cluster		Lender	

Notes: This table reports the 7v12 binary difference-in-differences estimate of the January 2014 collateral valuation reform on loan spreads, estimated separately by loan-size range. The dependent variable is the loan spread in percentage points, defined as the contract interest rate minus the maturity-matched Treasury yield. Column (1) restricts to loans between \$350K and \$1M ($N = 24{,}851$). Column (2) restricts to loans above \$1M through \$2M ($N = 11{,}068$). Column (3) restricts to loans above \$2M ($N = 6{,}888$). Treated states are seven states with unlimited homestead exemptions; control states are twelve low-exemption states (exemption $\leq$ \$23K). All specifications include controls for log loan amount, term, fixed-rate indicator, relationship indicator, previous SBA borrower indicator, collateral pledged indicator, corporation indicator, franchise indicator, and revolver indicator. Fixed effects: quarter, lender, NAICS two-digit industry × quarter, and county. Standard errors in parentheses are clustered at the lender level. The sample comprises SBA 7(a) Standard Guaranty loans, 2011Q1–2019Q4. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 10: Falsification Tests and No-Reallocation

	Panel A		Panel B				Panel C		Panel D	
	(1) Exp→Exp $\leq$350K	(2) Std→Std $\leq$350K	(3) Log count state-qtr	(4) Log volume state-qtr	(5) Cancel rate	(6) Repeat share	(7) P(switch) >350K	(8) P(fixed) >350K	(9) Log N lenders Binary	(10) Log N lenders Continuous
Treated × Post	-0.0165	-0.0607	0.0366	-0.0205	-0.0034	0.0033	-0.0147	-0.0136	0.0348	0.0059
	(0.0682)	(0.0976)	(0.0927)	(0.1062)	(0.0087)	(0.0113)	(0.0328)	(0.0223)	(0.0394)	(0.0046)
Observations	6,518	2,620	683	683	43,770	683	5,470	43,770	683	1,753
Lender FE	Yes	Yes	No	No	Yes	No	Yes	Yes	No	No
Quarter/Entry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State FE	No	No	Yes	Yes	No	Yes	No	No	Yes	Yes

Notes: Panel A: credit growth for Express→Express (col 1) and Standard→Standard (col 2) repeat borrowers with first loans $\leq$\$350K, 7v12 states. SE clustered at lender level. Panel B: state-quarter log loan count (col 3), log aggregate volume (col 4), loan-level cancellation rate (col 5), and repeat-borrower share (col 6) for >\$350K Standard Guaranty, 7v12 states. Cols (3)–(4),(6) use state and quarter FE, SE clustered at state level; col (5) uses lender and quarter FE, SE clustered at lender level. Panel C: P(switch to Express) (col 7) and P(fixed rate) (col 8), >\$350K, lender and quarter FE, SE clustered at lender level. Panel D: log number of distinct lenders per state-quarter; col (9) uses 7v12 binary, col (10) uses continuous $\log(1 + \text{HE})$ across all states. State and quarter FE, SE clustered at state level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 11: Design 2 Summary Statistics: UCC-Matched Sample

	FL (Treated)		CA+CO (Control)		Norm.
	Pre	Post	Pre	Post	Diff.
Panel A: Loan characteristics					
Loan amount ($1,000)	1,143.97	1,218.82	1,063.01	1,212.69	0.092
	(935.15)	(991.05)	(825.73)	(990.72)	
Term (months)	203.29	202.75	212.93	208.09	-0.105
	(91.32)	(90.34)	(93.01)	(92.16)	
Spread (pp)	4.97	4.91	4.96	4.58	0.005
	(1.09)	(1.06)	(1.01)	(1.12)	
Fixed rate	0.262	0.205	0.160	0.222	0.251
Relationship	0.048	0.066	0.088	0.082	-0.159
Previous SBA	0.101	0.111	0.167	0.156	-0.195
Collateral	0.927	0.929	0.918	0.922	0.036
Corporation	0.966	0.970	0.853	0.888	0.402
Franchise	0.094	0.185	0.075	0.149	0.069
Observations	1,143	4,284	2,818	10,135	
Panel B: Collateral characteristics (from UCC match)					
Blanket lien	0.840	0.824	0.847	0.775	-0.020
Scope score	2.51	2.66	1.97	1.80	0.453
	(1.31)	(1.42)	(1.08)	(1.22)	
Specificity score	0.70	0.83	1.22	1.17	-0.405
	(0.83)	(0.81)	(1.62)	(1.16)	
Blanket scope	0.696	0.735	0.240	0.409	1.028
Effort index	-0.49	-0.43	0.40	0.13	-0.999
	(0.84)	(0.94)	(0.93)	(0.98)	
N categories	5.09	4.88	4.83	4.42	0.117
	(2.39)	(2.46)	(1.94)	(2.19)	
Has equipment	0.862	0.815	0.878	0.832	-0.048
Has inventory	0.762	0.750	0.773	0.765	-0.027
Fixtures/RE-related	0.184	0.120	0.055	0.039	0.404
PMSI language	0.018	0.019	0.024	0.015	-0.040
Observations	1,143	4,284	2,818	10,135	

Notes: Panel A reports loan-level characteristics for all >$350K Standard Guaranty loans with UCC-classified collateral in FL (treated, unlimited homestead exemption) vs. CA+CO (control, limited exemption), 2011Q1–2019Q4. Panel B reports collateral characteristics from matched UCC filings. Effort index is computed within blanket lien filings only ($N_{\mathrm{BL}} = 14,727$) as the standardized sum of (1 − blanket scope), specificity score, and N categories. Normalized differences computed as $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$ using pre-reform observations.

Table 12: Collateral Composition DiD: FL vs CA+CO Around Jan 2014 Reform

	UCC Filing		Blanket Lien		N Categories		Scope Score		Equipment		Fixtures/RE-related	
	(1) Binary	(2) Cont.	(3) Binary	(4) Cont.	(5) Binary	(6) Cont.	(7) Binary	(8) Cont.	(9) Binary	(10) Cont.	(11) Binary	(12) Cont.
DiD	-0.1375^{***}	-0.0377^{***}	0.0542^{**}	0.0143^{**}	0.1461	0.0350	0.2763^{***}	0.0580^{***}	0.0197	0.0040	-0.0325	-0.0073
	(0.0255)	(0.0059)	(0.0248)	(0.0059)	(0.1599)	(0.0402)	(0.0888)	(0.0219)	(0.0241)	(0.0054)	(0.0257)	(0.0061)
Observations	29,618	29,618	18,341	18,341	18,341	18,341	18,341	18,341	18,341	18,341	18,341	18,341
Within R^2	0.0662	0.0709	0.0066	0.0070	0.0039	0.0039	0.0044	0.0049	0.0050	0.0050	0.0143	0.0143
Controls	Yes											
Lender FE	Yes											
Quarter FE	Yes											
Industry FE	Yes											
County FE	Yes											
Cluster	Lender											

Notes: Binary DiD uses Treated$_{FL}$ $\times$ Post$_{2014}$; Continuous uses $\log(1+\text{Exemption}) \times$ Post. Cols (1)–(2): dependent variable is an indicator for having a UCC filing (extensive margin); sample is all $>$ \$350K Standard Guaranty loans in CA, FL, CO ($N = 29{,}618$). Cols (3)–(12): dependent variable is a collateral characteristic; sample restricted to UCC-classified loans ($N = 18{,}341$). Controls: log amount, term, fixed-rate, relationship, previous SBA, corporation, franchise (plus collateral indicator for cols 3–12). FE as shown. SE clustered at lender level. ***p<0.01; **p<0.05; *p<0.10.

Table 13: Soft-Information Effort DiD: Within Blanket Lien Filings

	Blanket Scope		Specificity		Effort Index	
	(1) Binary	(2) Cont.	(3) Binary	(4) Cont.	(5) Binary	(6) Cont.
DiD	-0.1841^{***}	-0.0479^{***}	0.1447^{**}	0.0349^{**}	0.2775^{***}	0.0707^{**}
	(0.0674)	(0.0175)	(0.0602)	(0.0146)	(0.1010)	(0.0275)
Observations	14,688	14,688	14,688	14,688	14,688	14,688
Within R^2	0.0095	0.0129	0.0052	0.0053	0.0050	0.0060
Controls	Yes					
Lender FE	Yes					
Quarter FE	Yes					
Industry FE	Yes					
County FE	Yes					
Cluster	Lender					

Notes: Sample restricted to blanket lien filings (UCC-classified, CA+FL+CO, $>$ $350K Standard Guaranty). Effort Index = standardized composite of (1−blanket scope) + specificity score + N categories. ***p<0.01; **p<0.05; *p<0.10.

Table 14: Effort–Pricing Feedback: Information Acquisition and Spread Changes (FL vs CA+CO)

	(1) Baseline	(2) Blanket × DiD	(3) Effort Change × DiD
Treated$_{FL}$ × Post	0.1289^{*} (0.0679)	0.2391 (0.1560)	0.1980^{**} (0.0908)
Treated$_{FL}$ × Post × Blanket		-0.1392 (0.1255)	
Treated$_{FL}$ × Post × High Effort Δ			-0.1989 (0.1291)
Observations	8,762	8,762	8,389
Sample	Multi-state lenders, \$350K–\$1M, UCC-matched		
Controls	Yes	Yes	Yes
FE: Quarter + Lender + Industry×Quarter + County	Yes	Yes	Yes
Cluster	Lender		

Notes: Dependent variable: loan spread (pp). Sample: UCC-matched Standard Guaranty loans, \$350K–\$1M, multi-state lenders, FL vs CA+CO, 2011Q1–2019Q4. Col (1): baseline FL × Post. Col (2): triple with blanket lien indicator. Col (3): triple with high effort change (above-median pre-to-post lender effort index shift). Controls: log amount, term, fixed rate, relationship, previous SBA, collateral, corporation, franchise, revolver. FE: quarter, lender, NAICS2 × quarter, county. SE clustered at lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 15: Spread × Collateral Effort: Triple Interaction and Subsample Splits

	(1) Triple	(2) High Effort	(3) Low Effort
Treated × Post × High Effort	0.0168		
	(0.0858)		
Treated × Post	0.0757		
	(0.0566)		
Treated × Post		0.1179	
		(0.0845)	
Treated × Post			0.0093
			(0.0631)
Observations	14,688	7,226	7,402
Within R^2	0.4358	0.4531	0.4134
Controls		Yes	
Lender FE		Yes	
Quarter FE		Yes	
Industry FE		Yes	
Cluster		Lender	

Notes: Dependent variable: spread (pp). Sample: blanket lien filings with non-missing effort index, CA+FL+CO, > \$350K Standard Guaranty. High Effort = effort index > 0 (above median). Col (1): triple interaction; Cols (2)–(3): separate regressions by effort group. Controls: log amount, term, fixed-rate, relationship, previous SBA, collateral, corporation, franchise. FE and clustering as shown. ***p<0.01; **p<0.05; *p<0.10.

Table 16: Collateral Requirement Shift and Loan Counts by Homestead Exemption Regime

	(1)	(2)	(3)	(4)
	Pct. with Collateral		Log Loan Count	
	7v12 Binary	Continuous	7v12 Binary	Continuous
Treated × Post	-0.0491^{**}		0.0366	
	(0.0190)		(0.0927)	
$\log(1+\text{HE})$ × Post		-0.0037^{**}		0.0045
		(0.0017)		(0.0065)
State × Quarter Observations	683	1753	683	1753
State + Quarter FE	Yes	Yes	Yes	Yes
Cluster		State		

Notes: Unit of observation: state-quarter. Cols (1)–(2): fraction of >\$350K Standard Guaranty loans with collateral pledged. Cols (3)–(4): log loan count. Cols (1),(3): 7v12 binary (7 unlimited vs 12 low-exemption states). Cols (2),(4): continuous $\log(1+\text{HE})$ across all states. State and quarter FE. SE clustered at state level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 17: Credit Growth DiD: Continuous Homestead Exemption (All States)

	(1) All (log)	(2) All (rank)	(3) Prev SBA (log)	(4) First-time (log)
HE × Post	-0.0058^{**}	-0.0524^{*}	-0.0150^{**}	-0.0026
	(0.0023)	(0.0275)	(0.0058)	(0.0033)
Observations	3,131	3,131	777	2,254
Within R^2	0.2736	0.2735	0.2850	0.2739
Lender FE	Yes	Yes	Yes	Yes
Entry quarter FE	Yes	Yes	Yes	Yes
NAICS2 FE	Yes	Yes	Yes	Yes

Notes: Dependent variable is credit growth $= \log(L_2/L_1)$ for repeat borrowers with $\geq$2 SBA loans within the \$350K–\$1M tier, across all states. Col (3): experienced borrowers (PreviousSBA $= 1$ at entry, i.e., held an SBA relationship before the first observed loan). Col (4): first-time borrowers (PreviousSBA $= 0$ at entry, i.e., the first observed loan is their first SBA loan; credit growth is from this first loan to their second). The DiD compares borrower pairs whose first loan was approved before vs. after January 2014, across the homestead exemption gradient. Controls: first-loan log amount, term, fixed-rate, relationship, collateral, corporation, franchise (cols 3–4 drop PreviousSBA). SE clustered at lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

A Model Appendix

A.1 Proofs

Proof of Proposition 1. From equation (7), the net spread change is

$$\Delta s = -\frac{\pi}{1-\pi}(1-g)\frac{C}{L}\Big[(\bar{\theta}_{\text{std}} - \bar{\theta}) + \kappa\sigma^2 - \delta\Big].$$

The sensitivity $\partial s/\partial\theta_c = -[\pi/(1-\pi)](1-g)(C/L)$ is proportional to C/L. In unlimited-HE states, $E_a = 0$, so the borrower's effective collateral base is $\Omega = (\bar{\theta} + \delta)C$ and the equilibrium loan size satisfies $\partial L^*/\partial E_a > 0$ (additional implicit collateral relaxes the borrowing constraint). In zero-HE states, $\Omega = (\bar{\theta} + \delta)C + \theta_u E_a$, producing a larger equilibrium L^* for the same C. Therefore C/L is higher in unlimited-HE states, amplifying the spread response to any given change in θ_c.

The effect differs by collateral type because σ^2 differs. For immovable assets (real estate), pre-reform appraisals were already standardized through comparable-sales methodologies, so σ^2_{imm} is low; the preset haircut of 85% is generous relative to the conservative baseline ($\bar{\theta}^{\text{imm}}_{\text{std}} > \bar{\theta}^{\text{imm}}$); and δ is small (limited scope for private valuation advantage). The bracketed term is unambiguously positive, and the spread falls. For movable assets (equipment, inventory), σ^2_{mov} is high (few comparables, subjective condition assessment); the preset haircut (50–75%) may or may not exceed $\bar{\theta}^{\text{mov}} + \delta$ for relationship borrowers; and the lender's effort choice e^* introduces residual uncertainty $\sigma^2(1-e^*)$ and cost $c(e^*)$. The sign depends on the balance between uncertainty reduction and information loss, attenuated by effort. The spread reduction is therefore unambiguously larger for immovable-collateral loans than for movable-collateral loans, and the unlimited-HE differential $\Delta s|_{H=\infty} - \Delta s|_{H=0} < 0$ holds for both types but is more precisely estimated for immovable assets. □

Proof of Proposition 2. The lender's first-order condition for effort is equation (10):

$$\frac{\pi}{1-\pi}(1-g)\frac{C}{L}\left[\kappa\sigma_0^2 + \delta_c \cdot h'(e^*)\right] = c'(e^*).$$

The marginal benefit (LHS) is increasing in C/L and decreasing in e^* (because $h'' < 0$). The marginal cost (RHS) is increasing in e^* and does not depend on H. Define $\text{MB}(e; C/L) \equiv [\pi/(1-\pi)](1-g)(C/L)[\kappa\sigma_0^2 + \delta_c \cdot h'(e)]$ and $\text{MC}(e) \equiv c'(e)$. Both functions are continuous, MB is decreasing in e (concavity of h and the linearity of $\sigma_0^2(1-e)$), and MC is increasing in e (convexity of c). An interior solution e^* exists and is unique by the implicit function theorem.

Now consider the effect of H. In unlimited-HE states, $E_a = 0$, so the equilibrium loan size L^* is smaller for given wealth w and collateral C (no implicit collateral to expand the borrowing constraint). Therefore C/L is higher when $H = \infty$ than when $H = 0$. Since $\text{MB}(e; C/L)$ is increasing in C/L:

$$\text{MB}(e; C/L|_{H=\infty}) > \text{MB}(e; C/L|_{H=0}) \quad \text{for all } e.$$

The intersection of MB with the (unchanged) MC curve shifts rightward:

$$e^*_{H=\infty} > e^*_{H=0}.$$

The cost function $c(\cdot)$ is the same in both regimes; the asymmetry arises entirely from the marginal benefit. Lenders in unlimited-HE states face a higher return to effort because each dollar of improved asset valuation directly reduces the unguaranteed recovery shortfall, with no home-equity buffer to absorb errors.

The three empirical manifestations follow. (i) Higher effort implies more detailed collateral information acquisition (more asset categories, serial numbers, condition descriptions). (ii) In limited-HE states where e^* is low, the lender optimally shifts to generic blanket liens: the preset valuation is accepted and the filing serves only to perfect a broad security interest, not to document asset-specific recovery values. (iii) The information acquisition cost $c(e^*)$ is passed through to the spread (equation (4) with $\kappa\sigma_0^2(1-e^*)$ replacing $\kappa\sigma^2$). Within unlimited-HE states, lenders with higher effort changes exhibit smaller spread reductions because the information acquisition cost partially offsets the uncertainty-reduction benefit. □

Proof of Proposition 3. Credit growth $\log(L_2/L_1)$ depends on period-2 wealth $w_2 = w_1 + \Pi_1$ and the period-2 spread schedule. For relationship borrowers with total soft-information premium $\delta = \delta_c + \delta_r > 0$, the reform has three effects on credit growth.

Primary channel: borrowing constraint. The period-2 spread schedule shifts upward for borrowers whose pre-reform valuation $\bar{\theta} + \delta$ exceeded the post-reform standardized rate $\bar{\theta}_{\text{std}}$. The effective loss in recognized valuation, net of effort, is

$$\Delta\theta_{\text{eff}} = (\bar{\theta} + \delta) - \bar{\theta}_{\text{std}} - \tilde{\delta}_c(e^*) = \delta_r + \delta_c(1 - h(e^*)) + (\bar{\theta} - \bar{\theta}_{\text{std}}).$$

When $\delta > \Delta\bar{\theta} \equiv \bar{\theta}_{\text{std}} - \bar{\theta}$, the net effect is negative: recognized valuation falls, and the period-2 borrowing constraint tightens.

Retained earnings channel. From equation (IA.5), $\partial\Pi_1/\partial\theta_c = -L_1 \cdot (\partial s_1/\partial\theta_c) > 0$. Higher

θ_c lowers the period-1 spread, raises retained earnings Π_1, and increases wealth available for period-2 borrowing. For borrowers whose first loan was originated pre-reform at the relationship valuation $\bar{\theta} + \delta$, the reform does not retroactively change Π_1. But the period-2 valuation $\bar{\theta}_{\text{std}}$ (net of effort recovery $\tilde{\delta}_c(e^*)$) falls below the pre-reform valuation, restricting the period-2 loan directly.

Amplification by C/L. Both channels are proportional to C/L, which is higher when the homestead exemption is unlimited ($E_a = 0$). The credit growth reduction is therefore amplified where the exemption is higher.

Effort does not moderate the credit growth decline. The key separating prediction is that effort recovers δ_c but not δ_r. From equation (12):

$$\Delta \log(L_2/L_1) \propto -\big[\delta_r + \delta_c(1 - h(e^*))\big] \cdot \frac{C}{L}.$$

For experienced borrowers with $\delta_r > 0$, the term $\delta_r \cdot (C/L)$ ensures that credit growth declines even when $e^* = 1$ (maximal effort, full recovery of δ_c). The relationship premium, built through repeated interactions, cannot be substituted by a more detailed UCC filing.

First-time versus experienced borrowers. First-time borrowers have $\delta = \delta_c + \delta_r = 0$: the reform replaces $\bar{\theta}$ with $\bar{\theta}_{\text{std}} > \bar{\theta}$, a pure level improvement, and eliminates valuation uncertainty ($\kappa\sigma^2 \to 0$). Both effects improve the period-2 spread schedule for first-time borrowers. Credit growth may increase or remain unchanged, but it does not decline. Experienced borrowers with $\delta > \Delta\bar{\theta}$ face a net loss: the recognized valuation falls despite the baseline improvement. The credit growth reduction is therefore concentrated among experienced borrowers.

This pattern is inconsistent with an asset-shielding alternative (Gropp et al., 1997), under which the HE differential reflects lenders' response to the portion of borrower assets protected from creditor recovery. That mechanism operates on all borrowers with home equity to shield, regardless of relationship history; the observed concentration of the credit growth decline among experienced repeat borrowers is not predicted. The charge-off null (Section 6.4) additionally rules out differential default risk. □

A.2 Valuation Uncertainty: Derivation of $\lambda(\sigma^2)$

The main text writes the valuation uncertainty premium in the equilibrium spread (equation (4)) as $\kappa\sigma^2$, a linearization of the general convexity premium $\lambda(\sigma^2)$ derived here. This subsection derives $\lambda(\sigma^2)$ from the convexity of the lender's loss function in valuation errors and shows that the two forms coincide up to a curvature constant κ in the linearized regime used throughout the paper.

The lender observes a noisy signal of the true collateral valuation rate:

$$\hat{\theta}_s = \theta_c + \varepsilon, \qquad \varepsilon \sim F(0, \sigma^2),$$

where σ^2 is the signal variance, decreasing in the lender's information investment (pre-reform: decreasing in δ; post-reform: decreasing in effort e). The lender prices the loan using the signal $\hat{\theta}_s$. Upon default, the lender's unguaranteed loss from a valuation error is

$$\ell(\varepsilon) = (1-g) \cdot \max\bigl[\varepsilon \cdot C - \theta_u E_a,\ 0\bigr],$$

where the max operator reflects the implicit collateral buffer: in zero-HE states ($E_a > 0$), the real estate recovery absorbs small overestimates, so the error must exceed $\theta_u E_a / C$ before it generates a binding loss. In unlimited-HE states ($E_a = 0$), every overestimate translates dollar-for-dollar into a recovery shortfall.

The expected loss is

$$\mathbb{E}[\ell(\varepsilon)] = (1-g)C \int_{\theta_u E_a/C}^{\infty} \bigl(\varepsilon - \theta_u E_a/C\bigr)\, dF(\varepsilon).$$

This is a call-option-like payoff with strike $\theta_u E_a / C$. By standard option-pricing results, the expected value is convex in σ^2: increasing the variance of ε raises the expected loss because the upside of overestimation (the lender's uninsured recovery shortfall) is unbounded while the downside (underestimation, where the lender recovers more than expected) is bounded at zero.

Under normality, $\varepsilon \sim N(0, \sigma^2)$, the expected loss has a closed-form expression:

$$\mathbb{E}[\ell(\varepsilon)] = (1-g)C\Bigl[\sigma\phi\Bigl(\frac{\theta_u E_a}{\sigma C}\Bigr) - \frac{\theta_u E_a}{C}\Phi\Bigl(-\frac{\theta_u E_a}{\sigma C}\Bigr)\Bigr],$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal density and CDF. The valuation uncertainty premium is:

$$\lambda(\sigma^2) = \frac{1}{C} \cdot \frac{\partial}{\partial \sigma^2}\, \mathbb{E}[\ell(\varepsilon)] > 0. \tag{IA.1}$$

Three properties of λ matter for the comparative statics:

1. $\lambda(\sigma^2) > 0$: the premium is strictly positive whenever $\sigma^2 > 0$.

2. $\lambda'(\sigma^2) > 0$: the premium is increasing in the variance (convexity of the loss function).

3. $\lambda(\sigma^2)|_{E_a=0} > \lambda(\sigma^2)|_{E_a>0}$: the premium is larger in unlimited-HE states because the strike of the implicit call option is zero, making the loss more sensitive to variance.

Property 3 implies that the reform's uncertainty-reduction benefit ($\lambda(\sigma^2) \to 0$) is worth more in unlimited-HE states, reinforcing the spread differential in Proposition 1.

Connection to the soft-information premium. Pre-reform, the lender's information investment δ reduced σ^2 (higher $\delta \Rightarrow$ lower $\sigma^2(\delta)$, with $\sigma' < 0$) at cost $c(\delta)$. The lender chose δ to minimize total expected cost:

$$\min_{\delta} \ \pi \cdot \mathbb{E}[\ell(\varepsilon)] + c(\delta).$$

The first-order condition is

$$c'(\delta^*) = -\pi \cdot \frac{\partial}{\partial \delta} \mathbb{E}[\ell(\varepsilon)] = \pi(1-g)C \cdot \lambda(\sigma^2(\delta^*)) \cdot |\sigma'(\delta^*)|.$$

The RHS is the marginal benefit of information: the reduction in expected unguaranteed loss from a marginal decrease in signal variance. Since $\lambda(\sigma^2)|_{E_a=0} > \lambda(\sigma^2)|_{E_a>0}$, the marginal benefit is larger in unlimited-HE states, so $\delta^*_{H=\infty} > \delta^*_{H=0}$: lenders in unlimited-HE states invested more in soft information pre-reform. The reform ($\delta \to 0$, $\sigma^2 \to 0$) therefore destroys more information where more was developed.

Post-reform, the same mechanism applies to the effort choice e through equation (10), with $\sigma^2(1-e)$ replacing $\sigma^2(\delta)$. The derivation above establishes that the incentive asymmetry between HE regimes is a general property of the loss function's convexity, not specific to the pre- or post-reform information technology.

A.3 Robustness to Endogenous Borrower Effort

The model treats default probability π as exogenous to the collateral valuation methodology. This subsection addresses the concern that borrower moral hazard (effort to avoid default) may be affected by the reform, separately from the lender's collateral information acquisition formalized in the main text.

The distinction is between two types of effort that enter the model at different points:

- *Lender effort* (e in Section 3.2): the lender's post-reform investment in collateral information acquisition, which reduces valuation uncertainty $\sigma^2(1-e)$ and partially recovers δ_c. This effort is endogenous and central to the model.

- *Borrower effort*: the borrower's investment in project success, which determines $\pi = \pi(\tilde{e})$ where $\tilde{e}$ denotes borrower effort. This effort is held fixed in the model.

Even allowing endogenous borrower effort ($\pi = \pi(\tilde{e})$), the reform's impact on effort incentives is bounded and economically negligible. The borrower's first-order condition for effort balances the marginal cost against the marginal benefit of success, the good-state residual $f(I) - (1 + r)L$. The reform raises r (via the information loss channel) but also lowers L^* (via the tighter borrowing constraint), making the direction of the effort change ambiguous. The estimated spread changes are 10–17 bp on a $\sim$5 pp mean, a 2–3% shift in borrowing cost, economically negligible as an effort incentive. Under competitive lending, the zero-profit condition pins the lender's expected return at $(1 + \rho)L$ regardless of θ_c; the guarantee ensures that the borrower's cost absorption is quantitatively small.

The null charge-off DiD (Section 6.4) provides direct empirical support: if the reform materially changed borrower effort, charge-off rates would respond. The precise zero is consistent with π being invariant to the collateral valuation methodology.

A.4 Relationship to Rampini and Viswanathan (2025)

I draw on R&V for the conceptual framework, specifically the distinction between explicit and implicit collateralization and the collateralization pecking order, while departing from their model structure. R&V develop an enforcement-based dynamic model in which default does not occur in equilibrium; I use an expected-loss framework with exogenous default and government guarantees. The two approaches yield equivalent comparative statics in θ_c but differ in mechanism.

R&V Concept	This Paper	Relationship
Explicit collateral (perfected lien)	$\theta_c \cdot C$ (UCC lien on business assets)	Guarantee $g = 0.75$ adds government layer
Implicit collateral (unencumbered)	$\theta_u \cdot E_a$ (unprotected home equity)	HE determines availability
Collateralization pecking order	HE determines collateral base breadth	Narrow base (unlimited HE) = more sensitive
Recovery rates exogenous	θ_c endogenized via reform	$\bar{\theta} + \delta \to \bar{\theta}_{\text{std}}$ (key contribution)
Encumbrance cost $\kappa > 0$	Abstracted (κ not binding)	SBA borrowers sufficiently constrained
No default in equilibrium	π treated as invariant to the reform	Charge-off DiD null is consistent (not a formal test of exogeneity)
Life-cycle accumulation	Repeat borrowers	growth $= \log(L_2/L_1)$

R&V treat collateral recovery rates as exogenous and study the choice between secured and unsecured debt. This paper endogenizes θ_c through the government's choice between discretionary and standardized valuation rules, and provides causal evidence on how this choice affects the cost and availability of credit.

A.5 Collateral Shortfall Justification

Assumption 1 is empirically reasonable at the loan sizes in my sample ($>$ \$350K). Recovery on unprotected home equity through deficiency judgment is substantially lower than on explicitly liened assets ($\theta_u \in [0.10, 0.30]$ vs. $\theta_c \in [0.50, 0.85]$). At a \$500K loan, even with \$200K in business collateral and \$300K in home equity, total effective collateral is \$130–240K, well below the loan amount. The assumption is maintained in both periods: since $\varphi < 1$ (the fraction of investment that becomes pledgeable collateral), each dollar of investment generates less than one dollar of new pledgeable collateral, so the collateral-to-loan ratio is weakly declining for expanding borrowers ($L_2 > L_1$), making the shortfall assumption easier to satisfy in period 2 than in period 1.

A.6 Relationship Spread Discount and Optimal Loan Size

Pre-reform, a relationship borrower ($\delta > 0$) received a spread

$$s^{\text{pre}}(\delta) = \frac{\pi}{1-\pi}\Big[(1+\rho) - g - (1-g)\frac{(\bar{\theta}+\delta)C + \theta_u E_a}{L}\Big] + \frac{\pi}{1-\pi}(1-g)\lambda(\sigma^2(\delta))\frac{C}{L}. \quad \text{(IA.2)}$$

The spread benefit of the relationship was

$$\Delta s(\delta) = s^{\text{pre}}(0) - s^{\text{pre}}(\delta) = \frac{\pi}{1-\pi}(1-g)\cdot\frac{C}{L}\big[\delta + \lambda(\sigma^2(0)) - \lambda(\sigma^2(\delta))\big] > 0. \quad \text{(IA.3)}$$

The relationship discount has two components: a direct valuation premium ($\delta \cdot C/L$) and an uncertainty-reduction premium ($[\lambda(\sigma^2(0)) - \lambda(\sigma^2(\delta))] \cdot C/L$). Post-reform, all borrowers face $\theta_c = \bar{\theta}_{\text{std}}$ with $\sigma^2 = 0$; both components of the relationship discount are gone.

With decreasing returns, the optimal investment equates marginal return to marginal cost: $f'(I^*) = 1 + \rho + s(L^*)$. Since the spread is increasing in L for fixed Ω (as Ω/L falls with larger L), a higher θ_c raises Ω, shifting the spread schedule downward and inducing more borrowing: $\partial L^*/\partial\theta_c > 0$.

A.7 Retained Earnings and Credit Growth Algebra

If the period-1 project succeeds, retained earnings are

$$\Pi_1 = f(I_1) - (1+\rho+s_1)L_1 - (1+\rho)w_1. \quad \text{(IA.4)}$$

Net worth grows: $w_2 = w_1 + \Pi_1$. Business collateral grows: $C_2 = C_1 + \varphi \cdot I_1$. Retained earnings depend on the spread:

$$\frac{\partial \Pi_1}{\partial \theta_c} = -L_1 \cdot \frac{\partial s_1}{\partial \theta_c} > 0. \quad \text{(IA.5)}$$

Higher θ_c lowers the spread, raises retained earnings, and increases wealth available for period-2 borrowing.

A.8 Threshold Condition: Quantitative Plausibility

Proposition 3 requires that δ exceeds the average improvement $\Delta\bar{\theta} \equiv \bar{\theta}_{\text{std}} - \bar{\theta}$ for a non-trivial mass of relationship borrowers. This is quantitatively plausible: pre-reform OLV for used equipment was typically 30–50% of book value (conservative liquidation estimates), while the standardized schedule credits 50% of NBV, implying $\Delta\bar{\theta} \approx 0.00$–$0.20$.

A relationship lender who valued well-maintained equipment 15–25 percentage points above the conservative baseline, a modest informational advantage, would exceed the threshold.

A.9 Model Calibration

The spread estimate provides a back-of-envelope calibration of the implied valuation improvement $\Delta\theta_c$. From the spread equation, the sensitivity is $\partial s/\partial\theta_c = [\pi/(1-\pi)](1-g)(C/L) \approx [0.10/0.90](0.25)(0.60) = 0.0167$, using sample averages for the default rate ($\pi = 0.10$), unguaranteed share ($1 - g = 0.25$), and collateral-to-loan ratio ($C/L = 0.60$). The binary spread DiD of 10.6 basis points implies $\Delta\theta_c = 0.00106/0.0167 \approx 6.3$ percentage points. The actual schedule change from discretionary OLV (baseline $\sim 50\%$) to preset NBV ($\sim 75\%$) is approximately 25 percentage points, so the implied pass-through is $6.3/25 = 25\%$. This is consistent with voluntary pledging in non-Texas unlimited-exemption states attenuating the effective treatment to roughly one-quarter of the full schedule change. For δ, the experienced-borrower continuous coefficient (-1.50 pp per log-exemption) exceeds the first-time coefficient (-0.26 pp) by 1.24 pp, attributable to the soft-information channel. Translating this difference to the model's δ requires assumptions about the credit growth elasticity, so I treat the 1.24 pp excess as a lower bound on the information premium eliminated by standardization.

A.10 Discussion of Alternative Interpretations

Lender incentive compatibility. If $\bar{\theta}_{\text{std}} > \bar{\theta}$, why do lenders follow a more generous schedule? Three reasons: (i) pre-reform OLV was conservatively biased (lenders faced asymmetric regulatory risk from overvaluation and no penalty for undervaluation, so the standardized schedule likely corrects a downward bias); (ii) the collateral rules are binding, and competition forces pricing to converge to the mandated valuation; (iii) the guarantee limits the cost of any overvaluation to $(1-g) = 25\%$ of the gap.

Agency interpretation. If δ reflects lender overvaluation (agency relative to the SBA) rather than genuine soft information, standardization corrects moral hazard rather than destroying information. Two pieces of evidence weigh against this view: under agency, standardization should reduce defaults (the charge-off DiD is null, consistent with unchanged screening), and overvaluation incentives should be largest for new borrowers (the observed concentration in experienced borrowers is contrary to what agency predicts).

Competitive lending. The competitive benchmark yields maximum pass-through of the reform to spreads: with market power, lenders could partially absorb the effective guarantee improvement into margins. The attenuation of the credit growth estimate when restricting to borrowers at multi-state lenders (Section 8) is consistent with two inseparable channels: multi-state lenders may produce less δ (less relationship-intensive) or may exercise more market power (absorbing the reform into margins). The baseline DiD is conservative under either interpretation.

Extensive margin. The model is entirely intensive margin: the reform changes prices and loan sizes but does not generate credit rationing. This is consistent with the SBA guarantee structure: with $g = 0.75$, the SBA absorbs three-quarters of the collateral shortfall upon default, so the lender's participation constraint is easily satisfied for any borrower meeting the program's creditworthiness requirements. The guarantee effectively eliminates the extensive margin for qualified borrowers, which I confirm empirically: loan counts, cancellation rates, and borrower composition are unchanged at the reform boundary.

Connection to asset types. The reform's impact on θ_c varies by asset type through two channels formalized in Proposition 1. First, pre-reform valuation uncertainty (σ^2) was higher for movable assets: equipment and inventory lack standardized appraisal methodologies (no comparable-sales data, condition-dependent recovery), while real estate benefits from established appraisal practices and public transaction records. The uncertainty-reduction component $\kappa\sigma^2 \to 0$ is therefore larger for movable assets. Second, the soft-information premium (δ) was more valuable for movable assets: lender-specific knowledge of equipment condition, maintenance history, and redeployability generated a larger informational advantage than real estate appraisal expertise. The preset schedule therefore eliminates more private information for movable-asset loans. These two effects work in opposite directions for the net spread change: the uncertainty reduction lowers spreads (benefiting all borrowers), while the information loss raises spreads (harming relationship borrowers). The empirical results (Section 6.2) confirm this: immovable-collateral loans show a clean spread reduction, while movable-collateral loans show heterogeneous effects that depend on the lender's effort response (Section 7.3). The effort-adaptation mechanism (Proposition 2) resolves this heterogeneity: movable-asset lenders in unlimited-HE states invest in documentation to partially recover the private valuation advantage, attenuating the net spread change.

B Additional Tables and Figures

Tables and figures referenced in Section 6 are collected at the end of the manuscript per journal convention.

C Data Construction Details

This appendix describes the three data construction pipelines: SBA loan cleaning and borrower identification, UCC collateral classification, and the SBA-to-UCC fuzzy match. Full replication code is provided in the replication package.

C.1 SBA Borrower Cleaning and Identification

The raw SBA 7(a) dataset contains borrower names, addresses, and lender identifiers with substantial inconsistency: name variants for the same entity (e.g., "ACME INC" vs. "ACME INCORPORATED DBA BEST PIZZA"), address formatting differences, and missing fields. A four-step pipeline standardizes these records and assigns unique borrower identifiers.

Name and address standardization. Borrower names are uppercased, stripped of punctuation, and normalized for legal entity suffixes (LLC, INC, CORP $\rightarrow$ canonical short forms). DBA (doing-business-as) trade names are extracted and stored separately. Street addresses are parsed into number, street name, and unit components, with directional (N/S/E/W) and street type (ST/AVE/BLVD) standardization. Cities are cleaned for embedded state abbreviations and saint/fort/mount contractions.

Borrower identification via union-find. Unique borrower identities are constructed using a union-find algorithm with three blocking strategies and three matching passes of increasing aggressiveness. The blocking keys combine state with zip code, city, or street-level identifiers to generate candidate pairs for comparison:

1. *Strong matches:* Require name similarity ≥ 92 with address confirmation (same zip or street match). Exact address matches accept weaker name similarity (≥ 88).

2. *Weak matches:* Require very strong name similarity (≥ 96) with geographic overlap (same zip or city). Applied only to pairs not already merged, with a guard preventing merges across more than five disjoint zip codes.

3. *DBA cross-match:* Exact-matches DBA names to legal names within the same city and state, capturing entities that operate under trade names distinct from their legal identity.

No cross-state merges are permitted. Two relationship indicators are computed after borrower identification: an indicator for any prior SBA loan by the same borrower, and an indicator for any prior loan from the same lender.

Policy variable construction. The analysis-ready dataset merges three policy variables: (a) the SBA maximum allowable interest rate, assigned by loan type (rate type × maturity × amount tier) and month; (b) state-level homestead exemption amounts from a panel of state statutes; and (c) county-level exemptions for states with sub-state variation (California, New York, Washington).

C.2 UCC Collateral Classification

Each UCC filing's free-text collateral description is classified into one of 14 Article 9 asset categories using a hybrid pipeline combining sentence-transformer embeddings with deterministic rule-based overrides.

Text extraction. Collateral descriptions are extracted from UCC filing documents (PDF or text) using a three-tier approach: (a) regex-based exact field extraction from standardized form layouts (30% of filings); (b) fuzzy header matching that identifies "collateral description" fields with OCR-tolerant patterns (65%); and (c) a window-based fallback that extracts 2,850-character windows around collateral keywords for filings with non-standard layouts (5%). Fewer than 0.1% of filings fail extraction entirely.

Classification hierarchy. The classifier assigns one primary category per filing from the following hierarchy: *blanket lien* (detected via 30+ OCR-tolerant regex patterns for "all assets" constructions); *equipment*; *inventory*; *accounts receivable*; *chattel paper*; *vehicles* (identified by VIN regex); *aircraft*; *vessels*; *farm products*; *deposit accounts*; *real property related*; *intangibles* (patents, trademarks, copyrights); *fixtures*; and *unclassified* (default when confidence falls below threshold). The semantic backbone uses `all-mpnet-base-v2` sentence-transformer embeddings with cosine similarity against category-specific prototype sentences, augmented by deterministic boosts for explicit keyword cues (e.g., +0.08 for blanket language, +0.10 for aircraft-specific terms).

Scope and specificity metrics. Three metrics characterize each filing's collateral documentation quality:

- *Perfection scope score* (0–5): Weighted count of breadth indicators. Blanket language receives weight 2; after-acquired property, proceeds / products, location breadth, and accessions / replacements each receive weight 1.
- *Specificity score*: Ratio of capped identifier counts (VINs, serial numbers, account numbers, parcel IDs) to token count, scaled by 100. Type-adjusted by subtracting the expected specificity for the filing's primary category.
- *Blanket scope indicator*: Binary flag equal to 1 if the filing contains explicit "all assets" language or "all" plus at least one breadth cue ("wherever located," "hereafter acquired," "proceeds").

The composite effort index used in the regression analysis standardizes three measures within the blanket lien subsample: ($1 -$ blanket scope), specificity score, and number of distinct asset categories. An independent audit of the Colorado classification pipeline found a post-correction error rate below 0.5%.

C.3 SBA-to-UCC Fuzzy Matching

The SBA-to-UCC match links 117,000+ SBA loans to 5–10 million UCC filings per state using industrial-scale fuzzy name matching with geographic support.

Blocking. Candidate pairs are generated using three independent blocking keys: three-character name prefixes, per-word three-character prefixes, and character bigrams and trigrams of the standardized borrower name. Blocks exceeding 50,000 entries are skipped to prevent combinatorial explosion. Within each block, candidates are pre-screened using RapidFuzz WRatio with a cutoff of 55.

Scoring. Each candidate pair receives three scores: (a) a *name score*, the maximum RapidFuzz ratio across all SBA name variants (legal name, DBA, cleaned) and UCC debtor variants, penalized for short names; (b) a *geographic score* combining street similarity (weight 0.60), city similarity (0.10), and zip match (0.30); and (c) a *combined score* weighting name (0.90) and geographic (0.10) components.

Date window and acceptance gates. An asymmetric date window restricts matches to UCC filings within 180 days before and 365 days after the SBA approval date, reflecting the typical timing of lien perfection. Candidates are accepted if the adjusted name score exceeds 90, or if it exceeds 70 with geographic confirmation (city score $\geq$ 80 or exact zip match). The minimum combined score is 70.

Deduplication and confidence. Each SBA loan retains its single best UCC match, selected by confidence tier (HIGH: name $\geq$ 90; MEDIUM: all others passing the gate), then by lender match (SBA lender identified in UCC secured party via exact or fuzzy matching to FDIC/SBIC/SBLC reference lists), then by date proximity, then by combined score. Match precision, evaluated on 500 random samples in California with manual verification, exceeds 95%. Approximately 94% of retained matches are classified as HIGH confidence (combined score $\geq$ 90).

C.4 Within-Borrower Spread Test

Table IA.14 adds borrower fixed effects to the spread DiD, restricting identification to within-borrower variation across the reform boundary. The coefficient is a precise zero across all specifications: $+0.03$ ($p = 0.84$) for the 7v12 binary and $+0.003$ ($p = 0.69$) for the continuous measure. The result is robust to varying the fixed-effect structure (borrower + quarter only, borrower + lender, borrower + full FE set) and to dropping all controls. The sample collapses by approximately 89% after singleton removal, reflecting the rarity of repeat SBA borrowing.

C.5 Additional Figures

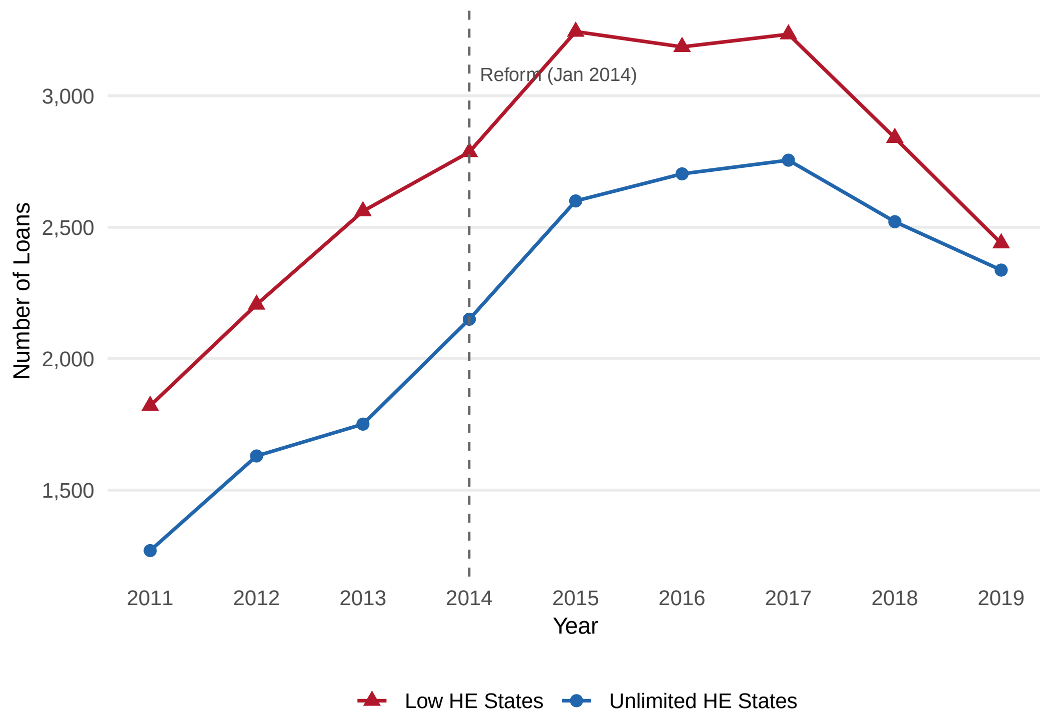

Figure IA.1: Loan Volume by Treatment Group and Year

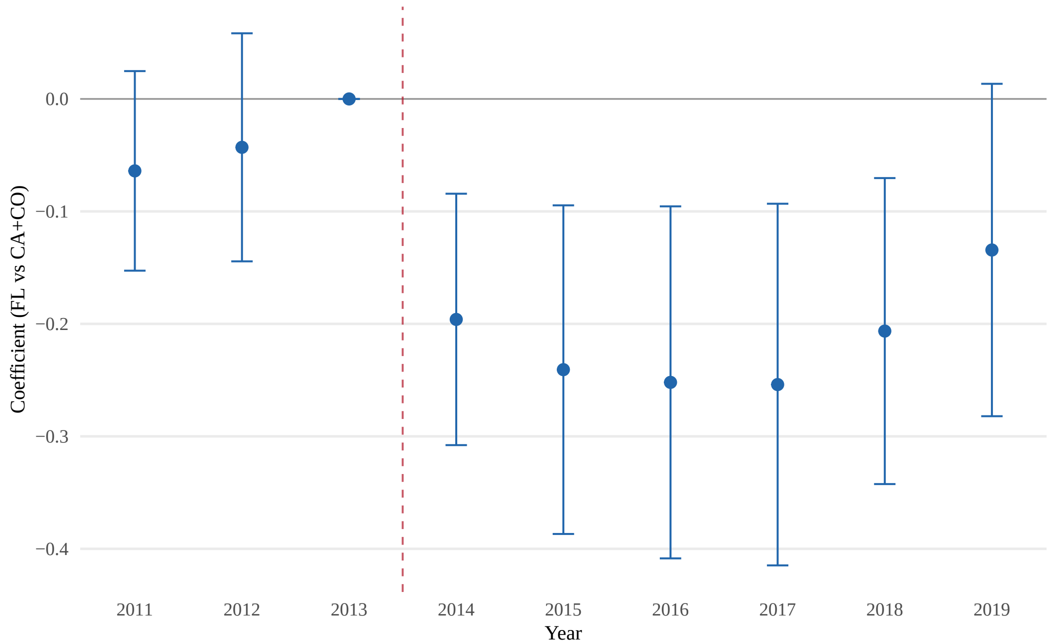

Figure IA.2: Event Study: Blanket Scope within Blanket Lien Filings. Coefficients on Year $\times$ Treated$_{FL}$ interactions from equation (15) with blanket scope as outcome, estimated on the blanket lien subsample. Reference year is 2013. Bars show 95% confidence intervals clustered at the lender level. Wald test for pre-trends: $F = 1.00$, $p = 0.369$.

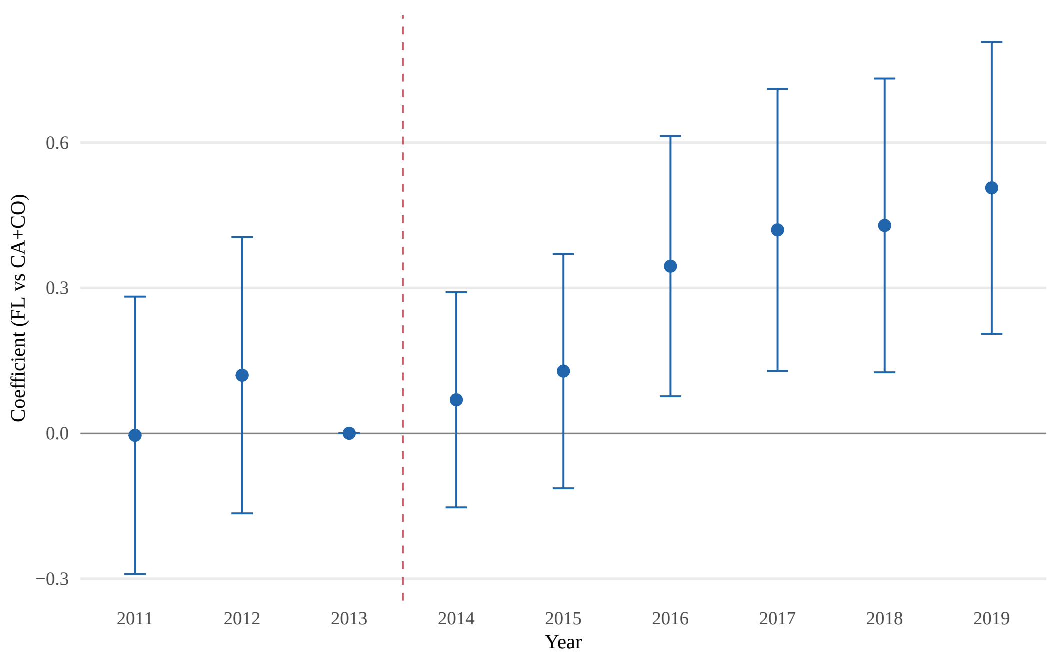

Figure IA.3: Event Study: Collateral Scope Score (All UCC-Classified). Coefficients on Year $\times$ Treated$_{FL}$ interactions with scope score as outcome, estimated on the full UCC-classified sample. Reference year is 2013. Wald test for pre-trends: $F = 0.69$, $p = 0.504$.

Table IA.1: Summary Statistics: Binary HE Design (7v12 States)

	Treated (Unlimited HE)		Control (Low HE)		Norm.
	Pre	Post	Pre	Post	Diff.
Panel A: All loans (>$350K Standard Guaranty)					
Loan amount (in $1,000)	1171.74	1290.29	1142.41	1246.63	0.044
	(925.08)	(1045.34)	(892.09)	(1012.33)	
Term (months)	206.39	207.78	204.78	204.74	0.030
	(90.19)	(90.58)	(90.95)	(88.94)	
Spread (pp)	5.02	4.71	5.18	4.89	-0.178
	(1.05)	(1.10)	(0.95)	(1.04)	
Fixed rate	0.230	0.223	0.157	0.178	0.132
Relationship	0.072	0.066	0.075	0.070	-0.017
Previous SBA	0.120	0.107	0.128	0.123	-0.043
Collateral	0.931	0.908	0.878	0.917	0.026
Corporation	0.905	0.934	0.929	0.944	-0.051
Franchise	0.107	0.218	0.116	0.209	0.021
Observations	4,651	15,066	6,588	17,729	
Panel B: Repeat borrowers ($350K–$1M)					
First loan amount (in $1,000)	606.71	608.84	598.76	616.34	-0.006
	(183.00)	(184.93)	(177.32)	(180.70)	
First loan term (months)	171.19	180.83	185.06	182.20	-0.068
	(91.08)	(91.64)	(91.13)	(87.56)	
First loan spread (pp)	5.02	4.93	5.32	5.06	-0.198
	(1.05)	(1.09)	(0.83)	(1.01)	
Fixed rate	0.227	0.191	0.108	0.177	0.146
Relationship	0.136	0.218	0.209	0.192	-0.028
Previous SBA	0.199	0.266	0.313	0.261	-0.094
Collateral	0.920	0.915	0.824	0.897	0.161
Corporation	0.926	0.935	0.942	0.963	-0.099
Franchise	0.045	0.212	0.097	0.182	0.004
Credit growth	-0.00	0.01	-0.00	-0.02	0.063
	(0.36)	(0.29)	(0.28)	(0.29)	
Number of loans	2.17	2.09	2.11	2.09	0.049
	(0.55)	(0.35)	(0.39)	(0.32)	
Observations	176	293	278	406	

Notes: Panel A reports loan-level statistics for all >$350K Standard Guaranty loans in 7 unlimited-homestead-exemption states (treated) vs. 12 low-exemption states (control, exemption $\leq$ $23K), 2011Q1–2019Q4. Panel B reports first-loan characteristics and credit growth for repeat borrowers ($350K–$1M). Normalized differences computed as $(\bar{x}_T - \bar{x}_C)/\sqrt{(s_T^2 + s_C^2)/2}$.

Table IA.2: Spread Difference-in-Differences: Binary HE Design (7v12 States)

	Panel A: Full Sample			Panel B: By Loan Size		
	(1) Baseline	(2) + HPrice	(3) NAICS×Qtr	(4) \$350K–500K	(5) \$500K–1M	(6) \$1M–2M
Treated × Post	-0.1063^{***}	-0.1092^{***}	-0.1095^{***}	-0.1082^{*}	-0.1078^{**}	-0.1063^{**}
	(0.0405)	(0.0402)	(0.0386)	(0.0605)	(0.0432)	(0.0478)
Observations	43,770	43,205	43,171	9,179	15,417	11,143
Within R^2	0.3828	0.3819	0.3802	0.3939	0.3865	0.3384
Lender FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
NAICS2 FE	No	No	×Qtr	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Home price control	No	Yes	Yes	No	No	No

Notes: Dependent variable is the loan spread (percentage points over matched-maturity Treasury). Treated = 7 unlimited homestead exemption states; Control = 12 low-exemption states (exemption $\leq$ \$23K). Sample: >\$350K Standard Guaranty loans, 2011Q1–2019Q4. Column (3) replaces additive NAICS2 FE with NAICS2 × quarter interactions, absorbing industry-specific time shocks. All columns include controls for log loan amount, term, fixed-rate indicator, relationship lending, previous SBA, collateral, corporation, and franchise indicators. Standard errors clustered at the lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.3: Pre-Trend Tests

	Panel A: Credit Growth		Panel B: Repeat Ratio		Panel C: Spread	
	Coef.	SE	Coef.	SE	Coef.	SE
2011	0.1315^{**}	(0.0530)	0.0018	(0.0136)	0.0175	(0.0338)
2012	−0.0086	(0.0495)	0.0415^{**}	(0.0193)	0.0731^{**}	(0.0339)
2014	0.0267	(0.0623)	0.0169	(0.0222)	−0.0455	(0.0383)
2015	−0.0170	(0.0382)	0.0179	(0.0212)	−0.0633	(0.0427)
2016	0.0114	(0.0550)	0.0270^{*}	(0.0148)	−0.0762	(0.0518)
2017	−0.0221	(0.0489)	0.0258	(0.0159)	−0.0555	(0.0533)
2018	0.0776	(0.0648)	0.0283	(0.0174)	-0.0986^{*}	(0.0584)
2019	−0.0672	(0.0648)	−0.0086	(0.0217)	−0.1178	(0.0714)
Pre-trend Wald F	3.10		2.54		2.68	
p-value	0.046		0.080		0.068	

Notes: Event study coefficients with 2013 as base year. Panel A: credit growth for \$350K–\$1M repeat borrowers (entry-year cohorts). Panel B: any-repeat share at state-quarter level. Panel C: loan-level spread. Pre-trend Wald test: joint significance of 2011 and 2012 coefficients. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.4: Robustness: Sample and Control Sensitivity

	(1) \$350K–1M	(2) \$350K–2M	(3) Full >350K	(4) No HPrice	(5) + HPrice	(6) Multi-state
Treated × Post	−0.0349	−0.0301	0.0430	−0.0349	−0.0392	−0.0108
	(0.0339)	(0.0421)	(0.0479)	(0.0339)	(0.0347)	(0.0339)
Observations	1,096	2,021	2,661	1,096	1,079	991

Notes: Dependent variable is credit growth for repeat borrowers. All specifications include lender and entry-quarter FE with lender-clustered SEs. Column (4) is identical to (1) for reference. Column (5) adds home price appreciation as control. Column (6) restricts to borrowers at lenders operating in both treated and control states. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.5: Leave-One-State-Out Robustness

	Spread			Credit Growth		
Dropped State	Coef.	SE	*N*	Coef.	SE	*N*
AL	−0.1037**	(0.0415)	42,841	−0.0027	(0.0419)	899
AR	−0.1005**	(0.0407)	43,132	−0.0032	(0.0419)	904
FL	−0.1653***	(0.0507)	37,433	−0.0009	(0.0547)	775
GA	−0.1073**	(0.0435)	38,218	−0.0278	(0.0413)	815
IA	−0.1037**	(0.0413)	43,367	−0.0034	(0.0423)	906
IL	−0.1034***	(0.0386)	40,211	−0.0488	(0.0512)	770
IN	−0.1120***	(0.0404)	42,033	−0.0124	(0.0426)	870
KS	−0.1078**	(0.0419)	43,101	−0.0064	(0.0425)	897
KY	−0.1031**	(0.0404)	43,067	−0.0005	(0.0420)	900
MD	−0.1045**	(0.0406)	42,514	−0.0070	(0.0419)	890
MO	−0.1100***	(0.0418)	42,193	−0.0065	(0.0428)	855
NJ	−0.1127***	(0.0418)	40,621	0.0220	(0.0429)	832
OK	−0.1048**	(0.0410)	42,958	0.0019	(0.0419)	896
PA	−0.0920**	(0.0399)	41,024	0.0059	(0.0434)	844
SD	−0.1056***	(0.0405)	43,623	−0.0104	(0.0420)	907
TN	−0.1040**	(0.0417)	42,507	−0.0011	(0.0430)	893
TX	−0.0141	(0.0430)	33,154	0.0434	(0.0546)	691
VA	−0.1024**	(0.0428)	42,219	0.0008	(0.0419)	897
WY	−0.1052**	(0.0409)	43,618	−0.0025	(0.0420)	909

Notes: Each row drops one state from the 7v12 binary sample. Spread: >\$350K Standard Guaranty, lender + quarter + NAICS + county FE. Growth: ≤\$1M repeat borrowers, lender + entry-quarter + NAICS FE. SE clustered at lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.6: Continuous Leave-One-State-Out: Spread

	Spread				Spread		
Dropped State	Coef.	SE	N	Dropped State	Coef.	SE	N
AL	-0.0091^{***}	(0.0034)	102,578	NC	-0.0092^{***}	(0.0034)	100,656
AR	-0.0088^{***}	(0.0034)	102,869	ND	-0.0091^{***}	(0.0034)	103,390
AZ	-0.0090^{***}	(0.0034)	100,373	NE	-0.0091^{***}	(0.0034)	103,133
CA	-0.0087^{**}	(0.0035)	83,804	NH	-0.0091^{***}	(0.0034)	103,276
CO	-0.0091^{***}	(0.0034)	99,890	NJ	-0.0133^{***}	(0.0045)	100,355
CT	-0.0091^{***}	(0.0034)	102,670	NM	-0.0091^{***}	(0.0034)	102,942
DC	-0.0093^{***}	(0.0034)	103,306	NV	-0.0090^{***}	(0.0034)	102,510
DE	-0.0091^{***}	(0.0034)	103,265	NY	-0.0093^{***}	(0.0034)	100,221
FL	-0.0118^{***}	(0.0037)	97,166	OH	-0.0093^{***}	(0.0034)	99,918
GA	-0.0091^{***}	(0.0034)	97,951	OK	-0.0091^{***}	(0.0034)	102,695
IA	-0.0090^{***}	(0.0034)	103,101	OR	-0.0092^{***}	(0.0034)	101,703
ID	-0.0092^{***}	(0.0034)	102,803	PA	-0.0073^{*}	(0.0038)	100,760
IL	-0.0089^{***}	(0.0033)	99,947	RI	-0.0091^{***}	(0.0034)	103,282
IN	-0.0092^{***}	(0.0034)	101,772	SC	-0.0091^{***}	(0.0034)	102,250
KS	-0.0093^{***}	(0.0034)	102,837	SD	-0.0092^{***}	(0.0034)	103,360
KY	-0.0090^{***}	(0.0033)	102,803	TN	-0.0090^{***}	(0.0034)	102,243
LA	-0.0091^{***}	(0.0034)	102,701	TX	-0.0051^{*}	(0.0029)	92,887
MA	-0.0093^{***}	(0.0034)	102,478	UT	-0.0093^{***}	(0.0034)	101,650
MD	-0.0091^{***}	(0.0034)	102,251	VA	-0.0090^{**}	(0.0035)	101,955
ME	-0.0091^{***}	(0.0034)	103,330	VT	-0.0091^{***}	(0.0034)	103,418
MI	-0.0091^{***}	(0.0033)	100,463	WA	-0.0091^{***}	(0.0034)	99,921
MN	-0.0091^{***}	(0.0034)	101,567	WI	-0.0091^{***}	(0.0034)	101,110
MO	-0.0091^{***}	(0.0034)	101,929	WV	-0.0091^{***}	(0.0034)	103,369
MS	-0.0091^{***}	(0.0034)	102,810	WY	-0.0091^{***}	(0.0034)	103,351
MT	-0.0091^{***}	(0.0034)	103,235				

Notes: Each row drops one state from the all-states continuous specification. Dep. var.: spread (pp). $\log(1 + \text{exemption}) \times \text{Post}$. Controls: log amount, term, fixed-rate, relationship, previous SBA, collateral, corporation, franchise. FE: quarter, lender, NAICS2, county. SE clustered at lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.7: Charge-Off Event Study

Year	Coefficient	SE
2011	-0.0185^{*}	(0.0101)
2012	-0.0003	(0.0104)
2014	-0.0022	(0.0087)
2015	0.0090	(0.0079)
2016	0.0097	(0.0085)
2017	0.0080	(0.0077)
2018	0.0097	(0.0091)
2019	0.0013	(0.0092)
Observations	43,770	

Notes: Binary charge-off event study with 2013 as base year. Sample: >$350K Standard Guaranty, 7v12 states. Lender FE + quarter FE, lender-clustered SEs. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.8: Symmetric Window Robustness (2011Q1–2016Q4)

	Spread DiD				Credit Growth	
	(1) Full	(2) $350K–500K	(3) $500K–1M	(4) $1M–2M	(5) All rates	(6) Variable
Treated × Post	-0.1010^{***}	-0.1290^{**}	-0.1104^{**}	-0.1008^{**}	-0.0208	-0.0354
	(0.0365)	(0.0533)	(0.0436)	(0.0425)	(0.0399)	(0.0459)
Observations	33,596	7,104	11,815	8,613	832	682
Lender FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter/Entry FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Symmetric 3+3 year window around the Jan 2014 reform (2011Q1–2016Q4). Columns (1)–(4): loan-level spread DiD. Columns (5)–(6): credit growth for ≤$1M repeat borrowers. All specs use lender FE + quarter/entry-quarter FE, lender-clustered SEs. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.9: Extensive Margin: Probability of Obtaining a Second SBA Loan

	(1) Baseline	(2) + Home Price
Treated × Post	-0.0026	-0.0054
	(0.0075)	(0.0081)
Observations	23,186	22,947
Within R^2	0.0229	0.0230
Dep. var. mean	0.0471	0.0471
Loan controls	Yes	Yes
Lender FE	Yes	Yes
Quarter FE	Yes	Yes
NAICS2 FE	Yes	Yes
County FE	Yes	Yes
Home price control	No	Yes

Notes: Dep. var.: P(second SBA loan). \$350K–\$1M Std. Guaranty, 7v12 states. SE clustered at lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.10: Credit Growth DiD: Same-Lender vs. Switcher Borrowers

	(1) Same Lender	(2) Pooled
Treated × Post	0.0222	-0.0028
	(0.0476)	(0.0419)
Observations	723	911
Within R^2	0.2808	0.3102
Dep. var. mean	-0.0157	-0.0037
Borrower controls	Yes	Yes
Entry quarter FE	Yes	Yes
Lender FE	Yes	Yes
NAICS2 FE	Yes	Yes
County FE	Yes	Yes

Notes: Dep. var.: credit growth $\log(L_2/L_1)$, \$350K–\$1M repeat borrowers. Col (1): same-lender pairs. Col (2): pooled. 7v12 states. Borrower controls, lender + entry-quarter + NAICS2 FE. SE clustered at lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.11: Wild Cluster Bootstrap Inference

Specification	$\hat{\beta}$	Lender p	WCB p	G_{county}
Panel A: Design 1 — Binary 7v12				
Spread DiD baseline	−0.106	0.009	0.000	1,140
Spread DiD + hprice	−0.109	0.007	0.000	1,128
Credit growth	−0.003	0.946	0.954	156
Credit growth prev-SBA	−0.178	0.006	0.001	128
Credit growth first-time	0.016	0.539	0.514	267
Panel B: Design 1 — Continuous HE				
Spread $\sim$ log(1+ex)	−0.009	0.008	0.000	2,129
Spread $\sim$ rank(ex)	−0.091	0.008	0.000	2,129
Growth prev-SBA	−0.012	0.001	0.003	334
Panel C: Falsification				
Exp→Exp growth	−0.017	0.809	0.745	440
Cancellation LPM	−0.003	0.700	0.677	1,140

Notes: Lender p: p-value from primary lender-clustered SE. WCB p: wild cluster bootstrap p-value resampled at the county level (Webb six-point distribution, 9,999 replications). G_{county}: number of county-level bootstrap groups. County-level clustering captures geographic correlation in local housing and lending markets. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.12: HonestDiD Sensitivity Analysis (Quarterly Event Study)

	Relative Magnitudes (RM)		Smoothness (SD)	
Parameter	95% CI	Includes 0?	95% CI	Includes 0?
Panel A: Spread (pp over Treasury)				
$\bar{M} = 0.0$	$[-0.228, 0.286]$	Yes		
$\bar{M} = 0.5$	$[-0.296, 0.359]$	Yes		
$\bar{M} = 1.0$	$[-0.385, 0.459]$	Yes		
$\bar{M} = 1.5$	$[-0.506, 0.595]$	Yes		
$\bar{M} = 2.0$	$[-0.637, 0.742]$	Yes		
$M = 0.00$			$[-0.364, 0.244]$	Yes
$M = 0.03$			$[-0.390, 0.268]$	Yes
$M = 0.05$			$[-0.409, 0.286]$	Yes
Original	$[-0.227, 0.286]$	Yes		
Panel B: Credit Growth				
$\bar{M} = 0.0$	$[-1.140, -0.520]$	No		
$\bar{M} = 0.5$	$[-1.418, -0.262]$	No		
$\bar{M} = 1.0$	$[-1.831, 0.145]$	Yes		
$\bar{M} = 1.5$	$[-2.271, 0.585]$	Yes		
$\bar{M} = 2.0$	$[-2.723, 1.030]$	Yes		
$M = 0.00$			$[-1.928, -0.546]$	No
$M = 0.03$			$[-1.953, -0.518]$	No
$M = 0.05$			$[-1.965, -0.502]$	No
Original	$[-1.147, -0.515]$	No		

Notes: HonestDiD robust confidence intervals from quarterly event study specifications. RM: $\bar{M}$ bounds the ratio of post- to pre-treatment trend violations. SD: M bounds the second difference of the bias. Implemented via `HonestDiD` R package.

Table IA.13: Clustering Robustness

		Standard errors / p-values				Clusters		
Specification	$\hat{\beta}$	Lender	County	Two-way	State	G_L	G_C	G_S
Spread baseline (T2)	-0.1063^{***}	0.009	0.000	0.009	0.111	315	1387	19
		(0.0405)	(0.0222)	(0.0407)	(0.0634)			
Spread + hprice (T2)	-0.1092^{***}	0.007	0.000	0.007	0.091	315	1370	19
		(0.0402)	(0.0217)	(0.0399)	(0.0612)			
Credit growth (T3)	-0.0028	0.946	0.956	0.949	0.953	189	347	19
		(0.0419)	(0.0515)	(0.0437)	(0.0469)			
Growth prev-SBA (T4)	-0.1467	0.406	0.333	0.227	0.339	111	153	18
		(0.1744)	(0.1495)	(0.1194)	(0.1476)			

Notes: Four clustering levels: Lender (baseline), County, Two-way (lender $\times$ county), State. SE in parentheses; p-values above. $G_L/G_C/G_S$ = lender/county/state cluster counts. Stars reflect lender-clustered significance. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table IA.14: Within-Borrower Spread DiD: Borrower Fixed Effects

	(1) Baseline	(2) +Borr FE (full)	(3) +Borr FE (qtr only)	(4) +Borr FE (+lender)	(5) +Borr FE (no ctrl)
Panel A: 7v12 Binary					
Treated $\times$ Post	-0.1072^{***}	0.0254	0.0687	0.0653	0.0211
	(0.0389)	(0.1273)	(0.0964)	(0.1092)	(0.1433)
Observations	43,733	5,557	5,800	5,761	5,800
Panel B: All-States Continuous					
Log(1+HE) $\times$ Post	-0.0091^{***}	0.0034	0.0032	0.0021	0.0017
	(0.0033)	(0.0087)	(0.0077)	(0.0085)	(0.0110)
Observations	103,476	15,102	15,267	15,235	15,267
Quarter FE	Yes	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes		Yes	
NAICS2 $\times$ Qtr	Yes	Yes			
County FE	Yes	Yes			
Borrower FE		Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	

Notes: Column (1) reproduces the baseline spread DiD. Columns (2)–(5) add borrower fixed effects under varying FE saturation. The sample collapses because borrower FE absorbs the state-level treatment indicator for single-loan borrowers; only repeat borrowers with pre- and post-reform loans contribute. Standard errors clustered at the lender level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.